MATHEMATICS

INDEFINITE INTEGRATION

THEORY SHEET

ENGLISH

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1.1 INTRODUCTION

Integration is a reverse process of differentiation. The **integral or primitive** of a function f(x) with respect to x is that function $\phi(x)$ whose derivative with respect to x is the given function f(x).

$$\int f(x)\,dx = \phi(x)$$

Thus $\int f(x)\,dx = \phi(x) \Leftrightarrow \frac{d}{dx}[\phi(x)] = f(x)$

The process of finding the **integral** of a function is called Integration and the given function is called **Integrand**. Now, it is obvious that the operation of integration is inverse operation of differentiation. Hence integral of a function is also named as **anti-derivative** of that function. Further we observe that-

$$\left.\begin{aligned}\frac{d}{dx}(x^2) &= 2x\\ \frac{d}{dx}(x^2+2) &= 2x\\ \frac{d}{dx}(x^2+k) &= 2x\end{aligned}\right\} \Rightarrow \int 2x\,dx = x^2 + \text{constant}$$

So we always add a constant to the integral of function, which is called the **constant of Integration**. It is generally denoted by c.

Due to presence of this constant such an integral is called an **Indefinite integral**.

$$\frac{d}{dx}\left(x^x\right) = x^x(1+l\,nx) \quad \Rightarrow \quad \int x^x(1+l\,nx).dx = x^x + c$$

1.2 STANDARD INTEGRALS

1.2.1 $\int x^n.dx = \frac{x^{n+1}}{n+1} + c \qquad \int (ax+b)^n dx = \frac{(ax+b)^{n+1}}{a(n+1)} + c;\ n \neq -1$

$$\int \frac{1}{x}.dx = l\,n\,|x| + c \qquad \int \frac{dx}{ax+b} = \frac{1}{a}\ell n\left|ax+b\right| + c$$

Questions

1. $\int e^{\ln\sqrt{x}}\,dx$

Ans. $\frac{2}{3}x^{3/2} + C$

Sol. $I = \int e^{\ell n\sqrt{x}}dx = \int x^{\frac{1}{2}}dx = \frac{x^{\frac{3}{2}}}{\frac{3}{2}} + C$

2. $\int e^{-\ln x^2}dx$

Ans. $-\frac{1}{x} + C$

Sol. $I = \int e^{-\ell nx^2} dx = \int \frac{1}{x^2} dx = -\frac{1}{x} + C$

3. $\int \ln\left(\frac{1}{e^x}\right) dx;$

Ans. $-\frac{x^2}{2} + C$

Sol. $I = \int \ell n\left(\frac{1}{e^x}\right) dx = \int -x dx = -\frac{x^2}{2} + C$

4. $\int \frac{dx}{2\sqrt{x}};$

Ans. $\sqrt{x} + C$

Sol. $I = \int \frac{1}{2\sqrt{x}} dx = \sqrt{x} + C$

5. $\int e^{\ln 2 + \ln x} dx$

Ans. $x^2 + C$

Sol. $I = \int e^{\ell n 2 + \ell nx} dx = \int 2x dx = x^2 + C$

6. $\int e^{m \ln x} dx$

Ans. $\frac{x^{m+1}}{m+1} + C$

Sol. $I = \int e^{m \ell nx} dx = \frac{x^{m+1}}{m+1} + C$

7. $\int 2^{\ln x} dx$

Ans. $\frac{x^{1+\ell n 2}}{1+\ell n 2} + C$

Sol. $I = \int 2^{\ell nx} dx = \int x^{\ell n 2} dx = \frac{x^{\ell n 2 + 1}}{\ell n 2 + 1} + C$

8. $\int \frac{dx}{3 - 2x}$

Ans. $-\frac{1}{2} \ln|3 - 2x| + C$

Sol. $I = \int \frac{1}{3 - 2x} dx = \frac{\ell n|3 - 2x|}{-2} + C$

9. $\int \frac{x}{x+3}dx$

Ans. $x - 3\ ln\ |x + 3| + C$

Sol. $I = \int \frac{x}{x+3}dx = \int\left(1-\frac{3}{x+3}\right)dx = x - 3\ell n\left|x+3\right| + C$

10. $\int \frac{(1+x)^3}{x}dx$

Ans. $\frac{x^3}{3} + ln\left|x\right| + \frac{3x^2}{2} + 3x + C$

Sol. $I = \int \frac{(1+x)^3}{x}dx = \int\left(\frac{1}{x} + 3 + 3x + x^2\right)dx = \ell n\left|x\right| + 3x + \frac{3x^2}{2} + \frac{x^3}{3} + C$

11. $\int \frac{x\,dx}{x^2+2x+1}$

Ans. $\frac{1}{x+1} + ln\left|x+1\right| + C$

Sol. $I = \int \frac{x}{x^2+2x+1}dx = \int\left(\frac{1}{x+1} - \frac{1}{(x+1)^2}\right)dx = \ell n\left|x+1\right| + \frac{1}{x+1} + C$

12. $\int \frac{2x\ +\ 3}{x^2\ +\ 3x\ -\ 10}dx$

Ans. $ln\left|x+5\right| + ln\left|x-2\right| + C$

Sol. $\Rightarrow \lambda\int \frac{2x+3}{x^2+3x-10}dx = \int \frac{(x+5)+(x-2)}{(x+5)(x-2)}dx = \ell n\left|x-2\right| + \ell n\left|x+5\right| + C$

13. $\int \frac{dx}{\sqrt{2x+3}\ +\ \sqrt{2x-3}}$

Ans. $\frac{(2x+3)^{3/2} - (2x-3)^{3/2}}{18}$

Sol. $\int \frac{1}{\sqrt{2x+3}+\sqrt{2x-3}}dx = \frac{1}{12}\left[\frac{(2x+3)^{\frac{3}{2}}}{\frac{3}{2}} - \frac{(2x-3)^{\frac{3}{2}}}{\frac{3}{2}}\right] + C$

14. $\int \frac{xdx}{x^2-5x+6}$

Ans. $3\ ln\left|x-3\right| - 2\ ln\left|x-2\right| + C$

Sol. $\int \frac{x}{x^2-5x+6}dx = \int \frac{3(x-2)-2(x-3)}{(x-2)(x-3)}dx$

$= 3\ell n|x-3| - 2\ell n|x-2| + C$

15. $\int \frac{dx}{x^2+7x+12}$

Ans. $ln|x+3| - ln|x+4| + C$

Sol. $\int \frac{1}{x^2+7x+12}dx = \int\left(\frac{1}{x+3} - \frac{1}{x+4}\right)dx$

$= \ell n|x+3| - \ell n|x+4| + C$

16. $\int \frac{x\,dx}{a+bx}$

Ans. $\frac{1}{b}\left(x - \frac{a}{b}\ln(a+bx)\right) + c$

Sol. $\int \frac{x}{a+bx}dx = \frac{1}{b}\int \frac{bx+a-a}{a+bx}dx = \frac{x}{b} - \frac{a}{b^2}\ell n|a+bx| + C$

1.2.2 $\int e^x dx = e^x + c$ $\qquad \int e^{px+q}dx = \frac{e^{px+q}}{p} + c$

$\int a^x dx = \frac{a^x}{\ell n a} + c,\ (a>0)$ $\qquad \int a^{px+q}dx = \frac{a^{px+q}}{p\ell n a} + c,\ (a>0)$

Questions

1. $\int (2^x + 3^x)^2 dx$

Ans. $\frac{4^x}{ln4} + \frac{9^x}{ln9} + \frac{2(6^x)}{ln6} + C$

Sol. $\int (2^x + 3^x)^2 dx = \int (4^x + 9^x + 2\cdot 6^x)dx = \frac{4^x}{\ell n4} + \frac{9^x}{\ell n9} + 2\frac{6^x}{\ell n6} + C$

2. $\int a^x \cdot e^{2x}\,dx$

Ans. $\frac{a^x e^{2x}}{2 + ln a} + C$

Sol. $\int a^x . e^{2x}dx = \frac{(ae^2)^x}{\ell n(ae^2)} + C$

3. $\int \frac{2^{x+1} - 5^{x-1}}{10^x}dx$

Ans. $\frac{-2(5)^{-x}}{ln5} + \frac{2^{-x}}{5ln2} + C$

Sol. $\int \frac{2^{x+1}-5^{x-1}}{10^x} = 2\cdot\frac{\left(\frac{1}{5}\right)^x}{-\ell n5} + \frac{\left(\frac{1}{2}\right)^x}{5\ell n2} + C$

4. $\int \frac{e^{3x}+e^{5x}}{e^x+e^{-x}}dx$

Ans. $\frac{e^{4x}}{4}+C$

Sol. $\int \frac{e^{4x}\left(e^{-x}+e^x\right)}{\left(e^{-x}+e^x\right)}dx = \frac{e^{4x}}{4}+C$

5. $\int a^{mx}\cdot b^{nx}\,dx$

Ans. $\frac{a^{mx}b^{nx}}{m\ell na+n\ell n b}$

Sol. $\int a^{mx}b^{nx}dx = \int\left(a^m b^n\right)^x dx = \frac{\left(a^m b^n\right)^x}{\ell n\left(a^m b^n\right)}+C = \frac{a^{mn}\cdot b^{nx}}{m\ell na+n\ell nb}+C$

1.2.3 $\int \sin x.dx = -\cos x + c$ $\qquad \int \sin(ax+b)dx = -\frac{1}{a}\cos(ax+b)+c$

$\int \cos x.dx = \sin x + c$ $\qquad \int \cos(ax+b)dx = \frac{1}{a}\sin(ax+b)+c$

Questions

1. $\int \sqrt{1-\cos 2x}\,dx$

Ans. $-\sqrt{2}\cos x + C$

Sol. $I = \int \sqrt{1-\cos 2x} = \int\left|\sqrt{2}\sin x\right|dx$

$\xrightarrow{\sin x = -ve} -\int \sqrt{2}\sin x dx = \sqrt{2}\cos x + C$

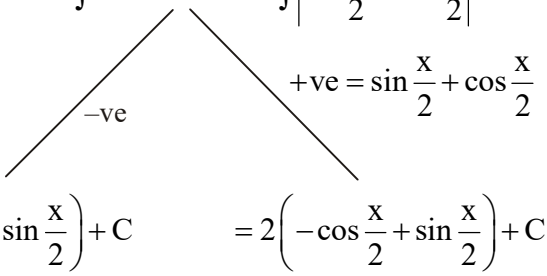
$\xrightarrow{\sin x = +ve} \int \sqrt{2}\sin x dx = -\sqrt{2}\cos x + C$

2. $\int \sqrt{1+\sin x}\,dx$

Ans. $2\left(\sin\frac{x}{2}-\cos\frac{x}{2}\right)+C$

Sol.

$$I = \int \sqrt{1+\sin x}\,dx = \int \left|\sin\frac{x}{2}+\cos\frac{x}{2}\right| dx$$



3. $\int \cos^2 x\,dx$

Ans. $\frac{x}{2}+\frac{\sin 2x}{4}+C$

Sol. $I = \int \cos^2 x dx = \int \frac{1+\cos 2x}{2} dx = \frac{x}{2}+\frac{\sin 2x}{4}+C$

4. $\int \sin^2 x\,dx$

Ans. $\frac{x}{2}-\frac{\sin 2x}{4}+C$

Sol. $I = \int \sin^2 x dx = \int \frac{1-\cos 2x}{2} dx = \frac{x}{2}-\frac{\sin 2x}{4}+C$

5. $\int \cos^3 x\,dx$

Ans. $\frac{3\sin x}{4}+\frac{\sin 3x}{12}+C$

Sol. $I = \int \cos^3 x dx = \int \left(\frac{\cos 3x+3\cos x}{4}\right) dx = \frac{\sin 3x}{12}+\frac{3}{4}\sin x+C$

6. $\int \sin^3 x\,dx$

Ans. $\frac{\cos 3x}{12}-\frac{3\cos x}{4}+C$

Sol. $I = \int \sin^3 x dx = \int \left(\frac{3\sin x-\sin 3x}{4}\right) dx = -\frac{3}{4}\cos x+\frac{\cos 3x}{12}+C$

7. $\int \cos 2x \cos 3x\,dx$

Ans. $\frac{\sin 5x}{10}+\frac{\sin x}{2}+C$

Sol. $I = \int \cos 2x \cdot \cos 3x dx$

$= \int \frac{\cos(5x)+\cos x}{2} dx$

$$= \frac{\sin 5x}{10} + \frac{\sin x}{2} + C$$

8. $\int \sin^4 x \, dx$

Ans. $\frac{3x}{8} - \frac{\sin 2x}{4} + \frac{\sin 4x}{32} + C$

Sol. $\int \sin^4 x dx = \int \left(\frac{1-\cos 2x}{2} \right)^2 dx = \frac{1}{4} \int \left(1 - 2\cos 2x + \frac{1+\cos 4x}{2} \right) dx$

$$= \frac{1}{4} \left[x - \sin 2x + \frac{x}{2} + \frac{\sin 4x}{8} \right] + C$$

9. $\int \frac{\cos^3 x + \sin^3 x}{\cos x + \sin x} dx$

Ans. $x + \frac{\cos 2x}{4} + C$

Sol. $\int \frac{\sin^3 x + \cos^3 x}{\sin x + \cos x} dx = \int \left(\sin^2 x + \cos^2 x - \sin x \cos x \right) dx = x + \frac{\cos 2x}{4} + C$

10. $\int \left(\cos^4(x) - \sin^4(x) \right) dx$

Ans. $\frac{\sin 2x}{2} + C$

Sol. $\int \left(\cos^4 x - \sin^4 x \right) dx = \int \cos 2x dx = \frac{\sin 2x}{2} + C$

11. $\int \cos^4(x) + \sin^4(x) \, dx$

Ans. $\frac{3x}{4} + \frac{\sin 4x}{16} + C$

Sol. $\int \left(\sin^4 x + \cos^4 x \right) dx = \int \left(1 - 2\sin^2 x \cos^2 x \right) dx = \int \left(1 - \frac{\sin^2 2x}{2} \right) dx$

$$= \left[x - \frac{1}{2} \left(x - \frac{\sin 4x}{4} \right) \right] + C$$

$$= \frac{x}{2} + \frac{\sin 4x}{8} + C$$

12. $\int \sin x^\circ \, dx$

Ans. $-\frac{180}{\pi} \cos(x^\circ) + C$

Sol. $\int \sin x^0 dx = \int \sin \frac{\pi x}{180} dx = -\frac{180}{\pi} \cos\left(x^0 \right) + C$

13. $\int \sin 2x \cdot \cos^2 x \, dx$

Ans. $-\frac{\cos 2x}{4} - \frac{\cos 4x}{16} + C$

Sol. $\int \sin 2x \cdot \cos^2 x dx = \int \sin 2x \frac{(1+\cos 2x)}{2} dx = -\frac{\cos 2x}{4} - \frac{\cos 8x}{32} + C$

14. $\int \frac{\cos x - \cos 2x}{1 - \cos x} dx$

Ans. $2 \sin x + x + C$

Sol. $\int \frac{\cos x - \cos 2x}{1 - \cos x} dx$

$I = \int \frac{\cos x - \cos^2 x + \sin^2 x}{1 - \cos x} dx$

$= \int (\cos x + 1 + \cos x) dx$

$= 2 \sin x + x + C$

15. $\int \frac{\cos 5x + \cos 4x}{1 - 2\cos 3x} dx$

Ans. $-\frac{\sin 2x}{2} - \sin x + C$

Sol. $\int \frac{\cos 5x + \cos 4x}{1 - 2\cos 3x} dx = \int \frac{2\cos \frac{9x}{2} \cos \frac{x}{2}}{1 - 2\cos 3x} dx$

$= \int \frac{2\left(4\cos^3 \frac{3x}{2} - 3\cos \frac{3x}{2}\right)}{1 - 2\left(2\cos^2 \frac{3x}{2} - 1\right)} dx$

$= \int \frac{2\cos \frac{x}{2} \cdot \cos \frac{3x}{2}\left(4\cos^2 \frac{3x}{2} - 3\right)}{-\left(4\cos^2 \frac{3x}{2} - 3\right)} dx$

$= \int -(\cos x + \cos 2x) dx$

$= -\left(\sin x + \frac{\sin 2x}{2}\right) + C$

1.2.4 $\int \sec^2 x.dx = \tan x + c$ $\qquad \int \sec^2 (ax + b) dx = \frac{1}{a} \tan(ax + b) + c$

$\int \operatorname{cosec}^2 x.dx = -\cot x + c$ $\qquad \int \operatorname{cosec}^2 (ax + b) dx = -\frac{1}{a} \cot(ax + b) + c$

$\int \sec x.\tan x.dx = \sec x + c$ $\int \sec(ax + b).\tan(ax + b)dx = \frac{1}{a}\sec(ax + b) + c$

$\int \operatorname{cosec} x.\cot x.dx = -\operatorname{cosec} x + c$ $\int \operatorname{cosec}(ax + b).\cot(ax + b)dx = -\frac{1}{a}\operatorname{cosec}(ax + b) + c$

Questions

1. $\int \frac{dx}{1+\cos x}$

Ans. $\tan\frac{x}{2} + C$

Sol. $= \int \frac{1}{2}\sec^2\frac{x}{2}dx = \tan\frac{x}{2} + C$

2. $\int \cot^2 x\,dx$

Ans. $-\cot x - x + C$

Sol. $\int \cot^2 x dx = \int \left(\operatorname{cosec}^2 x - 1\right)dx = -\cot x - x + C$

3. $\int \sec^2 x \operatorname{cosec}^2 x\,dx$

Ans. $\tan x - \cot x + C$

Sol. $\int \sec^2 x \cdot \operatorname{cosec}^2 x dx = \int \frac{\sin^2 x + \cos^2 x}{\sin^2 x \cos^2 x}dx = \tan x - \cot x + C$

4. $\int \tan^2 x\,dx$

Ans. $\tan x - x + C$

Sol. $\int \tan^2 x dx = \int \left(\sec^2 x - 1\right)dx = \tan x - x + C$

5. $\int \frac{\cos 2x + 2\sin^2 x}{\cos^2 x}dx$

Ans. $\tan x + C$

Sol. $\int \frac{\cos 2x + 2\sin^2 x}{\cos^2 x}dx = \int \sec^2 x dx = \tan x + C$

6. $\int \cot^2 x \cos^2 x\,dx$

Ans. $-\cot x - \frac{3x}{2} - \frac{\sin 2x}{4} + C$

Sol. $\int \cot^2 x \cdot \cos^2 x dx = \int \left(\cot^2 x - \cos^2 x\right)dx$

$= \int \left[\left(\operatorname{cosec}^2 x - 1\right) - \frac{1+\cos 2x}{2}\right]dx$

$= -\cot x - \frac{3x}{2} - \frac{\sin 2x}{4} + C$

7. $\int \tan^2 x \sin^2 x \, dx$

Ans. $\tan x - \frac{3x}{2} + \frac{\sin 2x}{4} + C$

Sol. $\int \tan^2 x \cdot \sin^2 x dx = \int \left(\tan^2 x - \sin^2 x\right) dx$

$= \tan x - \frac{3x}{2} + \frac{\sin 2x}{4} + C$

8. $\int \frac{1-\cos x}{1+\cos x} dx$

Ans. $2\tan\frac{x}{2} - x + C$

Sol. $\int \frac{1-\cos x}{1+\cos x} dx = \int \tan^2 \frac{x}{2} dx = \int \left(\sec^2 \frac{x}{2} - 1\right) dx = \frac{\tan \frac{x}{2}}{\frac{1}{2}} - x + C$

9. $\int \frac{dx}{1-\sin 3x}$

Ans. $\frac{\tan 3x + \sec 3x}{3} + C$

Sol. $\int \frac{1+\sin 3x}{\cos^2 3x} dx = \frac{\tan 3x}{3} + \frac{\sec 3x}{3} + C$

10. $\int \frac{a\sin^3 x + b\cos^3 x}{\sin^2 x \cos^2 x} dx$

Ans. $a \sec x - b \operatorname{cosec} x + C$

Sol. $I = \int \frac{a\sin^3 x + b\cos^3 x}{\sin^2 x \cdot \cos^2 x} dx$

$= a\int \left(\tan x \cdot \sec x\right) dx + \int \left(b \cot x \cdot \operatorname{cosec} x\right) dx$

$= a \sec x - b \operatorname{cosec} x + C$

11. $\int \frac{\operatorname{cosec} x + \tan^2 x + \sin^2 x}{\sin x} dx$

Ans. $-\cot x + \sec x - \cos x + C$

Sol. $= \int \frac{\operatorname{cosec} x + \tan^2 x + \sin^2 x}{\sin x} dx = \int \left(\operatorname{cosec}^2 x + \tan x \cdot \sec x + \sin x\right) dx$

$= -\cot x + \sec x - \cos x + C$

12. $\int \frac{\sin x}{\cos^2 x}(1 - 3\sin^3 x) dx$

Ans. $\sec x - 3\tan x + \frac{9x}{2} - \frac{3\sin 2x}{4} + C$

Sol. $\int \frac{\sin x}{\cos^2 x}\left(1-3\sin^3 x\right)dx = \int\left(\tan x\cdot\sec x - 3\sin^2 x\cdot\tan^2 x\right)dx$

$= \int\left(\tan x\cdot\sec x - 3\tan^2 x + 3\sin^2 x\right)dx$

$= \int\left[\tan x\cdot\sec x - 3\left(\sec^2 x - 1\right) + 3\frac{1-\cos 2x}{2}\right]dx$

$= \sec x - 3\tan x + \frac{9}{2}x - \frac{3}{4}\sin 2x + C$

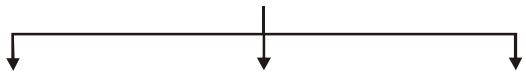
1.2.5 $\int \frac{dx}{\sqrt{1-x^2}} = \sin^{-1} x + c$; $\int \frac{dx}{\sqrt{a^2-x^2}} = \sin^{-1}\frac{x}{a} + c$

$\int \frac{dx}{1+x^2} = \tan^{-1}(x) + c$ $\int \frac{dx}{a^2+x^2} = \frac{1}{a}\tan^{-1}\frac{x}{a} + c$

$\int \frac{dx}{|x|\sqrt{x^2-1}} = \sec^{-1}(x) + c$ $\int \frac{dx}{|x|\sqrt{x^2-a^2}} = \frac{1}{a}\sec^{-1}\frac{x}{a} + c$

$\int \frac{dx}{x^2-a^2} = \frac{1}{2a}\ell n\left|\frac{x-a}{x+a}\right| + c$

Profile $\int \frac{dx}{\text{quad.}}$



D>0 D=0 D<0

$\int \frac{dx}{x^2-a^2}$ $\int \frac{dx}{(ax+b)^2}$ $\int \frac{dx}{x^2+a^2}$

Note: Complete the square and use corresponding formula

Ex.1 $\int \frac{dx}{x^2-10x+9}$

Ans. $\frac{1}{8}\ln\left|\frac{x-9}{x-1}\right| + C$

Ex.2 $\int \frac{dx}{x^2-6x+10}$

Ans. $\tan^{-1}(x-3) + C$

Questions

1. $\int \frac{x^2}{1+x^2}dx$

Ans. $x - \tan^{-1}(x) + C$

Sol. $\int \frac{x^2}{1+x^2}dx = \int\left(1-\frac{1}{1+x^2}\right)dx$

$= x - \tan^{-1}x + C$

2. $\int \frac{x^4}{1+x^2}dx$

Ans. $\frac{x^3}{3} - x + \tan^{-1}(x) + C$

Sol. $= \int \frac{x^4}{1+x^2}dx = \int\left[\left(x^2-1\right)+\frac{1}{1+x^2}\right]dx = \frac{x^3}{3} - x + \tan^{-1}x + C$

3. $\int \frac{x^2+\cos^2 x}{x^2+1}\cdot \operatorname{cosec}^2 x\,dx$

Ans. $-\cot x - \tan^{-1}(x) + C$

Sol. $\int \frac{x^2+\cos^2 x}{x^2+1}\operatorname{cosec}^2 x dx = \int \operatorname{cosec}^2 x dx - \int \frac{1}{1+x^2}dx$

$= -\cot x - \tan^{-1}x + C$

4. $\int \frac{dx}{\sqrt{9-4x^2}}$

Ans. $\frac{1}{2}\sin^{-1}\left(\frac{2x}{3}\right) + C$

Sol. $\int \frac{1}{\sqrt{9-4x^2}}dx = \frac{1}{2}\sin^{-1}\left(\frac{2x}{3}\right) + C$

5. $\int \frac{dx}{(x^2-4x+4)(x^2-4x+5)}$

Ans. $\frac{-1}{x-2} - \tan^{-1}(x-2) + C$

Sol. $\int\left(\frac{1}{x^2-4x+4} - \frac{1}{x^2-4x+5}\right)dx$

$= \int \frac{1}{(x-2)^2} - \frac{1}{(x-2)^2+1}dx$

$= -\frac{1}{x-2} - \tan^{-1}(x-2) + C$

6. $\int \frac{dx}{(2x-7)\sqrt{(x-3)(x-4)}}$

Ans. $\sec^{-1}(2x-7) + C$

Sol. $\int \frac{2}{(2x-4)\sqrt{(2x-7)^2-1}}dx = \sec^{-1}(2x-7)+C$

7. In each case, find a function f satisfying the given conditions.
(a) $f'(x^2) = 1/x$ for $x > 0$, $f(1) = 1$;
(b) $f'(\sin^2 x) = \cos^2 x$ for all x, $f(1) = 1$
(c) $f'(\sin x) = \cos^2 x$ for all x, $f(1) = 1$

Ans. **(a)** $f(t)=2\sqrt{t}-1$ if $t > 0$,

(b) $f(t)= t - \frac{1}{2}t^2+\frac{1}{2}$ if $0 \le t \le 1$,

(c) $f(t) = t - \frac{1}{3}t^3 + \frac{1}{3}$ if $|t| \le 1$]

2. INTEGRATION BY SUBSTITUTION

$\int f[\phi(x)]\ \phi'(x)\,dx$

Here we put $\phi(x) = t$ so that $\phi'(x)\,dx = dt$ and in that case the integrand is reduced to $\int f(t)\,dt$.

In this method the integrand is broken into two factors so that one factor can be expressed in terms of the function whose differential coefficient is the second factor.

Ex.1 $\int x\sin(x^2).dx$

Put $x^2 = t$ $\quad 2x.\,dx = dt$

$I = \frac{1}{2}\int \sin t.\,dt = \frac{-\cos t}{2}+c = -\frac{\cos(x^2)}{2}+c$

Questions

1. $\int \frac{x^5}{1+x^6}dx$

Ans. $\frac{1}{6}\ln(1+x^6)+C$

Sol. $I = \int \frac{x^5}{1+x^6}dx$

Put $1 + x^6 = t$

$x^5dx = \frac{dt}{6}$

$I = \frac{1}{6}\int \frac{1}{t}dt = \frac{1}{6}\ell n|1+x^6|+C$

2. $\int e^{\sin x}\cos x\,dx$

Ans. $e^{\sin x}+C$

Sol $I=\int e^{\sin x}\cos x dx$

$\sin x = t$

$\cos x.dx = dt$

$I=\int e^t.dt = e^t + c$

$= e^{\sin x}+c$

3. $\int \frac{\sec^2(\ell n\, x)}{x}dx$

Ans. $\tan(\ell n\, x)+C$

Sol. $I=\int \frac{\sec^2(\ell nx)}{x}dx$

Put $\ell nx = t$

$\frac{1}{x}dx = dt$

$= \tan(\ell nx) + C$

4. $\int \frac{x}{1+x^4}dx$

Ans. $\frac{1}{2}\tan^{-1}(x^2)+C$

Sol. $I=\int \frac{x}{1+x^4}dx$ $\qquad x^2 = t \Rightarrow xdx = \frac{dt}{2}$

$=\frac{1}{2}\tan^{-1}(x^2)+C$

5. $\boxed{\begin{array}{l}\int(\tan x)dx = \ln|\sec x|+C = -\ln|\cos x|+C;\\ \int(\cot x)dx = \ln|\sin x|+C\end{array}}$

Ans.

6. $\int \frac{\tan(\ell n\, x)}{x}dx$

Ans. $\ell n(\sec(\ell n\, x))+C$

Sol. $I=\int \frac{\tan(\ell nx)}{x}dx$ $\qquad \ell nx = t \Rightarrow \frac{1}{x}dx = dt$

$I=\int \tan t dt$

$\ell n|\sec t| + c$

$= \ell n|\sec(\ell nx)| + C$

7. $\int \frac{\tan(\sin^{-1} x)}{\sqrt{1-x^2}}dx$

Ans. $l\text{n}\left(\sec\left(\sin^{-1} x\right)\right)+C$

Sol. $I=\int \frac{\tan\left(\sin^{-1} x\right)}{\sqrt{1-x^2}}dx$ $\qquad \sin^{-1}x=t \Rightarrow \frac{1}{\sqrt{1-x^2}}dx=dt$

$=\ell n|\sec(\sin^{-1}x)|+C$

8. $\int \frac{\cos x}{\cos(x-a)}dx$

Ans. $x\cos a-\sin a\, l\text{n}\left(\sec\left(x-a\right)\right)+C$

Sol. $I=\int \frac{\cos x}{\cos\left(x-a\right)}dx$ $\qquad x-a=t \Rightarrow x=t+a \Rightarrow dx=dt$

$\int \frac{\cos\left(t+a\right)}{\cos t}dt$

$=\sin a\cdot t+\ell n|\sec t|\cdot\sin a+C$

$=\sin a\ (x-a)-\sin a\ \ell n|\sec(x-a)|+C$

9. $\int \frac{x\cos x}{(x\sin x+\cos x)^2}dx$

Ans. $-\frac{1}{x\sin x+\cos x}+C$

Sol. $I=\int \frac{x\cos x}{\left(x\sin x+\cos x\right)^2}dx$ $\qquad$ put $x\sin x+\cos x=t$

$=\int \frac{1}{t^2}dt$

$=-\frac{1}{x\sin x+\cos x}+C$

10. (i). $\int \frac{dx}{\cos(x-a)\cos(x-b)}$;

Ans. $\frac{1}{\sin\left(b-a\right)}\left(l\text{n}\frac{\sec\left(x-a\right)}{\sec\left(x-b\right)}\right)+C$

Sol. $I=\frac{1}{\sin\left(x-a\right)}\int \frac{\sin\left(x-a-\left(x-b\right)\right)}{\cos\left(x-a\right)\cos\left(x-b\right)}dx$

$$= \frac{1}{\sin(b-a)} \ell n \left| \frac{\sec(x-a)}{\sec(x-b)} \right| + C$$

10. (ii). $\int \frac{\sin 2x\, dx}{\sin 5x . \sin 3x}$

Ans. $\frac{\ell n \sin 3x}{3} - \frac{\ell n \sin 5x}{5} + C$

Sol. $I = \int (\cot 3x - \cot 5x)\, dx$

$$= \frac{\ell n \sin(3x)}{3} - \frac{\ell n \sin 5x}{5} + C$$

11. $\int (\sec^2 x \cdot \sqrt{5 + \tan x})\, dx$

Ans. $\frac{2}{3}(5 + \tan x)^{3/2} + C$

Sol. $I = \int \sec^2 x \sqrt{5 + \tan x} dx$

put $5 + \tan x = t \rightarrow \sec^2 x\, dx = dt$

$$= \frac{2}{3}(5 + \tan x)^{\frac{3}{2}} + C$$

12. $\int (\sin x - \cos x) \cdot (\sin x + \cos x)^5\, dx$

Ans. $-\frac{(\sin x + \cos x)^6}{6} + C$

Sol. $I = \int (\sin x - \cos x)(\sin x + \cos x)^5\, dx$

Put $\sin x + \cos x = t$

$$= \int -t^5 dt = -\frac{(\sin x + \cos x)^6}{6} + C$$

13. $\int \frac{\sec x\, dx}{\sqrt{\cos(2x + \alpha) + \cos \alpha}}$

Ans. $C - \sqrt{\frac{2(\cot \alpha - \tan x)}{\sin \alpha}}$

Sol. [**Hint:** Use cos C + cos D in D^r]

$$I = \int \frac{\sec x\ dx}{\sqrt{2\cos(x + \alpha)\cos x}} = \frac{1}{\sqrt{2}} \int \frac{\sec x\ dx}{\sqrt{(\cos x \cos \alpha - \sin x \sin \alpha)\cos x}}$$

$$= \frac{1}{\sqrt{2}} \int \frac{\sec^2 x\ dx}{\sqrt{\cos \alpha - \tan x \sin \alpha}} = \frac{1}{\sqrt{2 \sin \alpha}} \int \frac{\sec^2 x\ dx}{\sqrt{\cot \alpha - \tan x}}$$

put $\cot \alpha - \tan x = t^2 \quad \Rightarrow \quad -\sec^2 x\, dx = 2t\, dt$

$$I = \frac{1}{\sqrt{2\sin\alpha}}\int\frac{-2t\,dt}{t} = \frac{-2}{\sqrt{2\sin\alpha}}\int dt = -\frac{\sqrt{2}}{\sqrt{\sin\alpha}}\sqrt{\cot\alpha - \tan x} + C$$

$$= C - \sqrt{\frac{2(\cot\alpha - \tan x)}{\sin\alpha}}$$ **Ans.**

14. $\int e^x \sin e^x\,dx$;

Ans. $-\cos(e^x) + C$

Sol. $I = \int e^x \sin\left(e^x\right)dx$

put $e^x = t$

$= -\cos(e^x) + C$

15. $\int\frac{x^5}{1+x^{12}}dx$;

Ans. $\frac{1}{6}\tan^{-1}\left(x^6\right) + C$

Sol. $I = \int\frac{x^5}{1+x^{12}}dx$

Put $x^6 = t$

$= \frac{1}{6}\tan^{-1}\left(x^6\right) + C$

16. $\int\frac{x^2\tan^{-1}x^3}{1+x^6}dx$;

Ans. $\frac{\left(\tan^{-1}\left(x^3\right)\right)^2}{6} + C$

Sol. $I = \int\frac{x^2\tan^{-1}\left(x^3\right)}{1+x^6}dx$

put $\tan^{-1}(x^3) = t$

$= \frac{1}{6}\left\{\tan^{-1}\left(x^3\right)\right\}^2 + C$

17. $\int\frac{\tan\sqrt{x}\sec^2\sqrt{x}}{\sqrt{x}}dx$;

Ans. $\tan^2\sqrt{x} + C$

Sol. $I = \int\tan\sqrt{x}\cdot\frac{\sec^2\sqrt{x}}{\sqrt{x}}dx$

put $\tan\sqrt{x} = t$

$= \tan^2\sqrt{x} + C$

18. $\int \sec x \, ln(\sec x+\tan x)dx;$

Ans. $\frac{ln^2(\sec x+\tan x)}{2}+C$

Sol. $I=\int \sec x \cdot \ell n(\sec x+\tan x)dx$

put $\ell n(\sec x+\tan x)=t$

$I=\frac{\ell n^2(\sec x+\tan x)}{2}+C$

19. $\int \frac{\sqrt{\tan x}}{\sin 2x}dx$;

Ans. $\sqrt{\tan x}+C$

Sol. $I=\int \frac{\sqrt{\tan x}}{\sin 2x}dx$

$=\int \frac{1}{2\sqrt{\tan x}}\cdot \sec^2 x dx$

put $\tan x=t$

$=\sqrt{\tan x}+C$

20. $\int (\tan^3 x - x\tan^2 x)dx$

Ans. $\frac{(\tan x-x)^2}{2}+C$

Sol. $I=\int \tan^2 x(\tan x-x)dx$

put $\tan x-x=t$

$I=\frac{(\tan x-x)^2}{2}+C$

21. $\int \frac{\sec x \tan x}{\sqrt{2+\tan^2 x}}dx$

Ans. $ln|\sec x+\sqrt{\sec^2 x+1}|+C$

Sol. $I=\int \frac{\sec x\cdot \tan x}{\sqrt{2+\tan^2 x}}dx$

put $\sec x=t$

$=\ell n\left|\sec x+\sqrt{\sec^2 x+1}\right|+C$

22. $\int \frac{\ell n^2(\sec x)}{\cot x}dx$;

Ans. $\frac{\ell n^3(\sec x)}{3}+C$

Sol. $I=\int \frac{\ell n^2(\sec x)}{\cot x}dx$

put $\ell n \sec x = t$

$=\frac{\ell n^3(\sec x)}{3}+C$

23. $\int \frac{\sin 2x}{(a\sin^2 x + b\cos^2 x)^2}dx$;

Ans. $\frac{1}{(b-a)(a\sin^2 x+b\cos^2 x)}+C$

Sol. Put $a\sin^2x + b\cos^2x = t$

$I=\frac{1}{a-b}\left(-\frac{1}{a\sin^2 x+b\cos^2 x}\right)+C$

$=\frac{1}{(b-a)(a\sin^2 x+b\cos^2 x)}+C$

24. $\int \frac{\ell n^2 \frac{x}{x+1}}{x(x+1)}dx$

Ans. $\frac{\ell n^3\left(x/x+1\right)}{3}+C$

Sol. $I=\int \frac{\ell n^2\left(\frac{x}{x+1}\right)}{x(x+1)}dx$

put $\ell n\left(\frac{x}{x+1}\right)=t \qquad \rightarrow \frac{1}{x(x+1)}dx=dt$

$I=\frac{\ell n^3\left(\frac{x}{x+1}\right)}{3}+C$

25. $\int \frac{x+e^x(\sin x+\cos x)+\sin x\cos x}{(x^2+2e^x\sin x-\cos^2 x)^2}dx$

Ans. $C-\frac{1}{2(x^2+2e^x\sin x-\cos^2 x)}$

Sol. $I = \frac{1}{2}\int \frac{2x + 2e^x(\sin x + \cos x) + 2\sin x \cos x}{(x^2 + 2e^x \sin x - \cos^2 x)^2} dx$;

put $x^2 + 2e^x \sin x - \cos^2 x = t$

$[2x + 2(e^x \cos x + e^x \sin x) + 2 \sin x \cos x]dx = dt$

$[2x + 2e^x(\cos x + \sin x) + 2 \sin x \cos x]\, dx = dt$

$$I = \frac{1}{2}\int \frac{dt}{t^2} = C - \frac{1}{2}\cdot\frac{1}{t} = C - \frac{1}{2(x^2 + 2e^x \sin x - \cos^2 x)}$$

26. $\int x \cdot \sqrt{\frac{2\sin(x^2+1) - \sin 2(x^2+1)}{2\sin(x^2+1) + \sin 2(x^2+1)}}\, dx$

Ans. $\ln\left|\sec\left(\frac{x^2+1}{2}\right)\right| + C$

Sol $I = \int x\sqrt{\frac{2\sin\left(x^2+1\right) - \sin 2\left(x^2+1\right)}{2\sin\left(x^2+1\right) + \sin 2\left(x^2+1\right)}}dx$

$= \int x\sqrt{\frac{1-\cos\left(x^2+1\right)}{1+\cos\left(x^2+1\right)}}dx$

put $x^2 + 1 = t$

$I = \frac{1}{2}\int\sqrt{\frac{1-\cos t}{1+\cos t}}dt$

$I = \ell n\left|\sec\frac{\left(x^2+1\right)}{2}\right| + C$

27. $\int \sqrt{x}\sin^2\left(x^{3/2} - 1\right)dx$;

Ans. $\frac{x^{3/2}}{3} - \frac{\sin 2\left(x^{3/2}-1\right)}{6} + C$

Sol. $I = \int \sqrt{x}\sin^2\left(x^{\frac{3}{2}} - 1\right)dx$

put $x^{\frac{3}{2}} - 1 = t$

$\sqrt{x}dx = \frac{2}{3}dt$

$= \int \sin^2 t dt$

$$= \frac{x^{\frac{3}{2}}-1}{2} - \frac{\sin 2\left(x^{\frac{2}{3}}-1\right)}{4} + C$$

28. $\int \frac{\cos\sqrt{\theta}}{\sqrt{\theta}\sin^2\sqrt{\theta}}\,d\theta$

Ans. $\frac{-2}{\sin\sqrt{\theta}} + C$

Sol. $I = \int \frac{\cos\sqrt{\theta}}{\sqrt{\theta}\sin^2\sqrt{\theta}}\,d\theta$ put $\sin\sqrt{\theta} = t$

$$= -\frac{2}{\sin\sqrt{\theta}} + C$$

29. Formulae $\left.\begin{array}{l} \int \sec x\,dx = \ln|\sec x + \tan x| + C \text{ or } \ln\left|\tan\left(\frac{\pi}{4}+\frac{x}{2}\right)\right| + C; \\ \int \operatorname{cosec} x\,dx = \ln|\operatorname{cosec} x - \cot x| + c \text{ or } \ln\left|\tan\frac{x}{2}\right| + C \end{array}\right\}$ (Remember)

30. $\int \frac{dx}{\cos x(1-4\sin^2 x)}$;

Ans. $\frac{\ell n|\sec 3x + \tan 3x|}{3} + C$

Sol. $\int \frac{1}{4\cos^3 x - 3\cos x}\,dx = \int \sec 3x\,dx = \ell n|\tan 3x + \sec 3x| + C$

31. $\int \frac{\tan x+\sec x-1}{\tan x-\sec x+1}\,dx$;

Ans. $\ell n|\sec x| + \ell n|\sec x + \tan x| + C$

Sol. $I = \int \frac{\tan x + \sec x - 1}{\tan x - \sec x + 1}\,dx$

$I = \int (\sec x + \tan x)\,dx$

$= \ell n|\sec x + \tan x| + \ell n|\sec x| + C$

32. $\int \frac{\operatorname{cosec}(\tan^{-1} x)}{1+x^2}\,dx$;

Ans. $\ell n\left|\tan\left(\frac{\tan^{-1} x}{2}\right)\right| + C$

Sol. $I = \int \frac{\cos ec\left(\tan^{-1} x\right)}{1+x^2} dx$

put $\tan^{-1}x = t$

$= \ell n \tan\left(\frac{\tan^{-1} x}{2}\right) + C$

33. $\int \frac{\cos 2x}{\sin x} dx$;

Ans. $ln\left|\tan\frac{x}{2}\right| + 2\cos x + C$

Sol. $I = \int \frac{\cos 2x}{\sin x} dx$

$= \int \left(\cos ecx - 2\sin x\right) dx$

$= \ell n \tan\frac{x}{2} + 2\cos x + C$

34. $\int \frac{dx}{\sin x \cos^2 x}$

Ans. $ln\left|\tan\frac{x}{2}\right| + \sec x + C$

Sol. $\int \cos ec\, x\left(\tan^2 x + 1\right) dx = \int \left(\tan x \sec x + \cos ec\, x\right) dx$

$= \sec x + \ell n \tan\frac{x}{2} + C$

35. (i). $\int \frac{dx}{\sqrt{3}\sin x + \cos x}$;

Ans. $\frac{1}{2} ln\left|\tan\left(\frac{x}{2} + \frac{\pi}{12}\right)\right| + C$

Sol. (i) $I = \int \frac{1}{\sqrt{3}\sin x + \cos x} dx$

$= \frac{1}{2}\int \frac{1}{\sin\left(x + \frac{\pi}{6}\right)} dx$

$= \frac{1}{2} \ell n \tan\left(\frac{x}{2} + \frac{\pi}{2}\right) + C$

35. (ii). $\int \frac{dx}{a\sin x + b\cos x}$;

Ans. $\frac{1}{\sqrt{a^2+b^2}}\,ln\left|\tan\left(\frac{x}{2}+\frac{\theta}{2}\right)\right|+C$] [where $\tan\theta=\frac{b}{a}$

Sol. $I=\int\frac{1}{a\sin x+b\cos x}dx$

$=\frac{1}{\sqrt{a^2+b^2}}\int\frac{1}{\sin(x+\alpha)}dx$ where $\alpha=\tan^{-1}\left(\frac{b}{a}\right)$

$=\frac{1}{\sqrt{a^2+b^2}}\ell n\tan\left|\frac{x}{2}+\frac{\alpha}{2}\right|+C$

36. $\int\frac{dx}{\sec x+\cos ecx}$

Ans. $\frac{\sin x-\cos x}{2}-\frac{1}{2\sqrt{2}}\ln\left|\tan\left(\frac{x}{2}+\frac{\pi}{8}\right)\right|+C$

Sol. $I=\int\frac{\sin x\cos x}{(\sin x+\cos x)}dx$

$=\frac{1}{2}\int\frac{(\sin x+\cos x)^2-1}{\sin x+\cos x}dx$

$=\frac{1}{2}\left[\cos x+\sin x-\frac{1}{\sqrt{2}}\ell n\tan\left|\frac{x}{2}+\frac{\pi}{4}\right|\right]+C$

37. $\int\frac{dx}{\tan x+\cot x+\sec x+\operatorname{cosec} x}$

Ans. $\frac{1}{2}(\sin x-\cos x-x)+C$

Sol. $I=\int\frac{\sin x\cos x}{1+\sin x+\cos x}dx$

$=\frac{1}{2}\int\frac{(\sin x+\cos x)^2-1}{1+\sin x+\cos x}dx$

$=\frac{1}{2}(\sin x-\cos x-x)+C$

38. $\int e^{x^2+\ell nx}\,dx$;

Ans. $\frac{e^{x^2}}{2}+C$

Sol. $I=\int e^{x^2}\cdot x dx$ put $x^2=t$

$= \frac{1}{2} \cdot e^{x^2} + C$

39. $\int \frac{dx}{e^x + 1};$

Ans. $-\ell n\left|1 + e^{-x}\right| + C$

Sol. $I = \int \frac{1}{e^x + 1} dx$

$I = \int \frac{e^{-x}}{1 + e^{-x}} dx$ put $e^{-x} + 1 = t$

$= -\ell n\,|1 + e^{-x}| + C$

$= \ell n\left|\frac{e^x}{1 + e^x}\right| + C$

40. $\int \frac{e^x - 1}{e^x + 1} dx;$

Ans. $x + 2\,\ell n\left|1 + e^{-x}\right| + C$

Sol. $I = \int \frac{e^x - 1}{e^x + 1} dx$

$= \int 1 - \frac{2}{e^x + 1} dx$

$= x - 2\ell n\left|\frac{e^x}{1 + e^x}\right| + C$

41. $\int \frac{e^x (1 + x)}{\sin^2 (xe^x)} dx;$

Ans. $-\cot\left(xe^x\right) + C$

Sol. $I = \int \frac{e^x \left(1 + x\right)}{\sin^2\left(x \cdot e^x\right)} dx$

put $x \cdot e^x = t$

$= -\cot(x \cdot e^x) + C$

42. $\int (27e^{9x} + e^{12x})^{1/3}\, dx$

Ans. $\frac{(27 + e^{3x})^{4/3}}{4} + C$

Sol. $I = \int \left(27 \cdot e^{9x} + e^{12x}\right)^{\frac{1}{3}} dx$

$$I = \int e^{3x}\left(27 + e^{3x}\right)^{\frac{1}{3}} dx$$

put $27 + e^{3x} = t$

$$I = \frac{1}{3}\frac{\left(27 + e^{3x}\right)^{\frac{4}{3}}}{\frac{4}{3}} + C$$

43. $\int \frac{x\,dx}{(1+x^2)+\sqrt{(1+x^2)^3}}$

Ans. $\ell n \frac{\sqrt{1+x^2}}{\left(1+\sqrt{1+x^2}\right)} + C \quad \text{with } 1+x^2 = t^2$

Sol. $I = \int \frac{x}{\left(1+x^2\right)+\sqrt{\left(1+x^2\right)^3}} dx$

put $1 + x^2 = t^2$

$xdx = tdt$

$$= \int \frac{t}{t^2+t^3} dt = \int \frac{1}{t(t+1)} dt = \ell n\left|\frac{t}{t+1}\right| + C$$

$$= \ell n\left|\frac{\sqrt{1+x^2}}{\sqrt{1+x^2}+1}\right| + C$$

44. $\int \left(\frac{1-\sqrt{x}}{1+\sqrt{x}}\right)^{1/2} \frac{dx}{x}$

Ans. $2\cos^{-1}\sqrt{x} - 2\ell n\left(\frac{1+\sqrt{1-x}}{\sqrt{x}}\right) + C$

Sol. $I = \int \left(\frac{1-\sqrt{x}}{1+\sqrt{x}}\right)^{\frac{1}{2}} \frac{dx}{x}$

$$= \int \frac{1-\sqrt{x}}{\sqrt{1-x}} \frac{dx}{x} = \int \frac{1}{x\sqrt{1-x}} dx - \int \frac{1}{\sqrt{x}\sqrt{1-x}} dx$$

$$\downarrow \qquad\qquad \downarrow$$

Put $1 - x = u^2$ $\qquad$ $\sqrt{x} = t$

$$= \int \frac{-2udx}{\left(1-u^2\right)u} - 2\int \frac{dt}{\sqrt{1-t^2}}$$

$$= \frac{2}{2}\ell n\left|\frac{u-1}{u+1}\right| - 2\sin^{-1} t + C$$

$$= -2\sin^{-1}\sqrt{x} - \ell n\left|\frac{\sqrt{1-x}+1}{\sqrt{1-x}-1}\right| + C$$

3.1 INTEGRATION BY TRIGONOMETRIC SUBSTITUTION

$\int \frac{dx}{a^2+x^2}$ or $\int \sqrt{a^2+x^2}\,dx$; put $x = a\tan\theta$ or $x = a\cot\theta$

$\int \frac{dx}{\sqrt{a^2-x^2}}$ or $\int \sqrt{a^2-x^2}\,dx$; put $x = a\sin\theta$ or $x = a\cos\theta$

$\int \frac{dx}{\sqrt{x^2-a^2}}$ or $\int \sqrt{x^2-a^2}\,dx$; put $x = a\sec\theta$ or $x = a\,\text{cosec}\,\theta$

1. $\int \frac{dx}{\sqrt{a^2-x^2}} = \sin^{-1}\frac{x}{a} + C$

2. $\int \frac{dx}{\sqrt{x^2+a^2}} = \ell n\left|x+\sqrt{x^2+a^2}\right| + C$ 3. $\int \frac{dx}{\sqrt{x^2-a^2}} = \ell n\left|x+\sqrt{x^2-a^2}\right| + C$

Questions

1. $\int \frac{dx}{(1+x^2)^2}$

Ans. $\frac{\tan^{-1}x}{2} + \frac{x}{2(1+x^2)} + C$

Sol. $I = \int \frac{1}{(1+x^2)^2}dx$ put $x = \tan\theta$

$$\int \cos^{2+}\theta d\theta = \frac{\theta}{2} + \frac{\sin 2\theta}{4} + C = \frac{1}{2}\tan^{-1}x + \frac{x}{2(1+x^2)} + C$$

2. $\int \cos\left(2\cot^{-1}\sqrt{\frac{1-x}{1+x}}\right)dx$

Ans. $-\frac{x^2}{2} + C$

Sol. $I = \int \cos\left(\underbrace{2\cot^{-1}\sqrt{\frac{1-x}{1+x}}}_{\theta}\right)dx$

$$= \int \cos 2\theta dx$$

$$= \int \frac{\cot^2\theta - 1}{\cot^2\theta + 1} dx$$

$$= \int \frac{\frac{1-x}{1+x} - 1}{\frac{1-x}{1+x} + 1} dx$$

$$= \int -x dx = -\frac{x^2}{2} + C$$

3. $\int \sqrt{\frac{1-x}{1+x}} dx$

Ans. $-\cos^{-1}(x) + \sqrt{1-x^2} + C$

Sol. $I = \int \sqrt{\frac{1-x}{1+x}} dx$

put $x = \cos\theta \rightarrow dx = -\sin\theta\, d\theta$

$$I = \int \tan\frac{\theta}{2} \cdot (-\sin\theta d\theta)$$

$$I = \int (\cos\theta - 1) d\theta$$

$I = \sin\theta - \theta + C$

$= \sqrt{1-x^2} - \cos^{-1} x + C$

4. $\int \frac{dx}{\sqrt{-2x^2 + 3x + 5}}$;

Ans. $\frac{1}{\sqrt{2}} \sin^{-1}\left(\frac{4x-3}{7}\right) + C$

Sol. $I = \int \frac{2\sqrt{2}}{\sqrt{-16x^2 + 24x + 40}}$

$$= \int \frac{2\sqrt{2}}{\sqrt{-(4x-3)^2 + 49}} dx$$

$$= 2\sqrt{2} \sin^{-1}\left(\frac{4x-3}{7}\right)\frac{1}{4} + C$$

$$= \frac{1}{\sqrt{2}} \sin^{-1}\left(\frac{4x-3}{7}\right) + C$$

5. $\int \frac{\sqrt{(9-x^2)^3}}{x^6}dx$

Ans. $-\frac{1}{45}\left(\sqrt{\frac{9-x^2}{x^2}}\right)^5 + C$

Sol. $I = \int \frac{\sqrt{(9-x^2)^3}}{x^6}dx$

$x = 3\sin\theta$

$= \int \frac{\cos^3\theta}{\sin^6\theta}\frac{3}{27}\cos\theta d\theta$

$= \int \frac{(\cot\theta)^4 \operatorname{cosec}^2\theta d\theta}{9}$

$= -\frac{(\cot\theta)^5}{45} + C$

$= \frac{-\left(\sqrt{9-x^2}\right)^5}{45x^5} + C$

6. $\int \frac{\sqrt{x}\ dx}{\sqrt{a^3-x^3}}$

Ans. $\frac{2}{3}\sin^{-1}\left(\frac{x^{3/2}}{a^{3/2}}\right) + C$

Sol. $I = \int \frac{\sqrt{x}}{\sqrt{a^3-x^3}}d\theta$

$x = a\sin^{\frac{2}{3}}\theta$

$= \frac{2}{3}\theta + C$

$= \frac{2}{3}\sin^{-1}\left(\frac{x^{\frac{3}{2}}}{a^{\frac{3}{2}}}\right) + C$

7. $\int \frac{x^2}{\sqrt{a^6-x^6}}dx$

Ans. $\frac{1}{3}\sin^{-1}\left(\frac{x^3}{a^3}\right)+C$

Sol. put $x^3 = a^3 \sin\theta$

$$I = \frac{1}{3}\int d\theta$$

$$= \frac{1}{3}\sin^{-1}\left(\frac{x^3}{a^3}\right)+C$$

8. $\int \sqrt{\frac{x}{a-x}}.dx$

Ans. $-\sqrt{x}\sqrt{a-x} - a\tan^{-1}\left(\frac{\sqrt{a-x}}{\sqrt{x}}\right)+C$

Sol. $\int\sqrt{\frac{x}{a-x}}dx$

$x = a\sin^2\theta$

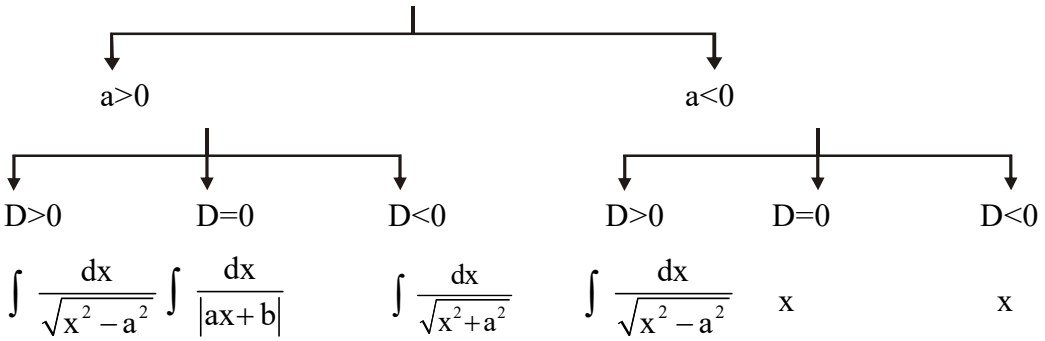
$$= a\int 1-\cos 2\theta d\theta$$

$$= a\left[\sin^{-1}\sqrt{\frac{x}{a}} - \sqrt{\frac{x}{a}}\sqrt{1-\frac{x}{a}}\right]+C$$

$$= a\sin^{-1}\sqrt{\frac{x}{a}} - \sqrt{x}\sqrt{a-x}+C$$

3.2 PROFILE

$\int \frac{dx}{\sqrt{\text{Quad.}}}$



Note: Complete the square and use corresponding formula

Questions

1. $\int \frac{dx}{25+16x^2}$

Ans. $\frac{1}{20}\tan^{-1}\left(\frac{4x}{5}\right)+C$

Sol. $I=\int\frac{dx}{25+16x^2}=\frac{1}{5}\cdot\frac{1}{4}\tan^{-1}\left(\frac{4x}{5}\right)+C$

2. $\int\frac{dx}{\sqrt{12x-4x^2}}$

Ans. $\frac{1}{2}\sin^{-1}\left(\frac{2x-3}{3}\right)+C$

Sol. $I=\int\frac{dx}{\sqrt{12x-4x^2}}=\int\frac{dx}{\sqrt{9-(2x-3)^2}}=\frac{1}{2}\sin^{-1}\left(\frac{2x-3}{3}\right)+C$

3. $\int\frac{dx}{\sqrt{16x^2-8x+26}}$

Ans. $\frac{1}{4}\,ln\left(4x-1+\sqrt{(4x-1)^2+25}\right)+C$

Sol. $I=\int\frac{dx}{\sqrt{16x^2-8x+26}}=\int\frac{dx}{\sqrt{(4x-1)^2+5^2}}=\frac{1}{4}\ell n\left|4x-1+\sqrt{16x^2-8x+26}\right|+C$

4. $\int\frac{dx}{2x^2-9x-5}$

Ans. $\frac{1}{11}\,ln\left|\frac{2x-10}{2x+1}\right|+C$

Sol. $\int\frac{dx}{2x^2-9x-5}=\int\frac{dx}{(2x+1)(x-5)}=\int\left(\frac{1}{x-5}-\frac{2}{2x+1}\right)\frac{1}{11}dx$

$=\frac{1}{11}\ell n\ell n|x-5|-\frac{1}{11}\ell n|2x+1|+C$

$=\frac{1}{11}\ell n\left|\frac{x-5}{2x+1}\right|+C$

5. $\int\frac{dx}{x^2+x+1}$

Ans. $\frac{2}{\sqrt{3}}\tan^{-1}\left(\frac{2x+1}{\sqrt{3}}\right)+C$

Sol. $I=\int\frac{1}{x^2+x+1}dx$

$$I = \int \frac{1}{\left(x+\frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2} dx$$

$$I = \frac{2}{\sqrt{3}} \tan^{-1}\left(\frac{2x+1}{\sqrt{3}}\right) + C$$

6. $\int \frac{\sin 2x}{\sqrt{9-\sin^4 x}} dx$;

Ans. $\sin^{-1}\left(\frac{\sin^2 x}{3}\right) + C$

Sol. $I = \int \frac{\sin 2x}{\sqrt{9-\sin^4 x}} dx$

put $\sin^2 x = t$

$$= \sin^{-1}\left(\frac{\sin^2 x}{3}\right) + C$$

7. $\int \frac{e^x dx}{\sqrt{e^{2x}-1}}$;

Ans. $\ln\left|e^x + \sqrt{e^{2x}-1}\right| + C$

Sol. put $e^x = t$

$$I = \int \frac{1}{\sqrt{t^2-1}} dt$$

$$I = \ell n\left|e^x + \sqrt{e^{2x}-1}\right| + C$$

8. $\int \frac{e^x}{4+e^{2x}} dx$;

Ans. $\frac{1}{2}\tan^{-1}\left(\frac{e^x}{2}\right) + C$

Sol. put $e^x = t$

$$I = \int \frac{1}{t^2+2^2} dt = \frac{1}{2}\tan^{-1}\left(\frac{t}{2}\right) + C = \frac{1}{2}\tan^{-1}\left(\frac{e^x}{2}\right) + C$$

9. $\int \frac{dx}{\sqrt{2ax-x^2}}$

Ans. $\sin^{-1}\left(\frac{x-a}{a}\right)+C$

Sol. $\int\frac{1}{\sqrt{2ax-x^2}}dx=\int\frac{1}{\sqrt{a^2-(x-a)^2}}dx=\sin^{-1}\left(\frac{x-a}{a}\right)+C$

10. $\int\frac{dx}{a^2e^x+b^2e^{-x}}$

Ans. $\frac{1}{ab}\tan^{-1}\left(\frac{ae^x}{b}\right)$

Sol. $\int\frac{1}{a^2e^x+b^2e^{-x}}dx=\int\frac{e^x}{\left(ae^x\right)^2+b^2}dx$ put $e^x=t$

$=\frac{1}{ab}\tan^{-1}\left(\frac{ae^x}{b}\right)+C$

11. $\int\left(\frac{x-1}{1+x^2}\right)^2dx$

Ans. $\tan^{-1}x+\frac{1}{1+x^2}$

Sol. $I=\int\frac{x^2+1-2x}{\left(1+x^2\right)^2}dx$

$=\int\frac{1}{\left(1+x^2\right)}-\frac{2x}{\left(1+x^2\right)^2}dx$

put $1+x^2=t$

$=\tan^{-1}x+\frac{1}{1+x^2}+C$

12. $\int\frac{\tan x}{\sqrt{b+a\tan^2 x}}dx,(b>a>0)$

Ans. $\frac{-1}{\sqrt{b-a}}ln\left|\sqrt{b-a}\,t+\sqrt{a+(b-a)}\,t^2\right|+C$, $t=\cos x$

Sol. $I=\int\frac{\tan x}{\sqrt{b+a\tan^2 x}}dx$

$$I = \int \frac{\sin x}{\sqrt{b\cos^2 x + a\sin^2 x}}\,dx$$

$$= \int \frac{\sin x}{\sqrt{(b-a)\cos^2 x + a}}\,dx$$

put $\cos x = t$

$$= \int \frac{-dt}{\sqrt{\left(\sqrt{a}\right)^2 + \left(\sqrt{b-a}\right)^2 (\cos x)^2}}$$

$$= -\ell n\left|\sqrt{b-a}\cos x + \sqrt{b\cos^2 x + a\sin^2 x}\right| + C$$

13. $\int \frac{e^x\,dx}{\sqrt{5-4e^x+e^{2x}}}$

Ans. $ln\left[\left(e^x-2\right)+\sqrt{e^{2x}-4e^x+5}\right]$

Sol. $I = \int \frac{e^x}{\sqrt{5-4e^x+e^{2x}}}\,dx$ put $e^x = t$

$$= \int \frac{1}{(t-2)^2+1}\,dt$$

$$= \ell n\left|e^x - 2 + \sqrt{e^{2x}-4e^x+5}\right| + C$$

14. $\int \sqrt{\frac{1-\sqrt{x}}{1+\sqrt{x}}}\;dx$

Ans. $\sqrt{1-x}\left(\sqrt{x}-2\right) - \sin^{-1}\sqrt{x} + C$

Sol. $I = \int \sqrt{\frac{1-\sqrt{x}}{1+\sqrt{x}}}dx$

$$I = \int \frac{1-\sqrt{x}}{\sqrt{1-x}}\,dx$$

$$= \int \frac{1}{\sqrt{1-x}}\,dx - \int \frac{\sqrt{x}}{\sqrt{1-x}}\,dx$$ put $x = \sin^2 2\theta$

$$= \frac{2\sqrt{1-x}}{-1} + \sqrt{x}\sqrt{1-x} - \sin^{-1}\sqrt{x} + C$$

$$= \sqrt{1-x}\left(\sqrt{x}-2\right) - \sin^{-1}\sqrt{x} + C$$

15.(A) $\int \tan\left(2\tan^{-1}\sqrt{\frac{\sqrt{1+\sqrt{x}}-1}{\sqrt{1+\sqrt{x}}+1}}\right)dx$ (B) $\int \cot\left(2\tan^{-1}\sqrt{\frac{\sqrt{1+\sqrt{x}}-\sqrt[4]{x}}{\sqrt{1+\sqrt{x}}+\sqrt[4]{x}}}\right)dx$ **[O]**

(C) $\int\left(\frac{1-\tan\left(\frac{1}{2}\sin^{-1}\left(\frac{1-\sqrt{x}}{1+\sqrt{x}}\right)\right)}{1+\tan\left(\frac{1}{2}\sin^{-1}\left(\frac{1-\sqrt{x}}{1+\sqrt{x}}\right)\right)}\right)dx$ (D) $\int\sqrt{x}\tan\left(2\tan^{-1}\left(\frac{\sqrt{\sqrt{1+\sqrt{x}}+1}-\sqrt{\sqrt{1+\sqrt{x}}-1}}{\sqrt{\sqrt{1+\sqrt{x}}+1}+\sqrt{\sqrt{1+\sqrt{x}}-1}}\right)\right)dx$

[Sol. Let $\sqrt{x}=\tan^2\theta$

$x=\tan^4\theta \Rightarrow \tan\theta = x^{1/4}$ $\left(\because x\in(0,1) \atop \therefore \theta\in(0,\pi/4)\right)$

$\therefore \quad dx = 4\tan^3\theta\sec^2\theta\,d\theta$

$\therefore \quad \sqrt{1+\sqrt{x}} = \sec\theta$

(A) $I=\int\tan\left(2\tan^{-1}\sqrt{\frac{\sqrt{1+\sqrt{x}}-1}{\sqrt{1+\sqrt{x}}+1}}\right)dx$

$\Rightarrow \quad \tan\left(2\tan^{-1}\sqrt{\frac{\sqrt{1+\sqrt{x}}-1}{\sqrt{1+\sqrt{x}}+1}}\right)=\tan\left(2\tan^{-1}\sqrt{\frac{\sec\theta-1}{\sec\theta+1}}\right)=\tan\left(2\tan^{-1}\left(\tan\frac{\theta}{2}\right)\right)=\tan\theta$

$\therefore \quad I=\int \tan\theta\cdot 4\tan^3\theta\sec^2\theta\,d\theta=\frac{4}{5}\tan^5\theta+c=\frac{4}{5}(x^{5/4})+c$ **Ans.**

(B) $\int\cot\left(2\tan^{-1}\sqrt{\frac{\sqrt{1+\sqrt{x}}-\sqrt[4]{x}}{\sqrt{1+\sqrt{x}}+\sqrt[4]{x}}}\right)dx$; $\therefore \cot\left(2\tan^{-1}\sqrt{\frac{\sqrt{1+\sqrt{x}}-\sqrt[4]{x}}{\sqrt{1+\sqrt{x}}+\sqrt[4]{x}}}\right)=\cot\left(2\tan^{-1}\sqrt{\frac{\sec\theta-\tan\theta}{\sec\theta+\tan\theta}}\right)$

$=\cot\left(2\tan^{-1}\sqrt{(\sec\theta-\tan\theta)^2}\right)$

$=\cot\left(2\tan^{-1}\frac{1-\sin\theta}{\cos\theta}\right)$ If $\theta\in(0,\pi/4)$ then $\sec\theta-\tan\theta>0$

$=\cot\left(2\tan^{-1}\tan\left(\frac{\pi}{4}-\frac{\theta}{2}\right)\right)=\cot\left(\frac{\pi}{2}-\theta\right)=\tan\theta$

$\therefore \quad I=\int\tan\theta\cdot 4\tan^3\theta\sec^2\theta\,d\theta=\frac{4}{5}\tan^5\theta+c=\frac{4}{5}(x^{5/4})+c$

(C) $\frac{1}{2}\sin^{-1}\left(\frac{1-\sqrt{x}}{1+\sqrt{x}}\right)=\frac{1}{2}\sin^{-1}\left(\frac{1-\tan^2\theta}{1+\tan^2\theta}\right)=\frac{1}{2}\sin^{-1}(\cos 2\theta)=\frac{1}{2}\sin^{-1}\sin\left(\frac{\pi}{2}-2\theta\right)=\frac{\pi}{4}-\theta$

$$\therefore \int \left(\frac{1-\tan\left(\frac{1}{2}\sin^{-1}\left(\frac{1-\sqrt{x}}{1+\sqrt{x}}\right)\right)}{1+\tan\left(\frac{1}{2}\sin^{-1}\left(\frac{1-\sqrt{x}}{1+\sqrt{x}}\right)\right)} \right) dx = \int \frac{1-\tan\left(\frac{\pi}{4}-\theta\right)}{1+\tan\left(\frac{\pi}{4}-\theta\right)} 4\tan^3\theta \sec^2\theta d\theta$$

$$= \int \tan\left(\frac{\pi}{4}-\left(\frac{\pi}{4}-\theta\right)\right) 4\tan^3\theta \sec^2\theta \, d\theta = \int 4\tan^4\theta \sec^2\theta \, d\theta = \frac{4}{5}\tan^5\theta + C = \frac{4}{5}(x^{5/4}) + C \;]$$

(D) Let $\sqrt{x} = \tan^2\theta \Rightarrow x = \tan^4\theta$ $\left(\because x \in (0,1), \therefore \theta \in (0, \pi/4)\right)$

$\therefore \quad dx = 4\tan^3\theta \sec^2\theta \, d\theta$

$\therefore \quad \sqrt{1+\sqrt{x}} = \sec\theta$

$$I = \int \sqrt{x} \tan\left(2\tan^{-1}\left(\frac{\sqrt{\sqrt{1+\sqrt{x}}+1}-\sqrt{\sqrt{1+\sqrt{x}}-1}}{\sqrt{\sqrt{1+\sqrt{x}}+1}+\sqrt{\sqrt{1+\sqrt{x}}-1}}\right)\right) dx$$

$$\therefore \quad \sqrt{x} \tan\left(2\tan^{-1}\left(\frac{\sqrt{\sqrt{1+\sqrt{x}}+1}-\sqrt{\sqrt{1+\sqrt{x}}-1}}{\sqrt{\sqrt{1+\sqrt{x}}+1}+\sqrt{\sqrt{1+\sqrt{x}}-1}}\right)\right)$$

$$= \tan^2\theta \cdot \tan\left(2\tan^{-1}\left(\frac{\sqrt{\sec\theta+1}-\sqrt{\sec\theta-1}}{\sqrt{\sec\theta+1}+\sqrt{\sec\theta-1}}\right)\right) = \tan^2\theta \cdot \tan\left(2\tan^{-1}\left(\frac{\cos\frac{\theta}{2}-\sin\frac{\theta}{2}}{\cos\frac{\theta}{2}+\sin\frac{\theta}{2}}\right)\right)$$

$$= \tan^2\theta \cdot \tan\left(2\tan^{-1}\tan\left(\frac{\pi}{4}-\frac{\theta}{2}\right)\right) = \tan^2\theta \cdot \tan\left(\frac{\pi}{2}-\theta\right) = \tan^2\theta \cdot \cot\theta = \tan\theta$$

$$\therefore I = \int \tan\theta \cdot 4\tan^3\theta \sec^2\theta \, d\theta = \frac{4\tan^5\theta}{5} + c = \frac{4}{5}(x^{5/4}) + c \;]$$

4.1 INTEGRATION BY PARTS

$(fg)' = f'(g) + fg$

$fg' = (fg)' - f'g$

$$\int \underbrace{f(x)}_{I} . \underbrace{g'(x)}_{II} dx = f(x).g(x) - \int f'(x).g(x).dx$$

$$\int I.II = I\int II - \int \left((I')\int II\right)$$

This is generally obtained by choosing first function as the function which comes first in the word **ILATE**, where; I- Inverse function, L-Logarithmic function, A-Algebraic function, T-Trigonometric function & E-Exponential function.

Ex.1 Evaluate : $\int x \sin x \, dx$

Ans $-x\cos x+\sin x+c$

Sol. Here integrating by parts by taking x as the first function and sin x as the second function.

$I=\int x.\sin x\,dx.$

$=x(-\cos x)-\int 1.(-\cos x)\,dx.$

$=-x\cos x+\int\cos x.\,dx$

$=-x\cos x+\sin x+c$

Questions

1. $\int x.e^x dx$

Ans. xe^x-e^x+C

Sol. $I=\int x\cdot e^x dx=x\cdot e^x-e^x+C$

2. $\int x\,\ell n\,x\,dx$

Ans. $\frac{x^2}{2}\ell n\,x-\frac{x^2}{4}+C$

Sol. $I=\int \underset{II}{x}\cdot\ell n\underset{I}{x}\,dx=\ell nx\cdot\frac{x^2}{2}-\frac{x^2}{4}+C$

3. $\int x^2\ e^x\ dx$

Ans. $x^2e^x-2xe^x+2e^x+C$

Sol. $I=\int \underset{I}{x^2}\ \underset{II}{e^x}dx=(x^2-2x+2)e^x+C$

Sol. Here integrating by parts by taking x^2 as the first function and e^x as the second function

$I=\int x^2\ e^x\ dx$

$=x^2\ e^x-\int 2x.e^x\ dx$

$=x^2\ e^x-2\,[x.e^x-\int 1.e^x\ dx]$ (taking x as first function)

$=x^2\ e^x-2xe^x+2e^x+c$

4. $\int x\tan^{-1}x dx$

Ans. $\frac{x^2}{2}\tan^{-1}(x)-\frac{x}{2}+\frac{\tan^{-1}(x)}{2}+C$

Sol. $I=\int \underset{II}{x}\tan^{-1}\underset{I}{x}\,dx=\tan^{-1}x\cdot\frac{x^2}{2}-\frac{1}{2}\int\frac{x^2}{1+x^2}dx$

$$= \frac{x^2 \cdot \tan^{-1} x}{2} - \frac{x}{2} + \frac{1}{2}\tan^{-1} x + C$$

5. $\int \ln x \, dx$

Ans. $x(\ell n\, x - 1) + C$

Sol. $I = \int \underset{II}{1} \underset{I}{\ell nx}\, dx$

$= \ell nx \cdot x - x + C$

6. $\int \sin^{-1} x \, dx$

Ans. $x \sin^{-1} x + \sqrt{1 - x^2} + C$

Sol. $I = \int 1 \sin^{-1} x dx$

$= \sin^{-1} x \cdot x - \int \frac{x}{\sqrt{1 - x^2}} dx$

$1 - x^2 = t$

$-x dx = \frac{dt}{2}$

$= x \sin^{-1} x + \sqrt{1 - x^2} + C$

7. $\int \tan^{-1} x \, dx$

Ans. $x.\ \tan^{-1} x - \frac{1}{2} \log (1 + x^2) + c$

Sol. Integrating it by parts by taking $\tan^{-1} x$ as the first function and 1 as the second function, we get

$I = \int \tan^{-1} x.\ 1\ dx$

$= \tan^{-1} x.\ x - \int \frac{1}{1 + x^2} .\ x\ dx$

$= x.\ \tan^{-1} x - \frac{1}{2} \log\ (1 + x^2) + c$

8. $\int \text{cosec}^2 x \cdot \ell n (\sec x)\, dx$

Ans. $-\cot x\, \ell n \sec x + x + c$

Sol. $I = \int \cos ec^2 x \cdot \ell n (\sec x)\, dx$

$= \ell n \sec x \cdot (-\cot x) + \int dx$

$= -\cot x \cdot \ell n \sec x + x + C$

9. $\int \sin x \cdot \ln(\sec x + \tan x)\, dx$

Ans. $-\cos x\, \ell n (\sec x + \tan x) + x + c$

Sol. $I = \int \sin x \cdot \ell n (\sec x + \tan x)\, dx$

$= -\cos x \cdot \ell n\left(\sec x + \tan x\right) + \int dx$

$= x - \cos x \cdot \ell n(\sec x + \tan x) + C$

10. $\int \frac{\sin^{-1} x}{(1-x^2)^{3/2}} dx$

Ans. $\frac{x \sin^{-1} x}{\sqrt{1-x^2}} + \ln\sqrt{1-x^2} + C$

Sol. $I = \int \sin^{-1} x \cdot \frac{1}{\sqrt{1-x^2}} \cdot \frac{1}{1-x^2} dx$

put $x = \sin\theta$

$= \int \frac{\theta}{\cos\theta} \cdot \frac{1}{\cos^2\theta} \cos\theta d\theta$

$= \int \theta \cdot \sec^2\theta d\theta$

$= \theta \cdot \tan\theta - \ell n|\sec\theta| + C$

11. $\int e^x (1+x) \cdot l n(xe^x) dx$

Ans. $xe^x\left(l n\left(xe^x\right) - 1\right) + C$

Sol. $I = \int e^x \left(1+x\right) \ell n\left(x \cdot e^x\right) dx$

$x \cdot e^x = t$

$= \int \ell n t dt = t\left(\ell n t - 1\right) + C$

$= xe^x (\ln(xe^x) - 1) + C$

12. $\int l n(1+x)^{1+x} dx$

Ans. $\frac{\left(x+1\right)^2 l n\left(x+1\right)}{2} - \frac{\left(x+1\right)^2}{4} + C$

Sol. $I = \int \ell n\left(1+x\right)^{(1+x)} dx$

$= \int \left(1+x\right) \ell n\left(1+x\right) dx = \ell n\left(1+x\right) \cdot \frac{\left(1+x\right)^2}{2} - \frac{\left(1+x\right)^2}{4} + C$

13. $\int \cos^{-1} \frac{1}{x} dx$

Ans. $x \sec^{-1} x - l n\left|x + \sqrt{x^2 - 1}\right| + C$

Sol. $I = \int 1 \cos^{-1}\left(\frac{1}{x}\right) dx$

$$= x\cos^{-1}\left(\frac{1}{x}\right) - \int \frac{(-1)}{\sqrt{1-\left(\frac{1}{x}\right)^2}} \cdot \left(-\frac{1}{x^2}\right) \cdot x\,dx$$

$$= x\cos^{-1}\left(\frac{1}{x}\right) \int \frac{1}{\sqrt{x^2-1}}\,dx$$

$$= c\cos^{-1}\left(\frac{1}{x}\right) - \ell n\left|x + \sqrt{x^2-1}\right| + C$$

$$= x\sec^{-1} x - \ell n\left|x + \sqrt{x^2-1}\right| + C$$

14. $\int \sin(\ln x)dx$

Ans. $\frac{x}{2}\left(\sin\left(\ln x\right) - \cos\left(\ln x\right)\right) + C$

Sol. $I = \int \sin\left(\ell nx\right)dx$

put $\ell nx = t$

$$= \int e^t \sin t\,dt$$

$$= \frac{e^t}{1+1}\left[1\cdot\sin t - 1\cdot\cos t\right] + C$$

$$= \frac{x}{2}\left[\sin x - \cos x\right] + C$$

15. $\int \sec^3 x\,dx$

Ans. $\frac{1}{2}\left(\sec x\tan x + \ln\left|\sec x + \tan x\right|\right) + C$

Sol. $I = \int \sec^3 x dx$

$$= \int \sec x\cdot\sec^2 x dx$$

$$= \sec x\cdot\tan x - \int \sec x\cdot\tan x\cdot\tan x dx$$

$$= \sec x\cdot\tan x - \int \sec^3 x dx + \int \sec x dx$$

$I = \sec x\cdot\tan x - I + \ell n(\sec x + \tan x) + C$

$$I = \frac{1}{2}\left|\sec x\cdot\tan x + \ell n\left(\sec x + \tan x\right)\right| + C$$

16. $\int \sqrt{x^2-a^2}\,dx$

Ans. $\frac{x}{2}\sqrt{x^2-a^2} - \frac{a^2}{2}\ln\left|x + \sqrt{x^2-a^2}\right| + C$

Sol. $I = \int 1\sqrt{x^2 - a^2}\,dx$

$$I = \sqrt{x^2 - a^2}\cdot x - \int \frac{x^2}{\sqrt{x^2 - a^2}}\,dx$$

$$= x\sqrt{x^2 - a^2} - \int \sqrt{x^2 - a^2}\,dx - a^2\int \frac{1}{\sqrt{x^2 - a^2}}\,dx$$

$$= x\sqrt{x^2 - a^2} - I - a^2 \ell n\left|x + \sqrt{x^2 - a^2}\right| + C$$

$$I = \frac{x}{2}\sqrt{x^2 - a^2} - \frac{a^2}{2}\ell n\left|x + \sqrt{x^2 - a^2}\right| + C$$

17. $\int \sqrt{x^2 + a^2}\,dx$

Ans. $\frac{x}{2}\sqrt{x^2 + a^2} + \frac{a^2}{2}\ln\left|x + \sqrt{x^2 + a^2}\right| + C$

Sol. $I = \int \sqrt{x^2 + a^2}\,dx$

$$I = x\cdot\sqrt{x^2 + a^2} - \int \frac{x^2}{\sqrt{x^2 + a^2}}\,dx$$

$$I = x\sqrt{x^2 + a^2} - \int \sqrt{x^2 + a^2}\,dx + a^2\int \frac{1}{\sqrt{x^2 + a^2}}\,dx$$

$$I = \frac{x}{2}\sqrt{x^2 + a^2} + \frac{a^2}{2}\ell n\left|x + \sqrt{x^2 + a^2}\right| + C$$

18. $\int x \sin x \cos^2 x\,dx$

Ans. $\dfrac{-x\cos x + \sin x}{4} + \dfrac{-\frac{x\cos 3x}{3} + \frac{\sin 3x}{9}}{4} + C$

Sol. $I = \int \underset{I}{x}\cdot\underbrace{\sin x\cdot\cos^2 x}_{II}\,dx$

$$I = x\cdot\left(-\frac{\cos^3 x}{3}\right) + \frac{1}{3}\int \cos^3 x\,dx$$

$$= -\frac{x}{3}\cos^3 x + \frac{1}{12}\left[\frac{\sin 3x}{3} + 3\sin x\right] + C$$

19. $\int \dfrac{x\ dx}{1 + \sin x}$

Ans. $x\tan x - \ell n\left(\sec x\right) - x\sec x + \ell n\left|\sec x + \tan x\right| + C$

Sol. $I = \int \frac{x}{1+\sin x}dx = \int \underset{I}{x}\cdot\underbrace{\frac{1-\sin x}{\cos^2 x}}_{II}dx$

$= x\cdot(\tan x - \sec x) - \ell n|\sec x| + \ell n|\sec x + \tan x| + C$

20. $\int x\cos x\ \cos 2x\ dx$

Ans. $\frac{x\sin 3x}{6} + \frac{\cos 3x}{18} + \frac{x\sin x}{2} + \frac{\cos x}{2} + C$

Sol. $I = \int \underset{I}{x}\cdot\underbrace{\cos x\cdot\cos 2x}_{II}dx$

$= x\left(\frac{\sin 3x}{6} + \frac{\sin x}{2}\right) + \frac{\cos 3x}{18} + \frac{\cos x}{2} + C$

21. $\int x^2 e^{3x}\,dx$

Ans. $e^{3x}\left(\frac{x^2}{3} - \frac{2x}{9} + \frac{2}{27}\right) + c$

Sol. $\int x^2\cdot e^{3x}dx$

$= x^2\cdot\left(\frac{e^{3x}}{3}\right) - (2x)\left(\frac{e^{3x}}{9}\right) + 2\left(\frac{e^{3x}}{27}\right) + C$

$= \frac{e^{3x}}{3}\left[x^2 - \frac{2x}{3} + \frac{2}{9}\right] + C$

22. $\int (x^3 + 3x + 1)e^{3x}\ dx$

Ans. $e^{3x}\left(\frac{x^3}{3} - \frac{x^2}{3} + \frac{33}{27}x - \frac{2}{27}\right) + C$

Sol. $I = \int \left(x^3 + 3x + 1\right)e^{3x}dx$

$I = \left(x^3 + 3x + 1\right)\frac{e^{3x}}{3} - \left(3x^2 + 3\right)\left(\frac{e^{3x}}{9}\right) + (6x)\left(\frac{e^{3x}}{27}\right) - 6\frac{e^{3x}}{81} + C$

$= \frac{e^{3x}}{3}\left[x^3 + 3x + 1 - x^2 - 1 + \frac{2}{3}x - \frac{2}{9}\right] + C$

$= \frac{e^{3x}}{3}\left[x^3 + \frac{11}{3}x - x^2 - \frac{2}{9}\right] + C$

23. $\int x^3 ln^2 x\,dx$

Ans. $x^4\left[\frac{\ln^2 x}{4}-\frac{\ln x}{8}+\frac{1}{32}\right]+C$

Sol. $I=\int x^3\cdot \ell n^2 x dx$

$=\ell n^2 x\cdot\frac{x^4}{4}-\int 2\ell nx\cdot\frac{x^3}{4}dx$

$=\frac{x^4}{4}\left(\ell nx\right)^2-\frac{1}{2}\int x^3\cdot\ell nx dx$

$=\frac{x^4}{4}\left(\ell nx\right)^2-\frac{1}{2}\ell nx\cdot\frac{x^4}{4}+\frac{1}{2}\int\frac{x^3}{4}+C$

$=\frac{x^4}{4}\left(\ell nx\right)^2-\ell nx\cdot\frac{x^4}{8}+\frac{x^4}{32}+C$

24. $\int \text{cosec}^3 x\, dx$

Ans. $-\frac{1}{2}\left(\cos ec\, x\cot x+l\, n\left|\cos ec\, x+\cot x\right|\right)+C$

Sol. $I=\int \cos ec^3 x dx$

$I=\int \cos ecx\cdot\cos ec^2 x dx$

$=-\cot x\cdot\cos ecx-\int \cos ecx\cdot\cot^2 x dx$

$=-\cot x\cdot\cos ecx-\int \cos ec^3 x dx+\int \cos ecx dx$

$=-\frac{\cot x\cdot\cos ecx}{2}+\ell n\left|\cos ecx-\cot x\right|+C$

25. $\int\frac{\cos^{-1}x}{x^3}dx$

Ans. $C-\frac{\cos^{-1}x}{2x^2}+\frac{\sqrt{1-x^2}}{2x}$

Sol. $I=\int\frac{\cos^{-1}x}{x^3}dx$

$=\cos^{-1}x\cdot\frac{x^{-2}}{-2}-\frac{1}{2}\int\frac{1}{\sqrt{1-x^2}}\frac{1}{x^2}dx$ put $x=\sin\theta$

$=-\frac{1}{2}\frac{\cos^{-1}x}{x^2}-\frac{1}{2}\int\frac{1}{\cos\theta}\frac{1}{\sin^2\theta}\cdot\cos\theta d\theta$

$=-\frac{1}{2}\frac{\cos^{-1}x}{x^2}+\frac{\cot\theta}{2}+C$

$$= -\frac{\cos^{-1} x}{2x^2} + \frac{1}{2}\frac{\sqrt{1-x^2}}{x} + C$$

4.2 THREE IMPORTANT RESULTS

1. $\int \sqrt{a^2 - x^2}\, dx = \frac{x}{2}\sqrt{a^2 - x^2} + \frac{a^2}{2} \sin^{-1}\frac{x}{a} + C$

2. $\int \sqrt{x^2 - a^2}\, dx = \frac{x}{2}\sqrt{x^2 - a^2} - \frac{a^2}{2}\ln\left|x + \sqrt{x^2 - a^2}\right| + C$

3. $\int \sqrt{x^2 + a^2}\, dx = \frac{x}{2}\sqrt{x^2 + a^2} + \frac{a^2}{2}\ln\left|x + \sqrt{x^2 + a^2}\right| + C$

4.3 IMPORTANT PROFILE

$$\int e^{ax}.\ \sin bx\, dx = \frac{e^{ax}}{a^2 + b^2}\ (a \sin bx - b \cos bx) + C$$

$$\int e^{ax.}\ \cos bx\, dx = \frac{e^{ax}}{a^2 + b^2}\ (a \cos bx + b \sin bx) + C$$

Q.1 Evaluate : $\int e^{2x} \sin 3x\ dx$

Sol. Let $I = \int e^{2x} \sin 3x\ dx$

$$= e^{2x}\ \left(-\frac{\cos 3x}{3}\right) - \int 2e^{2x}\ \left(-\frac{\cos 3x}{3}\right) dx$$

$$= -\frac{1}{3}\ e^{2x}\ \cos 3x + \frac{2}{3}\ \int e^{2x} \cos 3x\ dx$$

$$= -\frac{1}{3}\ e^{2x}\ \cos 3x + \frac{2}{3}\ \left[e^{2x}\frac{\sin 3x}{3} - \int 2e^{2x}\frac{\sin 3x}{3}\, dx\right]$$

$$= -\frac{1}{3}\ e^{2x}\ \cos 3x + \frac{2}{9}e^{2x}\ \sin 3x - \frac{4}{9}\ \int e^{2x} \sin 3x\ dx$$

$$\Rightarrow \quad I = -\frac{1}{3}\ e^{2x}\ \cos 3x + \frac{2}{9}e^{2x}\ \sin 3x - \frac{4}{9}\ I$$

$$\Rightarrow \quad I + \frac{4}{9}\ I = \frac{e^{2x}}{9}\ (2 \sin 3x - 3 \cos 3x)$$

$$\Rightarrow \quad \frac{13}{9}\ I = \frac{e^{2x}}{9}\ (2 \sin 3x - 3 \cos 3x)$$

$$\Rightarrow \quad I = \frac{e^{2x}}{13}\ (2 \sin 3x - 3 \cos 3x) + C$$

4.4 INTEGRATION OF RATIONAL FUNCTION

Profile:-1 $\int \frac{dx}{\text{linear}}$

Eg:- $\int \frac{dx}{3x+2} = \frac{1}{3} ln|3x+2| + c$

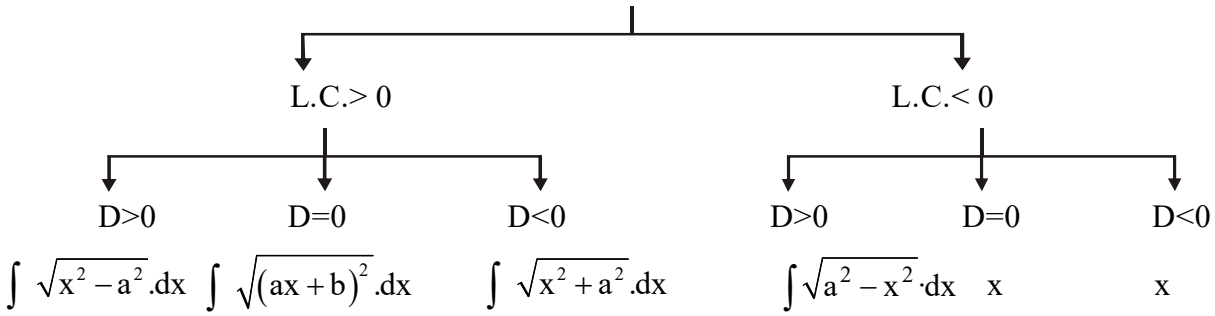
Profile:-2 $\int \frac{\text{poly}}{\text{linear}}.dx$

Eg:- $\int \frac{x^3+x^2+x+4}{x+1}dx = \int \frac{(x^2+1)(x+1)+3}{x+1}.dx = \int \left(x^2+1+\frac{3}{x+1}\right).dx$

$= \frac{x^3}{3} + x + 3ln|x+1| + c$

Profile:-3 $\int \frac{dx}{\text{Quad.}}$ Profile:-4 $\int \frac{dx}{\sqrt{\text{Quad.}}}$

Profile:-5 $\int \sqrt{\text{Quad}}.dx$



Profile:-6 $\int \frac{\text{linear}}{\text{Quad.}}$ or $\int \frac{L}{\sqrt{Q}}$ or $\int L\sqrt{Q}$

L=A(Q')+B

compare coeff. and get A and B

Eg: $\int \frac{4x+3}{\sqrt{x^2-2x+2}}dx$

$4x + 3 = A(2x - 2) + B$

$\left.\begin{matrix} 2A = 4 \\ 3 = B - 2A \end{matrix}\right] \begin{matrix} A = 2 \\ B = 7 \end{matrix}$

$$I = \int \frac{2(2x-2)+7}{\sqrt{x^2-2x+2}}.dx$$

$$= 2\int \frac{(2x-2)dx}{\sqrt{x^2-2x+2}} + 7\int \frac{dx}{\sqrt{x^2-2x+2}}$$

Profile:-$7\int \frac{poly}{Quad.}$

Eg:- $\int \frac{x^4+x^2+2x+4}{x^2-x+1}.dx \quad = \int \frac{(x^2+x+1)(x^2-x+1)+2x+3}{x^2-x+1}.dx \; = \int \left(x^2+x+1+\frac{2x+3}{x^2-x+1}\right).dx$

5. TWO CLASSIC INTEGRALS

$\int e^x [f(x) + f'(x)]dx = e^x f(x) + c$

$\int [x f'(x) + f(x)]dx = x f(x) + c$

Ex.1 $\int e^x (\sin x + \cos x)\, dx$

Sol. $I = \int e^x (\sin x + \cos x)\, dx$

This is of the form

$\int e^x [f(x) + f'(x)]\, dx = e^x f(x) + c$

Now here $f(x) = \sin x$

$\therefore\ I = e^x \sin x + c$

Ex.2 $\int (x \sec^2 x + \tan x)\, dx$

Sol. Here $I = \int (x \sec^2 x + \tan x)\, dx$

$= \int [x f'(x) + f(x)]dx$ where $f(x) = \tan x$

$= x\ f(x) + c$

$= x.\ \tan x + c$

Questions

1. $\int e^x \left(x^2 + 2x\right)dx$

Ans. $e^x x^2 + C$

Sol. $\int e^x \left(x^2 + 2x\right)dx = x^2 \cdot e^x + C$

2. $\int e^x \left(\sec^2 x + \tan x\right)dx$

Ans. $e^x \tan x + C$

Sol. $\int e^x \left(\sec^2 x + \tan x\right)dx = e^x \tan x + C$

3. $\int e^x \left(l\text{n } x + \frac{1}{x} \right) dx + C$

Ans. $e^x \, l\text{n } x + C$

Sol. $I = \int (e^x \log x + e^x . 1/x) \, dx$

$= e^x \log x + c \left[\begin{matrix} \text{Here } f(x) = \log x \\ \& f'(x) = 1/x \end{matrix} \right]$

4. $\int e^x \left(\frac{2x+1}{2\sqrt{x}} \right) dx$

Ans. $e^x \sqrt{x} + C$

Sol. $I = \int e^x \left(\frac{2x+1}{2\sqrt{x}} \right) dx$

$= \int e^x \left(\sqrt{x} + \frac{1}{2\sqrt{x}} \right) dx = e^x \sqrt{x} + C$

5. $\int \frac{x e^x}{(1+x)^2} dx$

Ans. $\frac{e^x}{1+x} + C$

Sol. $I = \int e^x \cdot \frac{x}{(1+x)^2} dx = \int e^x \left[\frac{1+x-1}{(1+x)^2} \right] dx$

$= \int ehx \left[\frac{1}{1+x} - \frac{1}{(1+x)^2} \right] dx$

$= e^x \cdot \frac{1}{1+x} + C$

Sol. Given integral $= \int e^x \, \frac{x+1-1}{(x+1)^2} \, dx = \int e^x \left(\frac{1}{(x+1)} - \frac{1}{(x+1)^2} \right) dx = \frac{e^x}{(x+1)} + C$

6. $\int e^x \left(\sec x + l\text{n } \left(\sec x + \tan x \right) \right) dx$

Ans. $e^x \, l\text{n} \left(\sec x + \tan x \right) + C$

Sol. $I = \int e^x \cdot \left(\sec x + \ell\text{n} \left(\sec x + \tan x \right) \right) dx$

$= e^x \cdot \ell\text{n}(\sec x + \tan x) + C$

7. $\int e^{2x} \left(\sin x + \frac{\cos x}{2} \right) dx$

Ans. $\frac{e^{2x} \sin x}{2} + C$

Sol. $I = \int e^{2x}\left(\sin x + \frac{\cos x}{2}\right)dx$

$I = e^{2x}\frac{\sin x}{2} + C$

8. $\int \frac{e^x}{x}(1 + x \cdot \ell n\, x)dx$

Ans. $e^x\, \ell n\, x + C$

Sol. $I = \int \frac{e^x}{x}(1 + x\cdot \ell nx)\,dx = \int e^x\left(\frac{1}{x} + \ell nx\right)dx$

$= e^x \cdot \ell nx + C$

9. $\int e^x \frac{1+\sin x}{1+\cos x}dx$

Ans. $e^x \tan\frac{x}{2} + C$

Sol. $I = \int e^x \cdot \left(\frac{1+\sin x}{1+\cos x}\right)dx$

$I = \int e^x\left(\frac{1 + 2\sin\frac{x}{2}\cdot\cos\frac{x}{2}}{2\cos^2\frac{x}{2}}\right)dx$

$= \int e^x\left(\frac{1}{2}\cdot\sec^2\frac{x}{2} + \tan\frac{x}{2}\right)dx$

$= e^x \cdot \tan\frac{x}{2} + C$

10. $\int \frac{e^x(1-\sin x)}{1-\cos x}dx$

Ans. $-e^x \cot\left(\frac{x}{2}\right) + C$

Sol. Given integral $= \int e^x\left(\frac{1 - 2\sin\frac{x}{2}\cos\frac{x}{2}}{2\sin^2\frac{x}{2}}\right)dx$

$= \int e^x\left(\frac{1}{2}\text{cosec}^2\frac{x}{2} - \cot\frac{x}{2}\right)dx = -e^x \cot\frac{x}{2} + C$

11. $\int \frac{e^{\tan^{-1}x}(1+x+x^2)}{1+x^2}dx$

Ans. $x\,e^{\tan^{-1}x} + C$

Sol. $I = \int e^{\tan^{-1}x}\left(1+\frac{x}{1+x^2}\right)$

$= \int\left(e^{\tan^{-1}x} + x\cdot\frac{e^{\tan^{-1}x}}{1+x^2}\right)dx$

$= x\cdot e^{\tan^{-1}x} + C$

12. $\int \frac{e^x(x^2+5x+7)}{(x+3)^2}dx$

Ans. $e^x\left(\frac{x+2}{x+3}\right)+C$

Sol. $I = \int e^x\cdot\frac{\left(x^2+5x+7\right)}{\left(x+3\right)^2}dx$

$I = \int e^x\left(\frac{(x+2)(x+3)}{(x+3)^2}+\frac{1}{(x+3)^2}\right)dx$

$= \int e^x\left(\frac{x+2}{x+3}+\frac{1}{(x+3)^2}\right)dx$

$= e^x\cdot\frac{x+2}{x+3}+C$

13. $\int \frac{x^2e^x}{(x+2)^2}dx$

Ans. $e^x\ \frac{x-2}{x+2}\ + C$

Sol. $I = \int \frac{x^2\cdot e^x}{(x+2)^2}dx$

$I = \int e^x\left(\frac{x-2}{x+2}+\frac{4}{(x+2)^2}\right)dx$

$I = e^x\frac{(x-2)}{x+2}+C$

14. $\int \frac{e^x(1+x+x^3)}{(1+x^2)^{3/2}}dx$

Ans. $\dfrac{e^x x}{\sqrt{x^2+1}} + C$

Sol. $I = \int e^x \dfrac{1+x+x^3}{\left(1+x^2\right)^{\frac{3}{2}}} dx$

$= \int e^x \left[\dfrac{x}{\sqrt{1+x^2}} + \dfrac{1}{\left(1+x^2\right)^{\frac{3}{2}}} \right] dx$

$= e^x \cdot \dfrac{x}{\sqrt{1+x^2}} + C$

15. $\int \dfrac{e^x (x-1)}{(x+1)^3} dx$ [**JEE 83**]

Ans. $\dfrac{e^x}{(1+x)^2} + C$

Sol. $I = \int \dfrac{e^x (x-1)}{(x+1)^3} dx$

$= \int e^x \left[\dfrac{1}{(x+1)^2} - \dfrac{2}{(x+1)^3} \right] dx$

$= \dfrac{e^x}{(x+1)^2} + C$

16. $\int \dfrac{\sqrt{1+\sin 2x}}{(1+\cos 2x)e^{-x}} dx$

Ans. $\dfrac{e^x \sec x}{2} + C$

Sol. $I = \int \dfrac{\sqrt{1+\sin 2x}}{(1+\cos 2x)e^{-x}} dx$

$= \int \dfrac{e^x \sqrt{1+\sin 2x}}{1+\cos 2x} dx$

$= \int e^x \left(\dfrac{\sin x + \cos x}{2\cos^2 x} \right) dx$

$= \dfrac{1}{2} e^x \cdot \sec x + C$

17. $\int \frac{e^x(x^2+1)}{(x+1)^2}dx$

Ans. $\frac{e^x(x-1)}{x+1} + C$

Sol. $I = \int \frac{e^x\left(x^2+1\right)}{\left(x+1\right)^2}dx$

$I = \int ehx\left[\frac{x-1}{x+1} + \frac{2}{\left(x+1\right)^2}\right]dx$

$= e^x\left(\frac{x-1}{x+1}\right) + C$

18. $\int \frac{x+\sin x}{1+\cos x}dx$

Ans. $x\tan\frac{x}{2} + C$

Sol. $I = \int \frac{x+\sin x}{1+\cos x}dx = \int\left(\frac{x+\sin x}{2\cos^2\frac{x}{2}}\right)dx = \int\left(x\frac{1}{2}\sec^2\frac{x}{2} + \tan\frac{x}{2}\right)dx = x\tan\frac{x}{2} + c$

19. $\int [\sin(\ell n\, x) + \cos(\ell n\, x)dx$

Ans. $x\sin(\ell n\, x) + C$

Sol. $\int\left(\sin\left(\ell nx\right) + \cos\left(\ell nx\right)\right)dx$

put $\ell nx = t$

$x = e^t$

$dx = e^t$

$I = \int e^t\left(\sin t + \cos t\right)dt$

$I = x\sin(\ell nx) + C$

20. $\int \frac{e^{2x}(\sin 4x - 2)}{1-\cos 4x}dx$

Ans. $\frac{1}{2}e^{2x}\cot 2x + C$

Sol. $I = \int e^{2x}.\frac{\sin 4x - 2}{1-\cos 4x}dx$

$= \frac{1}{2}\int e^{2x}\left(2\cot 2x - \operatorname{cosec}^2 2x\right)dx$

$$= \frac{e^{2x} \cdot \cot 2x}{2} + C$$

21. $\int \frac{x - \sin x}{1 - \cos x}\, dx$

Ans. $-x \cot \frac{x}{2} + C$

Sol. $I = \int \frac{x - \sin x}{1 - \cos x} dx$

$$= \int \left(x \frac{1}{2} \operatorname{cosec}^2 \frac{x}{2} - \cot \frac{x}{2} \right) dx$$

$$= -x \cot \frac{x}{2} + C$$

22. $\int \frac{\ell n x}{(1 + \ell n x)^2}\, dx$

Ans. $\frac{x}{1 + \ell n x} + C$

Sol. $\int \frac{\ell n x}{(1 + \ell n x)} dx = \int \left[\frac{1}{1 + \ell n x} + x \cdot \frac{1}{x} \frac{1}{(1 + \ell n x)^2} \right] dx$

$$= \frac{x}{1 + \ell n x} + C$$

23. $\int e^{\cos^{-1} x} \cdot \frac{(x+1) + \sqrt{1 - x^2}}{(x+1)^2 \sqrt{1 - x^2}} dx$

Ans. $C - \frac{e^{\cos^{-1} x}}{1 + x} + C$

Sol. Let $\cos^{-1} x = t \Rightarrow \quad - \frac{dx}{\sqrt{1 - x^2}} = dt$

$$I = - \int e^t \frac{(1 + \cos t) + \sin t}{(1 + \cos t)^2} dt = - \int e^t \left(\frac{1}{1 + \cos t} + \frac{\sin t}{(1 + \cos t)^2} \right) dt$$

$$= C - \frac{e^t}{1 + \cos t} = C - \frac{e^{\cos^{-1} x}}{1 + x} \text{ Ans.}$$

24. $\int e^x \left(\frac{x^4 + 2}{(1 + x^2)^{5/2}} \right) dx$

Ans. $\frac{e^x(1+x+x^2)}{(1+x^2)^{3/2}}+C$

Sol. $I=\int e^x\left(\frac{(1+x^2)^2+1-2x^2}{(1+x^2)^{5/2}}\right)dx=\int e^x\left(\frac{1}{\sqrt{1+x^2}}-\frac{x}{(1+x^2)^{3/2}}\right)+\left(\frac{x}{(1+x^2)^{3/2}}+\frac{1-2x^2}{(1+x^2)^{5/2}}\right)dx$

$=e^x\left(\frac{1}{\sqrt{1+x^2}}+\frac{x}{(1+x^2)^{3/2}}\right)+C$

25. $\int\left(\ell n(\ell nx)+\frac{1}{\ell n^2 x}\right)dx$

Ans. $x\,ln\,(ln\,x)-\frac{x}{ln\,x}+C$

Sol. Let $I=\int\left[\ell n\,(\ell nx)+\frac{1}{(\ell nx)^2}\right]dx$ {put $x=e^t\Rightarrow e^t\,dt$}

$\therefore I=\int e^t\left(\ell nt+\frac{1}{t^2}\right)dt=\int e^t\left(\ell nt-\frac{1}{t}+\frac{1}{t}+\frac{1}{t^2}\right)dt$

$=e^t\left(\ell nt-\frac{1}{t}\right)+C=x\left[\ell n\,(\ell nx)-\frac{1}{\ell nx}\right]+C$

26. $\int\sin 4x\cdot e^{\tan^2 x}\,dx$

Ans. $C-2e^{\tan^2 x}\cos^4 x+C$

Sol. $\int 2\cos 2x\cdot 2\frac{\sin x}{\cos x}\cos^2 x\,e^{\tan^2 x}\,dx$; $\int 2\cos 2x\cos^2 x\cdot 2\tan x\cdot e^{\tan^2 x}\,dx$

$\int 2\cos 2x\cos^4 x\cdot 2\tan x\sec^2 x\cdot e^{\tan^2 x}dx$; put $\tan^2x=t\Rightarrow 2\tan t\sec^2 t\,dx=dt$

$=2\int\frac{(1-t)e^t\,dt}{(1+t)^2(1+t)}=2\int\frac{e^t(1-t)\,dt}{(1+t)^3}=-2\int\frac{e^t[t+1-2]dt}{(1+t)^3}$

$=-2\int e^t\left[\frac{1}{(t+1)^2}-\frac{2}{(t+1)^3}\right]dt=-2\left[e^t\frac{1}{(t+1)^2}\right]+C=C-2e^{\tan^2 x}\cos^4 x$ Ans.

27. $\int e^{g(x)}\left(f(x)g'(x)+f'(x)\right)dx$

Ans. $e^{g(x)}\cdot f(x)+C$

Sol. $I=\int\underbrace{e^{g(x)}}_{II}\underbrace{f(x)}_{I}\underbrace{g'(x)}_{II}dx+\int e^{g(x)}f'(x)\,dx$

$=f(x)\cdot e^{g(x)}-\int f'(x)\cdot e^{g(x)}dx+\int e^{g(x)}\cdot f'(x)\,dx=f(x)\cdot e^{g(x)}$

28. $\int e^{(x\sin x+\cos x)}\left(\frac{x^2\cos^2 x-(x\sin x+\cos x)}{x^2}\right)dx$

Ans. $e^{(x\sin x+\cos x)}\cdot\frac{\cos x}{x}+C$

Sol $\int e^{(x\sin x+\cos x)}\left(\cos^2 x-\frac{x\sin x+\cos x}{x^2}\right)dx$

$\int e^{(x\sin x+\cos x)}\left(x\cos x\left(\frac{\cos x}{x}\right)+\left(\frac{\cos x}{x}\right)'\right)dx$

$e^{(x\sin x+\cos x)}\cdot\frac{\cos x}{x}+C$

29. $\int e^{\tan x}(\sin x-\sec x)dx$

Ans. $-e^{\tan x}\cdot\cos x+C$

Sol $=-e^{\tan x}\cdot\cos x+\int e^{\tan x}\sec^2 x\cos x\,dx-\int e^{\tan x}\sec x\,dx$

$=-e^{\tan x}\cdot\cos x$

30. If the primitive of the function $f(x)=\frac{x^{2009}}{(1+x^2)^{1006}}$ w.r.t. x is equal to $\frac{1}{n}\left(\frac{x^2}{1+x^2}\right)^m+C$ then find $(m+n)$ (where $m, n\in N$).

Ans. 3015

[Sol. $f(x)=\int\frac{x^{2009}}{(1+x^2)^{1006}}dx$

Put $1+x^2=t \Rightarrow 2x\,dx=dt$

$I=\frac{1}{2}\int\frac{(t-1)^{1004}\,dt}{t^{1006}}=\frac{1}{2}\int\left(1-\frac{1}{t}\right)^{1004}\cdot\frac{1}{t^2}dt$

put $1-\frac{1}{t}=y \Rightarrow \frac{1}{t^2}\,dt=dy$

$\therefore I=\frac{1}{2}\int y^{1004}\,dy=\frac{1}{2}\cdot\frac{y^{1005}}{1005}+C=\frac{1}{2010}\cdot\left(\frac{t-1}{t}\right)^{1005}+C$

$=\frac{1}{2010}\cdot\left(\frac{x^2}{1+x^2}\right)^{1005}+C \Rightarrow m=1005, n=2010 \Rightarrow m+n=3015$

31. If $\int\frac{4e^x+6e^{-x}}{9e^x-4e^{-x}}dx=Ax+B\log(9e^{2x}-4)+C$, then A = __ , B = __ and C = __

Ans. $A=-\frac{3}{2}, B=\frac{35}{36}$

Sol. $\dfrac{4e^x+6e^{-x}}{9e^x-4e^{-x}}=\dfrac{4e^{2x}+6}{9e^{2x}-4}=A+B\cdot\dfrac{18e^2x}{9e^{2x}-4}$

$\Rightarrow 4\cdot e^{2x}+6=(9A+18B)\,e^{2x}-4x$

$4A=-6 \qquad 9A+18B=4$

$A=\dfrac{-3}{2} \qquad B=\dfrac{35}{36}$

$$\int\frac{4e^x+6e^{-x}}{9e^x-4e^{-x}}dx=\int\left[-\frac{3}{2}+\frac{35}{36}\frac{18e^{2x}}{9e^{2x}-4}\right]dx$$

$$=-\frac{3}{2}x+\frac{35}{36}\ell n\left|9e^{2x}-4\right|+C$$

$A=-\dfrac{3}{2} \qquad B=\dfrac{35}{36}$

6.1 INTEGRATION OF TRIGONOMETRIC FUNCTIONS

Profile 1 $\int \sin^m x.\cos^n x\,dx$

Case - I If at least one of m or n is odd natural number then if m is odd put cosx = t and vice-versa.

Ex.1 Evaluate : $\int \sin^5 x\cos^4 x\,dx$

Sol. Let $I=\int \sin^5 x\cos^4 x\,dx$ put $\cos x=t \Rightarrow -\sin x\,dx=dt$

$$\Rightarrow I=-\int(1-t^2)^2\,.\,t^4\,.\,dt=-\int(t^4-2t^2+1)\,t^4\,dt=-\int(t^8-2t^6+t^4)\,dt$$

$$=-\frac{t^9}{9}+\frac{2t^7}{7}-\frac{t^5}{5}+C=-\frac{\cos^9 x}{9}+2\,\frac{\cos^7 x}{7}-\frac{\cos^5 x}{5}+C$$

Case - II : If m and n are even natural number then converts higher power into higher angles.

Ex.2 Evaluate : $\int \sin^2 x\cos^4 x\,dx$

Sol. $\dfrac{1}{8}\int\sin^2 2x(1+\cos 2x)dx=\dfrac{1}{8}\int\sin^2 2x\,dx+\dfrac{1}{8}\int\sin^2 2x\cos 2x\,dx$

$$=\frac{1}{16}\int(1-\cos 4x)\,dx+\frac{1}{16}\left(\frac{\sin^3 2x}{3}\right)=\frac{x}{16}-\frac{\sin 4x}{64}+\frac{\sin^3 2x}{48}+C$$

Case - III : When m + n is a negative even integer then put tan x = t.

Ex.3 Evaluate : $\int(\sin x)^{1/3}(\cos x)^{-7/3}dx$

Sol. Here $m+n=\dfrac{1}{3}-\dfrac{7}{3}=-2$ (a negative integer)

$$\therefore\ \int(\sin x)^{1/3}(\cos x)^{-7/3}dx=\int(\tan x)^{1/3}\ \frac{1}{\cos^2 x}\,dx \quad \{\text{put } \tan x=t \Rightarrow \sec^2 x\,dx=dt\}$$

$$= \int t^{1/3}\, dt = \frac{3}{4}\, t^{4/3} + C = \frac{3}{4}\, (\tan x)^{4/3} + C$$

Profile 2: $\int \frac{dx}{a \sin^2 x + b \sin x \cos x + c \cos^2 x + d}$

Multiply Nr & Dr by $\sec^2 x$ & put $\tan x = t$.

Profile 3 $\int \frac{dx}{a + b \sin x + c \cos x}$

Convert sines & cosines into their respective tangents of half the angles and then, put $\tan \frac{x}{2} = t$

Ex.4 Evaluate : $\int \frac{1}{1 + \sin x + \cos x}\, dx$

Sol. $I = \int \frac{1}{1 + \sin x + \cos x}\, dx = \int \frac{1}{1 + \frac{2\tan x/2}{1 + \tan^2 x/2} + \frac{1 - \tan^2 x/2}{1 + \tan^2 x/2}}\, dx$

$$= \int \frac{1 + \tan^2 x/2}{1 + \tan^2 x/2 + 2\tan x/2 + 1 - \tan^2 x/2}\, dx = \int \frac{\sec^2 x/2}{2 + 2\tan x/2}\, dx$$

Putting $\tan \frac{x}{2} = t$ and $\frac{1}{2} \sec^2 \frac{x}{2}\, dx = dt$, we get

$$I = \int \frac{1}{t+1}\, dt = \ell n \,|t + 1| + C = \ell n \left| \tan \frac{x}{2} + 1 \right| + C$$

Profile 4 $\int \frac{l.\cos x + m.\sin x + n}{a.\cos x + b.\sin x + c}$ **dx.**

Express Nr ≡ A(Dr) + B $\frac{d}{dx}$ (Dr) + C & proceed.

Ex.5 $\int \frac{7 + 3\sin x + 14\cos x}{3 + 4\sin x + 5\cos x}.dx$

$7+3\sin x+14\cos x = A(3+4\sin x+5\cos x) + B(4\cos x-5\sin x)+D$

compare coeff.

$$\left.\begin{array}{l} 3 = 4A - 5B \\ 14 = 5A + 4B \\ 7 = D + 3A \end{array}\right| \begin{array}{l} A = 2 \\ B = 1 \\ D = 1 \end{array}$$

$$I = \int \frac{2(3 + 4\sin x + 5\cos x) + (4\cos x - 5\sin x) + 1}{3 + 4\sin x + 5\cos x}.dx$$

$$= 2\int dx + \int \frac{4\cos x - 5\sin x}{3 + 4\sin x + 5\cos x}.dx + \int \frac{dx}{3 + 4\sin x + 5\cos x}$$

Profile:-5 When there is (sinx + cosx)/(sinx–cosx) and sin2x in N^r & D^r

Ex.6 $I_1 = \int \frac{(\sin x + \cos x)}{8 + \sin 2x}.dx$

$\sin x - \cos x = t \quad \Rightarrow 1 - \sin 2x = t^2$

$(\cos x + \sin x)\, dx = dt$

$I_1 = \int \frac{dt}{8 + 1 - t^2}$

Ex.7 $I_2 = \int \frac{\sin x - \cos x}{8 + \sin 2x}.dx$

$\sin x + \cos x = t \Rightarrow 1 + \sin 2x = t^2$

$(\cos x - \sin x)dx = dt$

$I_2 = -\int \frac{dt}{8 + t^2 - 1}$

Ex.8 $I_3 = \int \frac{\sin x.dx}{8 + \sin 2x}$

$= \frac{1}{2}\int \frac{(\sin x + \cos x) + (\sin x - \cos x)}{8 + \sin 2x}$

$= \frac{1}{2}(I_1 + I_2)$

Ex.9 $I_4 = \int \frac{\cos x.dx}{8 + \sin 2x}$

$= \frac{1}{2}\int \frac{(\sin x + \cos x) - (\sin x - \cos x)}{8 + \sin 2x}$

$= \frac{1}{2}(I_1 - I_2)$

Ex.10 $I_5 = \int \frac{a \sin x + b \cos x}{8 + \sin 2x}$

$= a\, I_3 + b\, I_4$

Ex.11 $I = \int \frac{8 + \sin 2x}{\sin x + \cos x}.dx$

$= \int \frac{8 + (\sin x + \cos x)^2 - 1}{(\sin x + \cos x)}.dx$

$= \int \left(\frac{7}{\sin x + \cos x} + (\sin x + \cos x)\right).dx$

Ex.12 $I = \int \frac{8 + \sin 2x}{\sin x - \cos x}.dx$

$$= \int \frac{8+1-(\sin x-\cos x)^2}{(\sin x-\cos x)}.dx$$

$$= \int \left(\frac{9}{\sin x-\cos x}-(\sin x-\cos x)\right).dx$$

Questions

1. $\int \frac{dx}{4-5\sin^2 x}$

Ans. $\frac{1}{4}\ln\left|\frac{\tan x+2}{\tan x-2}\right|+C$

Sol. $I = \int \frac{1}{4-5\sin^2 x}$

$$I = \int \frac{\sec^2 x}{4-\tan^2 x}dx$$

$$= \frac{1}{4}\cdot \ell n\left|\frac{\tan x+2}{\tan x-2}\right|+C$$

2. $\int \frac{dx}{(3\sin x-4\cos x)^2}$

Ans. $-\frac{\cot(x-\theta)}{25}+C$; where $\theta = \tan^{-1}\left(\frac{4}{3}\right)$

Sol. $\int \frac{dx}{(3\sin x-4\cos x)^2}$

$$= \int \frac{1}{5\sin^2(x-\theta)}dx \text{ where } \tan\theta = \frac{4}{3}$$

$$= -\frac{1}{25}\cot(x-\theta)+C \text{ where } \theta = \tan^{-1}\left(\frac{4}{3}\right)$$

3. $\int \frac{dx}{3+2\sin x+\cos x}$

Ans. $\tan^{-1}\left(1+\tan\frac{x}{2}\right)+C$

Sol. $I = \int \frac{1}{3+2\sin x+\cos x}dx$

$$= \int \frac{\frac{1}{2}\sec^2\frac{x}{2}}{\tan^2\frac{x}{2}+2\tan\frac{x}{2}+2}dx$$

put $\tan\frac{x}{2}+1=t$

$=\tan^{-1}\left(\tan\frac{x}{2}+1\right)+C$

4. $\int\frac{dx}{5+3\cos x}$

Ans. $\frac{1}{2}\tan^{-1}\left(\frac{1}{2}\tan\frac{x}{2}\right)+C$

Sol. $I=\int\frac{dx}{5+3\cos x}$

$I=\int\frac{\frac{1}{2}\sec^2\frac{x}{2}}{\tan^2\frac{x}{2}+4}dx$

put $\tan\frac{x}{2}=t$

$I=\frac{1}{2}\tan^{-1}\left(\frac{\tan\frac{x}{2}}{2}\right)+C$

5. $\int\frac{dx}{3+\cos^2 x}$

Ans. $\frac{1}{2\sqrt{3}}\tan^{-1}\left(\frac{\sqrt{3}\tan x}{2}\right)+C$

Sol. $I=\int\frac{dx}{3+\cos^2 x}$

$I=\int\frac{\sec^2 x}{3\tan^2 x+4}dx$

put $\sqrt{3}\tan x=t$

$=\frac{1}{2\sqrt{3}}\tan^{-1}\left(\frac{\sqrt{3}}{2}\tan x\right)+C$

6. $\int\frac{\cos x\,dx}{5-3\cos x}$

Ans. $-\frac{x}{3}+\frac{5}{6}\tan^{-1}\left(2\tan\frac{x}{2}\right)+C$

Sol. $I = \int \frac{\cos x}{5 - 3\cos x} dx$

$I = \frac{1}{3}\int \frac{5 - (5 - \cos x)}{5 - 3\cos x} dx$

$= \frac{1}{3}\int \frac{5}{5 - 3\cos x} dx - \frac{1}{3}\int dx$

$= -\frac{x}{3} + \frac{5}{3}\int \frac{1}{5 - 3\cos x} dx$

$= -\frac{x}{3} + \frac{5}{3}\int \frac{\frac{1}{2}\sec^2\frac{x}{2}}{4\tan^2\frac{x}{2} + 1} dx$

put $\tan\frac{x}{2} = t$

$= -\frac{x}{3} + \frac{5}{6}\tan^{-1}\left(2\tan\frac{x}{2}\right) + C$

7. $\int \frac{\sin 2x}{(a + b\cos x)^2} dx$

Ans. $\frac{-2}{b^2}\ln|a + b\cos x| - \frac{2a}{b^2(a + b\cos x)} + C$

Sol. $I = \int \frac{\sin 2x}{(a + b\cos x)} dx$

$I = \int \frac{2\sin x \cdot \cos x}{(a + b\cos x)^2} dx$

put $a + b\cos x = t$

$\sin x dx = -\frac{dt}{b}$

$= \int \frac{-\frac{2}{b}\cdot\frac{t - a}{b}}{t^2} dt$

$= \frac{2}{b^2}\int \frac{a - t}{t^2} dt = -\frac{2a}{b^2}\frac{1}{t} - \frac{2}{b^2}\ell n|t| + C$

$= -\frac{2a}{b^2(a + b\cos x)} - \frac{2}{b^2}\ell n|a + b\cos x| + C$

8. $\int \frac{6 + 3\sin x + 14\cos x}{3 + 4\sin x + 5\cos x} dx$

Ans. $2x + \ln|3 + 4\sin x + 5\cos x| + C$

Sol. $I = \int \frac{6 + 3\sin x + 14\cos x}{3 + 4\sin x + 5\cos x} dx$

Let $6 + 3\sin x + 14\cos x = A(3 + 4\sin x + 5\cos x) + B(4\cos x - 5\sin x) + C$

Compare coefficient

$A = 2 \quad B = 1 \quad C = 0$

$$I = \int \left[2 + \frac{(4\cos x - 5\sin x)}{3 + 4\sin x + 5\cos x} \right] dx$$

$= 2x + \ell n|3 + 4\sin x + 5\cos x| + C$

9. $$\int \frac{\sin 2x}{\sin^4 x + \cos^4 x} dx$$

Ans. $c - \tan^{-1}(\cot^2 x)$

Sol. $$\int \frac{\sin 2x}{\sin^4 x + \cos^4 x} dx = \int \frac{2\tan x \sec^2 x}{1 + \tan^4 x} dx$$ (multiply N^r & D^r by $\sec^4 x$ and put $\tan^2 = t$)

$$\int \frac{dt}{1 + t^2} dx = \tan^{-1}(\tan^2 x) + c = \tan^{-1}\left(\frac{1 - \cos 2x}{1 + \cos 2x}\right) + c = \tan^{-1} 1 - \tan^{-1}(\cos 2x) + c$$

$= c - \tan^{-1}(\cos 2x)$

Also $$\tan^{-1}(\tan^2 x) = \cot^{-1}\left(\frac{1}{\tan^2 x}\right) = \cot^{-1}(\cot^2 x) \left[\text{using } \tan^{-1} x = \cot^{-1}\left(\frac{1}{x}\right)\right]$$

$$\frac{\pi}{2} - \tan^{-1}(\cot^2 x) + c = c - \tan^{-1}(\cot^2 x)$$

10. $$\int \frac{dx}{\sin^4 x + \cos^4 x}$$

Ans. $$\frac{1}{\sqrt{2}} \tan^{-1}\left(\frac{\tan(2x)}{\sqrt{2}}\right) + C$$

Sol. $$I = \int \frac{1}{\sin^4 x + \cos^4 x} dx$$

$$= \int \frac{1}{1 - \frac{1}{2}\sin^2 2x} dx$$

$$= 2\int \frac{\sec^2 2x}{2\sec^2 2x - \tan^2 2x} dx$$

$$= \int \frac{2\sec^2 2x}{\tan^2 2x + 2} dx$$

put $\tan 2x = t$

$$= \frac{1}{\sqrt{2}} \tan^{-1}\left(\frac{\tan 2x}{\sqrt{2}}\right) + C$$

11. $\int \frac{\sin x}{e^x - \sin x - \cos x} dx$

Ans. $\frac{1}{2} ln\left|(e^x - \sin x - \cos x)\right| - \frac{1}{2} x + C$

Sol. $I = -\frac{1}{2}\int \frac{(e^x - \sin x - \cos x) - (e^x - \cos x + \sin x)}{e^x - (\sin x + \cos x)} dx$

$$= \frac{1}{2}\int \frac{\frac{d}{dx}(e^x - \sin x - \cos x) + \cos x + \sin x - e^x}{e^x - (\sin x + \cos x)} dx = \frac{1}{2} ln\left|(e^x - \sin x - \cos x)\right| - \frac{1}{2} x \text{ Ans.}$$

12. $\int \frac{\cos^3 x + \cos^5 x}{\sin^2 x + \sin^4 x} dx =$

Ans. $\sin x - 2\,(\sin x)^{-1} - 6\tan^{-1}(\sin x) + c$

Sol. $I = \int \frac{\cos^3 x + \cos^5 x}{\sin^2 x + \sin^4 x} dx$

put $\sin x = t$

$$= \int \frac{1 - t^2 + \left(1 - t^2\right)^2}{t^2 + t^4} dt$$

$$= \int \frac{2 - 3t^2 + t^4}{t^2 + t^4} dt$$

$$= \int 1 + \frac{2 - 4t^2}{t^2\left(t^2 + 1\right)} dt$$

$$= t + 2\int \left(\frac{1}{t^2} - \frac{3}{t^2 + 1}\right) dt$$

$$= t - \frac{2}{t} - 6\tan^{-1} t + C$$

$= (\sin x) - 2(\sin x)^{-1} - 6\tan^{-1}(\sin x) + C$

13. $\int \frac{\sin^{-1}\sqrt{x} - \cos^{-1}\sqrt{x}}{\sin^{-1}\sqrt{x} + \cos^{-1}\sqrt{x}} dx$. **[JEE – 86]**

Ans. $\frac{2}{\pi}\left(\sqrt{x(1-x)} - \frac{(1-2x)}{2}\left(2\sin^{-1}\sqrt{x} - \frac{\pi}{2}\right)\right) + C$

Sol. $I = \int \frac{\sin^{-1}\sqrt{x} - \cos^{-1}\sqrt{x}}{\sin^{-1}\sqrt{x} + \cos^{-1}\sqrt{x}} dx$

$$I = \frac{2}{\pi}\int\left(2\sin^{-1}\sqrt{x} - \frac{\pi}{2}\right)dx$$

$$= -x + \frac{4}{\pi}\int \sin^{-1}\sqrt{x}\,dx$$

$$= -x + \frac{4}{\pi}\sin^{-1}\sqrt{x}\cdot x - \frac{4}{\pi}\int \frac{x}{2\sqrt{x}\sqrt{1-x}}dx$$

$$= -x + \frac{4x}{\pi}\sin^{-1}\sqrt{x} + \frac{2}{\pi}\int \frac{\sqrt{x}}{\sqrt{1-x}}dx$$

$$= -x + \frac{4x}{\pi}\sin^{-1}\sqrt{x} + \frac{2}{\pi}\int \frac{\sqrt{x}}{\sqrt{1-x}}dx$$

put $x = \sin^2\theta$

$dx = 2\sin\theta\cdot\cos\theta\,d\theta$

$$= -x + \frac{4x}{\pi}\sin^{-1}\sqrt{x} + \frac{2}{\pi}\int \frac{\sin\theta}{\cos\theta}\cdot 2\sin\theta\cdot\cos\theta d\theta$$

$$= -x + \frac{4x}{\pi}\sin^{-1}\sqrt{x} + \frac{4}{\pi}\int (1-\cos 2\theta)\,d\theta$$

$$= -x + \frac{4x}{\pi}\sin^{-1}\sqrt{x} + \frac{2}{\pi}\sin^{-1}\sqrt{x} - \frac{2}{\pi}\sqrt{x}\sqrt{1-x} + C$$

$$= \frac{2}{\pi}\left[-\frac{\pi}{2} + (1+2x)\sin^{-1}\sqrt{x} - \sqrt{x}\sqrt{1-x}\right]$$

7. INTEGRATION BY PARTIAL FRACTIONS

Form $\frac{P(x)}{Q(x)}$, where P(x) and Q(x) are polynomials in x and $Q(x) \neq 0$.

Case 1:

If the degree of P(x) is greater than/equal to the degree of Q(x), it can be reduced to the proper rational functions by long division process.

$$\frac{P(x)}{Q(x)} = T(x) + \frac{P_1(x)}{Q(x)},$$

Case 2 :

Degree of P(x) is less than the degree of Q(x)

Profile 1

When denominator is expressible as the product of non-repeating linear factors.

$Q(x) = (x - a_1)(x - a_2)\ldots.(x - a_n)$. Then, we assume that

$$\frac{P(x)}{Q(x)} \equiv \frac{A_1}{x-a_1} + \frac{A_2}{x-a_2} + \ldots.. + \frac{A_n}{x-a_n}$$

where $A_1, A_2, \ldots.. A_n$ are constants and can be determined by equating the numerator on R.H.S. to the numerator on L.H.S. and then substituting $x = a_1, a_2, \ldots\ldots, a_n$.

Ex. 1 $\int \frac{dx}{(x-1)(x-2)(x-3)}$.

Sol. Put $\frac{1}{(x-1)(x-2)(x-3)} = \frac{A}{(x-1)} + \frac{B}{(x-2)} + \frac{C}{(x-3)}$

$\Rightarrow 1 = A(x-2)(x-3) + B(x-1)(x-3) + C(x-1)(x-2)$

Put x = 1, we get, $A = \frac{1}{2}$ x = 2, we get, B = – 1 x = 3, we get, $C = \frac{1}{2}$

So integral = $\frac{1}{2}\int \frac{dx}{x-1} - \int \frac{dx}{x-2} + \frac{1}{2}\int \frac{dx}{x-3}$ $= \ell n\left(\frac{\sqrt{x^2-4x+3}}{|x-2|}\right) + c$

Profile 2

When the denominator g(x) is expressible as the product of the linear factors such that some of them are repeating.

Example $\frac{1}{g(x)} = \frac{1}{(x-a)^k(x-a_1)(x-a_2).....(x-a_r)}$ this can be expressed as

$$\frac{A_1}{x-a} + \frac{A_2}{(x-a)^2} + \frac{A_3}{(x-a)^3} + ...+ \frac{A_k}{(x-a)^k} + \frac{B_1}{(x-a_1)} + \frac{B_2}{(x-a_2)} + ... + \frac{B_r}{(x-a_r)}$$

Now to determine constants we equate numerators on both sides. Some of the constants are determined by substitution as in case I and remaining are obtained by equating the coefficient of same power of x.

Profile 3

When some of the factors of denominator g(x) are quadratic

Eg:- $\frac{1}{(x-1)(x-2)(x^2+x+2)} = \frac{A}{x-1} + \frac{B}{x-2} + \frac{Cx+D}{x^2+x+2}$

Eg:- $\frac{1}{(x-1)(x-2)^2(x^2+x+2)^2} = \frac{A}{x-1} + \frac{B}{x-2} + \frac{C}{(x-2)^2} + \frac{Dx+E}{x^2+x+2} + \frac{Fx+G}{\left(x^2+x+2\right)^2}$

Questions

1. $\int \frac{x\,dx}{\left(x^2+2\right)\left(x^2+1\right)}$

Ans. $\frac{1}{2}ln\left|\frac{x^2+1}{x^2+2}\right| + C$

Sol. $I = \frac{1}{2}\int \frac{2x}{\left(x^2+2\right)\left(x^2+1\right)}dx$

put $x^2 + 1 = t$

$2xdx = dt$

$= \frac{1}{2}\int \frac{dt}{t(t+1)}$

$$= \frac{1}{2}\ell n\left|\frac{t}{t+1}\right| + C$$

$$= \frac{1}{2}\ell n\left|\frac{x^2+1}{x^2+2}\right| + C$$

2. $\int \frac{2x^2-5}{x^4-5x^2+6}$

Ans. $\frac{1}{2\sqrt{2}} ln\left|\frac{x-\sqrt{2}}{x+\sqrt{2}}\right| + \frac{1}{2\sqrt{3}} ln\left|\frac{x-\sqrt{3}}{x+\sqrt{3}}\right| + C$

Sol. $I = \int \frac{2x^2-5}{(x^2-2)(x^2-3)}dx$

$$= \int \frac{x^2-3+x^2-2}{(x^2-2)(x^2-3)}dx = \int\left(\frac{1}{x^2-2}+\frac{1}{x^2-3}\right)dx$$

$$= \frac{1}{2\sqrt{2}}\ell n\left|\frac{x-\sqrt{2}}{x+\sqrt{2}}\right| + \frac{1}{2\sqrt{3}}\ell n\left|\frac{x-\sqrt{3}}{x+\sqrt{3}}\right| + C$$

3. $\int \frac{4x-4}{x(x^2+4)}dx$

Ans. $2\tan^{-1}\left(\frac{x}{2}\right) - ln|x| + \frac{1}{2} ln|x^2+4| + C$

Sol $I = \int \frac{4x-4}{x(x^2+4)}dx$

$$= 4\int \frac{1}{x^2+4}dx - \int\left(\frac{1}{x}-\frac{x}{x^2+4}\right)dx$$

$$= 2\tan^{-1}\left(\frac{x}{2}\right) - \ell n|x| + \frac{1}{2}\ell n|x^2+4| + C$$

4. $\int \frac{2x^2+41x-91}{(x-1)(x+3)(x-4)}dx$

Ans. $\left[4\ln|x-1| - 7\ln|x+3| + 5\ln|x-4| + C\right]$

Sol. $I = \int \frac{2x^2+41x-91}{(x-1)(x+3)(x-4)}dx$

$$\frac{2x^2+41x-91}{(x-1)(x+3)(x-4)} = \frac{A}{x-1}+\frac{B}{x+3}+\frac{C}{x-4}$$

A = 4 B = –7 C = 5

$$I = 4\int \frac{1}{x-1}dx - 7\int \frac{1}{x+3}dx + 5\int \frac{1}{x-4}dx$$

$= 4\ \ell n|x-1| - 7\ \ell n|x+3| + 5\ \ell n|x-4| + C$

5. $\int \frac{x^3-2x^2+4}{x^2(x-2)^2}dx$

Ans. $\ln|x| - \frac{1}{x} - \frac{1}{(x-2)} + C$

Sol. $I = \int \frac{x^3-2x^2+4}{x^2(x-2)^2}dx$

$$= \frac{x^3-2x^2+4}{x^2(x-2)^2} = \frac{A}{x} + \frac{B}{x^2} + \frac{C}{x-2} + \frac{D}{(x-2)^2}$$

$A = 1 \qquad B = 1 \qquad C = 0 \qquad D = 1$

$$I = \int\left(\frac{1}{x} + \frac{1}{x^2} + \frac{1}{(x-2)^2}\right)dx$$

$$= \ell n|x| - \frac{1}{x} - \frac{1}{x-2} + C$$

6. $\int \frac{x}{(x-1)(x^2+4)}dx$

Ans. $\frac{1}{5}\ln|x-1| - \frac{1}{10}\ln|x^2+4| + \frac{2}{5}\tan^{-1}\left(\frac{x}{2}\right) + C$

Sol. $I = \int \frac{x}{(x-1)(x^2+4)}dx \qquad \frac{x}{(x-1)(x^2+4)} = \frac{A}{x-1} + \frac{Bx+C}{x^2+4}$

$$A = \frac{1}{5} \qquad B = -\frac{1}{5} \qquad C = \frac{4}{5}$$

$$I = \int\left(\frac{1}{5}\frac{1}{x-1} - \frac{1}{5}\frac{x}{x^2+4} + \frac{4}{5}\frac{1}{x^2+4}\right)dx$$

$$= \frac{1}{5}\ell n|x-1| - \frac{1}{10}\ell n|x^2+4| + \frac{2}{5}\tan^{-1}\left(\frac{x}{2}\right) + C$$

7. $\int \frac{x^3dx}{x^4+3x^2+2}$ $(x^2=t)$

Ans. $\ln|x^2+2| - \frac{1}{2}\ln|1+x^2| + C$

Sol. $I = \int \frac{x^3}{x^4+3x^2+2}dx$

put $x^2 = t$

$= \frac{1}{2}\int \frac{tdt}{(t+1)(t+2)} = \frac{1}{2}\int\left(\frac{2}{t+2} - \frac{1}{t+1}\right)dt$

$= \frac{1}{2}\left(2\ell n\left|x^2+2\right| - \ell n\left|x^2+1\right|\right) + C$

$= \ell n|x^2+2| - \frac{1}{2}\ \ell n|x^2+1| + C$

8. $\int \frac{(x^2-x)\,dx}{(x^2+2)(x^2+1)}dx$

Ans. $2\sqrt{2}\tan^{-1}\left(\frac{x}{\sqrt{2}}\right) - 2\tan^{-1}(x) - \frac{1}{2}\ell n\left|\frac{x^2+1}{x^2+2}\right| + C$

Sol. $2\int \frac{x^2dx}{(x^2+2)(x^2+1)} - \int \frac{x\,dx}{(x^2+2)(x^2+1)}$

Alt.

$I = \int \frac{x^2}{(x^2+2)(x^2+1)}dx - \int \frac{x}{(x^2+2)(x^2+1)}dx$

put $x^2 = t$

$\int\left[\frac{2}{x^2+2} - \frac{1}{x^2+1}\right]dx - \frac{1}{2}\int \frac{1}{(t+2)(t+1)}dt$

$= 2\frac{1}{\sqrt{2}}\tan^{-1}\left(\frac{x}{\sqrt{2}}\right) - \tan^{-1}(x) - \frac{1}{2}\ell n\left|\frac{t+1}{t+2}\right| + C$

$= \sqrt{2}\tan^{-1}\left(\frac{x}{\sqrt{2}}\right) - \tan^{-1}x - \frac{1}{2}\ell n\left|\frac{x^2+1}{x^2+2}\right| + C$

9. $\int \frac{dx}{x^3+1}$

Ans. $\frac{1}{3}ln\left|x^3+1\right| - \frac{1}{2}ln\left|x^2-x+1\right| + \frac{1}{\sqrt{3}}\tan^{-1}\left(\frac{2x-1}{\sqrt{3}}\right) + C$

Sol. $I = \int \frac{x^2-(x^2-1)}{x^3+1}dx$

$= \int \frac{x^2}{x^3+1}dx - \int \frac{(x-1)(x+1)}{(x+1)(x^2-x+1)}dx$

put $x^3 + 1 = t$

$$= \frac{1}{3}\ell n\left|x^3+1\right| - \int \frac{x-1}{x^2-x+1}dx$$

$$= \frac{1}{3}\ell n\left|x^3+1\right| - \frac{1}{2}\int \frac{2x-1}{x^2-x+1} + \frac{1}{2}\int \frac{1}{x^2-x+1}dx$$

$$= \frac{1}{3}\ell n\left|x^3+1\right| - \frac{1}{2}\ell n\left|x^2-x+1\right| + \frac{1}{\sqrt{3}}\tan^{-1}\left(\frac{2x-1}{\sqrt{3}}\right) + C$$

10. $\int \frac{dx}{x^3-1}$

Ans. $\frac{1}{3}ln\left|x^3-1\right| - \frac{1}{2}ln\left|x^2+x+1\right| - \frac{1}{\sqrt{3}}\tan^{-1}\left(\frac{2x+1}{\sqrt{3}}\right) + C$

Sol $\int \frac{x^2-(x^2-1)dx}{x^3-1}$ [Hint: add and sub. x in N^r]

11. $\int \frac{dx}{\sin x(3+2\cos x)}$

Ans. $\frac{1}{10}ln\left|1-\cos x\right| - \frac{1}{2}ln\left|1+\cos x\right| + \frac{2}{5}ln\left|2\cos x+3\right| + C$

Sol. $I = \int \frac{\sin x}{\sin^2 x(3+2\cos x)}dx$

put $\cos x = t$

$$= \int \frac{1}{(t^2-1)(2t+3)}dt = \int \frac{1}{(t-1)(t+1)(2t+3)}dt$$

$$= \frac{1}{10}\ell n\left|-\cos x+1\right| - \frac{1}{2}\ell n\left|1+\cos x\right| + \frac{2}{5}\ell n\left|2\cos x+3\right| + C$$

12. $\int \frac{dx}{\sin 2x - 2\sin x}$

Ans. $\frac{-1}{8}ln\left|1+\cos x\right| + \frac{1}{8}ln\left|1-\cos x\right| + \frac{1}{4(\cos x-1)} + C$

Sol. $\int \frac{\sin x}{(1-\cos^2 x)(2\cos x-2)}dx$

put $\cos x = 1$

$1 = A(t-1)^2 + B(t^2-1) + C(t+1)$

$t = 1 \qquad C = 1/2$

$t = -1 \qquad A = 1/4$

Coefficient $t^2 \Rightarrow A + B = 0$

$B = -1/4$

$$= \frac{1}{2}\int \frac{1}{(t-1)(t-1)(t+1)}\,dt$$

$$= \frac{1}{2}\left[\frac{1}{4}\ell n|1+\cos x| - \frac{1}{4}\ell n|\cos x - 1| - \frac{1}{2}\frac{1}{\cos x - 1}\right] + C$$

$$= \frac{1}{8}\ell n|1+\cos x| - \frac{1}{8}\ell n|\cos x - 1| - \frac{1}{4}\frac{1}{\cos x - 1} + C$$

13. $\int \frac{\sin x}{\sin 4x}\,dx$

Ans. $-\frac{1}{4\sqrt{2}}\ell n\left|\frac{\sqrt{2}\sin x - 1}{\sqrt{2}\sin x + 1}\right| + \frac{1}{8}\ell n\left|\frac{1-\sin x}{1+\sin x}\right| + C$

Sol. $I = \int \frac{\sin x}{2\sin 2x \cdot \cos 2x}\,dx$

$$= \int \frac{\cos x}{4\cos^2 x \cdot (1 - 2\sin^2 x)}\,dx$$

put $\sin x = t$

$$= \frac{1}{4}\int \frac{dt}{(2t^2-1)(t^2-1)}$$

$$= \frac{1}{4}\int \left(\frac{1}{t^2-1} - \frac{2}{2t^2-1}\right)dt$$

$$= \frac{1}{4}\left\{\frac{1}{2}\ell n\left|\frac{\sin x - 1}{\sin x + 1}\right| - \frac{1}{\sqrt{2}}\ell n\left|\frac{\sqrt{2}\sin x - 1}{\sqrt{2}\sin x + 1}\right|\right\} + C$$

$$= \frac{1}{8}\ell n\left|\frac{\sin x - 1}{\sin x + 1}\right| - \frac{1}{4\sqrt{2}}\ell n\left|\frac{\sqrt{2}\sin x - 1}{\sqrt{2}\sin x + 1}\right| + C$$

14. $\int \frac{1 + x\cos x}{x(1 - x^2 e^{2\sin x})}\,dx$

Ans. $\ell n|t| - \frac{1}{2}\ell n|1-t| - \frac{1}{2}\ell n|1+t| + C$, where $t = xe^{\sin x}$

[Hint $I = \int \frac{(1 + x\cos x)e^{\sin x}}{e^{\sin x} x[1 - (x e^{\sin x})^2}\,dx$;

put $x e^{\sin x} = t$; $e^{\sin x} + x e^{\sin x}\cos x$; $e^{\sin x}(1 + x\cos x)\,dx = dt$; $I \int \frac{dt}{t(1-t^2)}$]

8.1 MISCELLANEOUS - TYPE 1 (KUTUR PUTUR)

(i) $\int \frac{dx}{x(x^n+1)}$, $n \in N$, take x^n common & put $1 + x^{-n} = t$.

(ii) $\int \frac{dx}{x^2\left(x^n+1\right)^{(n-1)/n}}$, $n \in N$, take x^n common & put $1 + x^{-n} = t^n$

(iii) $\int \frac{dx}{x^n(1+x^n)^{1/n}}$, take x^n common and put $1 + x^{-n} = t^n$.

Ex.1 Evaluate : $\int \frac{dx}{x^n(1+x^n)^{1/n}}$

Ans $\frac{-1}{n-1}\left(1+\frac{1}{x^n}\right)^{\frac{n-1}{n}} + c$

Sol. Let $I = \int \frac{dx}{x^n(1+x^n)^{1/n}} = \int \frac{dx}{x^{n+1}\left(1+\frac{1}{x^n}\right)^{1/n}}$

Put $1 + \frac{1}{x^n} = t^n$, then $\frac{1}{x^{n+1}}dx = -t^{n-1}dt$

$I = -\int \frac{t^{n-1}dt}{t} = -\int t^{n-2}dt = -\frac{t^{n-1}}{n-1} + c = \frac{-1}{n-1}\left(1+\frac{1}{x^n}\right)^{\frac{n-1}{n}} + c$

Questions

1. $\int \frac{dx}{x^2\left(x^n+1\right)^{(n-1/n)}}$

Ans. $-\left(1+\frac{1}{x^n}\right)^{1/n} + C$

Sol. $I = \int \frac{1}{x^2\left(x^n+1\right)^{\frac{n-1}{b}}}dx$

$= \int \frac{1}{x^{n+1}\left(1+x^{-n}\right)^{\frac{n-1}{n}}}$

put $1 + x^{-n} = t$

$-nx^{-n-1}dx = dt$

$= -\frac{1}{n}\int \frac{1}{t^{\frac{n-1}{n}}}dt$

$= -\left(1+\frac{1}{x^n}\right)^{\frac{1}{n}} + C$

2. $\int \frac{dx}{x^2(x^4+1)^{3/4}}$. [IIT – 84]

Ans. $-\left(1+\frac{1}{x^4}\right)^{1/4}+C$

Sol. put n = 4 in question.

3. $\int \frac{dx}{x(x^n+1)}$

Ans. $-\frac{1}{n}\ell n\left|1+\frac{1}{x^n}\right|+C$

Sol. $I=\int \frac{1}{x\left(x^n+1\right)}dx$ put $x^{-n}+1=t$

$=\int \frac{1}{x^{n+1}\left(1+x^{-n}\right)}dx=-\frac{1}{n}\ell n\left|1+x^{-n}\right|+C$

4. $\int \frac{dx}{x(x^2+1)}$

Ans. $-\frac{1}{2}\ell n\left|1+\frac{1}{x^2}\right|+C$

Sol. Put n = 2 in question 3.

5. $\int \frac{x^7}{(1-x^2)^5}dx$

Ans. $\frac{1}{8\left(\frac{1}{x^2}-1\right)^4}+C$

Sol. $I=\int \frac{x^7}{\left(1-x^2\right)^5}dx$

$=\int \frac{1}{x^3\left(x^{-2}-1\right)^5}dx$

put $x^{-2}-1=t$

$=-\frac{1}{2}\int t^{-5}dt$

$=\frac{1}{8}\frac{1}{\left(\frac{1}{x^2}-1\right)^4}+C$

6. $\int \frac{x\,dx}{(1-x^4)^{3/2}}$

Ans. $\frac{1}{2\sqrt{\frac{1}{x^4}-1}}+C$

Sol. $I=\int \frac{x}{\left(1-x^4\right)^{\frac{3}{2}}}dx$

$=\int \frac{1}{x^5\left(x^{-4}-1\right)^{\frac{3}{2}}}dx$ Put $x^{-4}+1=t$

$=\frac{1}{2}\frac{1}{\left(x^{-4}-1\right)^{\frac{1}{2}}}+C=\frac{1}{2\left(\frac{1}{x^4}-1\right)^{\frac{3}{2}}}+C$

7. $\int \frac{dx}{x^4(x^3+1)^2}$

Ans. $-\frac{1}{3}\left(t-2\ln|t|-\frac{1}{t}\right)+C$; Where $t=1+\frac{1}{x^3}$

Sol. $I=\int \frac{1}{x^4\left(x^3+1\right)^2}dx$

$=\int \frac{\left(x^{-3}\right)^2}{x^4\left(1+x^{-3}\right)^2}dx$

Put $1+x^{-3}=t$

$\frac{1}{x^4}dx=\frac{dt}{-3}$

$I=\int \frac{(t-1)^2}{-3t^2}dt$

$=-\frac{1}{3}\left[t-2\ell n|t|-\frac{1}{t}\right]$ when $t=1+x^{-3}$

8. $\int \frac{x^3-1}{4x^3-x}dx$

Ans. $\frac{1}{4}\left[x+4\ln|x|-\frac{7}{4}\ln|2x-1|-\frac{9}{4}\ln|2x+1|\right]+C$

Sol. $I = \int \frac{x^3 - 1}{4x^3 - x} dx$

$= \frac{1}{4}\int \frac{4x^3 - x + x - 4}{4x^3 - x} dx$

$= \frac{x}{4} + \frac{1}{4}\int \frac{x - 4}{4x^3 - x} dx$

$= \frac{x}{4} + \frac{1}{4}\int \frac{1}{(2x)^2 - 1} dx - \int \frac{1}{x^3\left(4 - x^{-2}\right)} dx$

put $4 - x^{-2} = t$

$= \frac{x}{4} + \frac{1}{4}\frac{1}{2}\ell n\left|\frac{2x-1}{2x+1}\right| + \frac{1}{2}\ell n\left|4 - x^{-2}\right| + C$

9. $\int \frac{x^2 - x^{-2}}{x\sqrt{x^2 + x^{-2} + 1}} dx$

Ans. $\frac{\sqrt{x^4 + x^2 + 1}}{x}$

Sol. $I = \int \frac{x^2 - x^{-2}}{x\sqrt{x^2 + x^{-2} + 1}} dx$

$I = \int \frac{x - x^{-3}}{\sqrt{x^2 + x^{-2} + 1}} dx$

put $x^2 + x^{-2} + 1 = t$

$\left(x - x^{-3}\right) dx = \frac{dt}{2}$

$I = \int \frac{dt}{2\sqrt{t}} = \frac{\sqrt{x^4 + x^2 + 1}}{x} + C$

10. $\int \frac{(x\cos\alpha + 1) dx}{(x^2 + 2x\cos\alpha + 1)^{3/2}}$

Ans. $\frac{x}{\sqrt{x^2 + 2x\cos\alpha + 1}}$

Sol. $I = \int \frac{x\cos\alpha + 1}{\left(x^2 + 2x\cos\alpha + 1\right)^{\frac{3}{2}}} dx$

$= \int \frac{x^{-2}\cos\alpha + x^{-3}}{\left(1 + 2\cos\alpha x^{-1} + x^{-2}\right)^{\frac{3}{2}}} dx$

put $2\cos\alpha\, x^{-1} + x^{-2} = t^2$

$(\cos\alpha\, x^{-2} + x^{-3})dx = t\,dt$

$$\Rightarrow I = -\frac{x}{(2\cos\alpha x + 1)^{\frac{1}{2}}} + C$$

11. $\int (x^6 + x^4 + x^2)\cdot(2x^4 + 3x^2 + 6)^{1/2}\, dx$

Ans. $\dfrac{(2x^6 + 3x^4 + 6x^2)^{3/2}}{18}$

Sol. $\int (x^5 + x^3 + x)\cdot(2x^6 + 3x^4 + 6x^2)^{1/2}\, dx$;
put $2x^6 + 3x^4 + 6x^2 = t^2$; $12[x^5 + x^3 + x]dx = 2t\, dt$

$(x^5 + x^3 + x)dx = \dfrac{t\ dt}{6}$; $\dfrac{1}{6}\cdot\dfrac{1}{3}\cdot t^3 = \dfrac{(2x^6 + 3x^4 + 6x^2)^{3/2}}{18}$ Ans.

12. $\int \dfrac{dx}{x^2\left(x+\sqrt{1+x^2}\right)}$

Ans. $ln\left(\dfrac{x+\sqrt{1+x^2}}{x}\right) - \dfrac{\sqrt{x^2+1}}{x} + C$

Sol. $\int \dfrac{x^{-3}\, dx}{\left(1+\sqrt{x^{-2}+1}\right)}$; put $1 + x^{-2} = t^2 \Rightarrow -2\, x^{-3}\, dx = 2t\, dt$

$$\int \frac{t\,dt}{1+t} = \int\left(1 - \frac{1}{1+t}\right) = t - ln(1+t) = -\sqrt{1+\frac{1}{x^2}} - ln\left(1+\sqrt{1+\frac{1}{x^2}}\right)$$

$= ln\left(\dfrac{x+\sqrt{1+x^2}}{x}\right) - \dfrac{\sqrt{x^2+1}}{x} + C$ Ans.

13. $\int \dfrac{x-1\, dx}{x^2\sqrt{2x^2-2x+1}}$

Ans. $\dfrac{\sqrt{2x^2-2x+1}}{x} + C$

14. $\int \dfrac{(ax^2-b)\, dx}{x\sqrt{c^2x^2-(ax^2+b)^2}}$

Ans. $\sin^{-1}\left(\dfrac{ax^2+b}{cx}\right) + C'$

Sol. $I = \int \frac{a - bx^{-2}}{\sqrt{c^2 - \left(ax + bx^{-1}\right)^2}}$

$ax + bx^{-1} = t$

$\Rightarrow (a - bx^{-2})dx = dt$

$$I = \sin^{-1}\left(\frac{ax + bx^{-1}}{c}\right) + C_1$$

$$= \sin^{-1}\left(\frac{ax^2 + b}{cx}\right) + C_1$$

8.2 MISCELLANEOUS - TYPE 2

Q.1 $I_1 = \int \frac{x^2 + 1}{x^4 + kx^2 + 1}.dx$

$\div$ by x^2

$$= \int \frac{1 + \frac{1}{x^2}}{x^2 + \frac{1}{x^2} + k}$$

$$x - \frac{1}{x} = t \quad \Rightarrow \quad x^2 + \frac{1}{x^2} = t^2 + 2$$

$$\left(1 - \frac{1}{x^2}\right)dx = dt$$

$$I_1 = \int \frac{dt}{t^2 + 2 + k}$$

Q.2 $I_2 = \int \frac{\left(x^2 - 1\right)}{x^4 + kx^2 + 1}$

$$= \int \frac{1 - \frac{1}{x^2}}{x^2 + \frac{1}{x^2} + k}.dx$$

$$x + \frac{1}{x} = t$$

$$I_2 = \int \frac{dt}{t^2 - 2 + k}$$

Q.3 $I_3 = \int \frac{x^2.dx}{x^4 + kx^2 + 1}$

$$= \frac{1}{2}\int \frac{\left(x^2+1\right)+\left(x^2-1\right)}{x^4+kx^2+1}.dx$$

$$= \frac{1}{2}\left(I_1+I_2\right)$$

Q.4 $I_4 = \int \frac{dx}{x^4+kx^2+1}$

$$= \frac{1}{2}\int \frac{\left(x^2+1\right)-\left(x^2-1\right)}{x^4+kx^2+1}.dx$$

$$= \frac{1}{2}\left(I_1-I_2\right)$$

Q.5 $I = \int \frac{ax^2+bx+c}{x^4+kx^2+1}$

$$= aI_3 + cI_4 + \int \frac{bx.dx}{x^4+kx^2+1}$$

$x^2 = t$

Q.6 $\int \frac{x^2+4}{x^4+kx^2+16}.dx$

same as I_1

$x - \frac{4}{x} = t$

Q.7 $\int \frac{ax^2+bx+c}{x^4+kx^2+16}.dx$

Eg:-

$$\int \frac{x^2+3x+5}{x^4+kx^2+16}.dx$$

$$= \frac{1}{2}\int \frac{\left(x^2+4\right)+\left(x^2-4\right)}{x^4+kx^2+16}.dx + 3\int \frac{x.dx}{x^4+kx^2+16} + \frac{5}{8}\int \frac{\left(x^2+4\right)-\left(x^2-4\right)}{x^4+kx^2+16}.dx$$

8.3 MISCELLANEOUS - TYPE 3(INTEGRATION OF IRRATIONAL ALGEBRAIC FUNCTIONS)

P-1 $\int \sqrt{\frac{x-\alpha}{\beta-x}}\ dx$ or $\int \sqrt{(x-\alpha)(\beta-x)}\,dx$; $\beta > x > \alpha > 0$

put $x = \alpha\cos^2\theta + \beta\sin^2\theta$

$dx = (\beta-\alpha)\sin 2\theta\ .\ d\theta$

$x-\alpha = \alpha\cos^2\theta + \beta\sin^2\theta - \alpha = (\beta-\alpha)\sin^2\theta$

$\beta - x = \beta - \alpha\cos^2\theta - \beta\sin^2\theta = (\beta-\alpha)\cos^2\theta$

$\int \sqrt{\frac{x-\alpha}{x-\beta}}\,dx$ or $\int \sqrt{(x-\alpha)(x-\beta)}\,dx$; put $x = \alpha\sec^2\theta - \beta\tan^2\theta$

$\int \frac{dx}{\sqrt{(x-\alpha)(x-\beta)}}$; put $x-\alpha = t^2$ or $x-\beta = t^2$.

P-2 $\int \frac{dx}{(ax+b)\sqrt{px+q}}$

Put $px + q = t^2$.

Q.1 Evaluate : $\int \frac{1}{(x-3)\sqrt{x+1}}\,dx$

Ans $\frac{1}{2}\,\ell n\left|\frac{\sqrt{x+1}-2}{\sqrt{x+1}+2}\right| + C.$

Sol. Let $I = \int \frac{1}{(x-3)\sqrt{x+1}}\,dx$ {Put $x + 1 = t^2 \Rightarrow dx = 2t\,dt$}

$\therefore I = \int \frac{1}{(t^2-1-3)}\frac{2t}{\sqrt{t^2}}\,dt$

$\Rightarrow I = 2\int \frac{dt}{t^2-2^2} = 2\cdot\frac{1}{2(2)}\,\ell n\left|\frac{t-2}{t+2}\right| + C \quad \Rightarrow \quad I = \frac{1}{2}\,\ell n\left|\frac{\sqrt{x+1}-2}{\sqrt{x+1}+2}\right| + C.$

P-3 $\int \frac{dx}{(ax^2+bx+c)\sqrt{px+q}}$.

Put $px + q = t^2$.

Q.2 $\int \frac{dx}{(x^2+5x+2)\sqrt{x-2}}$

P-4 $\int \frac{dx}{(ax+b)\sqrt{px^2+qx+r}}$,

put $ax + b = \frac{1}{t}$;

Q.3 $\int \frac{dx}{(x+1)\sqrt{x^2+x+1}}$

Ans $-\ell n\left|t-\frac{1}{2}+\sqrt{\left(t-\frac{1}{2}\right)^2+\frac{3}{4}}\right| + C$ where $t = \frac{1}{x+1}$

Sol. Let $I = \int \frac{dx}{(x+1)\sqrt{x^2+x+1}}$ $\{$put $x + 1 = \frac{1}{t} \Rightarrow dx = -\frac{1}{t^2}\, dt\}$

$$\Rightarrow I = \int \frac{-dt}{t^2\left(\frac{1}{t}\right)\sqrt{\left(\frac{1}{t}-1\right)^2+\frac{1}{t}}} = \int \frac{-dt}{t\sqrt{\frac{1}{t^2}-\frac{1}{t}+1}} = \int \frac{-dt}{\sqrt{t^2-t+1}}$$

$$= \int \frac{-dt}{\sqrt{\left(t-\frac{1}{2}\right)^2+\frac{3}{4}}} = -\ell n\left|t-\frac{1}{2}+\sqrt{\left(t-\frac{1}{2}\right)^2+\frac{3}{4}}\right| + C,$$

where $t = \frac{1}{x+1}$

P-5 $\int \frac{dx}{(ax^2 + b)\sqrt{px^2+q}}$,

put $\mathbf{x = \frac{1}{t}}$

Q.4 $\int \frac{dx}{(1+x^2)\sqrt{1-x^2}}$

Ans $-\frac{1}{\sqrt{2}} \tan^{-1}\left(\frac{\sqrt{1-x^2}}{\sqrt{2}x}\right) + C$

Sol. Put $x = \frac{1}{t} \Rightarrow dx = -\frac{1}{t^2}\, dt \Rightarrow I = \int \frac{dt}{(t^2+1)\sqrt{t^2-1}}$ $\{$put $t^2 - 1 = y^2 \Rightarrow tdt = ydy\}$

$$\Rightarrow I = -\int \frac{y\, dy}{(y^2+2)\, y} = -\frac{1}{\sqrt{2}} \tan^{-1}\left(\frac{y}{\sqrt{2}}\right) + C = -\frac{1}{\sqrt{2}} \tan^{-1}\left(\frac{\sqrt{1-x^2}}{\sqrt{2}x}\right) + C$$

8.4 MISCELLANEOUS -TYPE 4 (FORCING INTEGRAND BY PARTS)

1. $\int \frac{dx}{(x^4-1)^2} = \int \underbrace{\frac{4x^3}{(x^4-1)^2}}_{II} \cdot \underbrace{\frac{1}{4x^3}}_{I}\, dx$

8.5 SPECIAL INTEGRANDS [O]

1. $\int \frac{x^2}{(x\sin x+\cos x)^2}\,dx$

Ans. $\tan(x-\tan^{-1}x)+c$

2. $\int \frac{dx}{(a+b\cos x)^2}$ $(a>b)$

3. $\int \frac{dx}{(\sin x+2\sec x)^2}$

4. $\int e^{\sin x}\left(\frac{x\cos^3 x-\sin x}{\cos^2 x}\right)dx$

5. $\int \sqrt[3]{\frac{1-x}{1+x}}\,\frac{1}{x}\,dx$

Sol start $\frac{1-x}{1+x}=t^3$ or $t^3+t^3x=1-x$ or $x(1+t^3)=1-t^3$ or $x=\frac{1-t^3}{1+t^3}$

6. $\int \frac{dx}{\sqrt[4]{(x-1)^3\,(x+2)^5}}$

7. $\int \frac{e^x\,dx}{(\sin e^x+e^{-x}\cos e^x)^2}$

Ans. $\tan(e^x-\tan^{-1}e^x)+C]$

[Sol. $I=\int \frac{dx}{e^{-x}(\sin e^x+e^{-x}\cos e^x)^2}=\int \frac{dx}{e^{-3x}(e^x\sin e^x+\cos e^x)^2}=\int \frac{e^{2x}\cdot e^x\,dx}{(e^x\sin e^x+\cos e^x)^2}$;

Let $e^x=t \Rightarrow e^x\,dx=dt$;

$I=\int \frac{t^2\,dt}{(t\sin t+\cos t)^2}$; put $t=\tan\theta$

$I=\int \frac{\tan^2\theta\sec^2\theta\,d\theta}{(\tan\theta\sin(\tan\theta)+\cos(\tan\theta))^2}=\int \frac{\tan^2\theta\cos^2\theta\sec^2\theta\,d\theta}{(\cos(\tan\theta)\cos\theta+\sin(\tan\theta)\sin\theta)^2}$

$= \int \frac{(\sec^2\theta - 1)\,d\theta}{\cos^2(\tan\theta - \theta)}$; $\tan\theta - \theta = y \Rightarrow (\sec^2\theta - 1)d\theta = dy$

$I = \int \frac{dy}{\cos^2 y} = \int \sec^2 y\, dy$

$= \tan y = \tan(\tan\theta - \theta) = \tan(t - \tan^{-1}t) + C = \tan(e^x - \tan^{-1} e^x) + C$ Ans.]

8.6 OTHER EXAMPLES [O]

1. $\int \frac{x^2}{x^4+1}dx;$

Ans. $\frac{1}{2\sqrt{2}}\left[\tan^{-1}\left(\frac{x^2-1}{\sqrt{2}x}\right)+\frac{1}{2}\ln\left|\frac{x^2-\sqrt{2}x+1}{x^2+\sqrt{2}x+1}\right|\right]+C$

2. $\int \frac{dx}{x^4+1}$

Ans. $\frac{1}{2\sqrt{2}}\left[\tan^{-1}\left(\frac{x^2-1}{\sqrt{2}x}\right)-\frac{1}{2}\ln\left|\frac{x^2-\sqrt{2}x+1}{x^2+\sqrt{2}x+1}\right|\right]+C$

3. $\int \frac{(x+1)^2}{x^4+x^2+1}dx$

Ans. $\sqrt{3}\tan^{-1}\left(\frac{2x-1}{\sqrt{3}}\right)-\frac{1}{\sqrt{3}}\tan^{-1}\left(\frac{2x+1}{\sqrt{3}}\right)+C$

4. $\int \frac{x^{17}}{1+x^{24}}dx$

Hint put $x^6 = t$

5. $\int \frac{x^2+3}{x^4+8x^2+9}dx$

Ans. $\frac{1}{\sqrt{14}}\tan^{-1}\left(\frac{t}{\sqrt{14}}\right)$ where $t = x - \frac{3}{x}$

6. $\int \frac{dx}{x^4+a^4}$

Ans.

7. $\int \sqrt{\tan\theta}\,d\theta$

Ans. $\frac{1}{\sqrt{2}}\tan^{-1}\left(\frac{t-\frac{1}{t}}{\sqrt{2}}\right)+\frac{1}{2\sqrt{2}}\log\left|\frac{t+\frac{1}{t}-\sqrt{2}}{t+\frac{1}{t}+\sqrt{2}}\right|+C$, where $t=\sqrt{\tan x}$

8. $\int\sqrt{\cot\theta}\,d\theta$

Ans. $-\left[\frac{1}{\sqrt{2}}\tan^{-1}\left(\frac{\cot\theta-1}{\sqrt{2\cot\theta}}\right)+\frac{1}{2\sqrt{2}}\ln\left|\frac{\cot\theta-\sqrt{2\cot\theta}+1}{\cot\theta+\sqrt{2\cot\theta}+1}\right|\right]+C$

9. $\int\left(\sqrt{\cot\theta}\pm\sqrt{\tan\theta}\right)d\theta$ **[JEE – 89]**

Ans. $\sqrt{2}\tan^{-1}\left(\frac{\tan\theta-1}{\sqrt{2\tan\theta}}\right)+C$

10. $\int\frac{\cos x}{\sqrt{8-\sin 2x}}dx$

11. $\int\frac{dx}{\operatorname{cosec}x+\cos x}$

Ans. $\frac{1}{2\sqrt{3}}\log\left[\frac{\sqrt{3}+\sin x-\cos x}{\sqrt{3}-\sin x+\cos x}\right]-\tan^{-1}(\sin x+\cos x)+C$

12. $\int\frac{\cos x}{10+\sin 2x}dx$

Ans.

MATHEMATICS

DEFINITE INTEGRATION

THEORY SHEET

ENGLISH

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1.1 INTRODUCTION

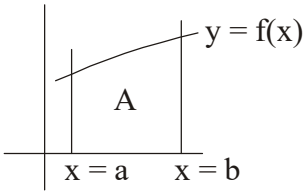
If $\frac{d}{dx}[f(x)] = \phi(x)$ and a and b, are two values independent of variable x,

$\int_a^b \phi(x)\,dx = [f(x)]_a^b = f(b) - f(a)$ is called **Definite Integral** of $\phi(x)$ within limits a and b.

Here **a** is called the **lower limit** and **b** is called the **upper limit** of the integral.

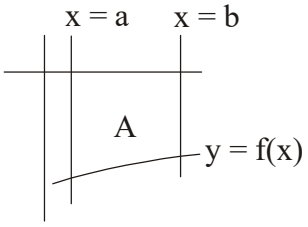
Let f(x) be a function defined on a closed interval [a, b]. Then $\int_a^b f(x)dx$ represents the algebraic sum of the areas of the region bounded by the curve y = f(x), x–axis and the lines x = a, x = b. Here algebraic sum means that area which is above the x–axis will be added in this sum with + sign and area which is below the x–axis will be added in this sum with – sign. So value of the definite integral may be positive, zero or negative.

(i)



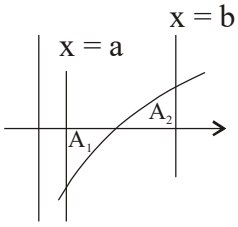
$$\int_a^b f(x)\cdot dx = A$$

(ii)



$$\int_a^b f(x)\cdot dx = -A$$

(iii)

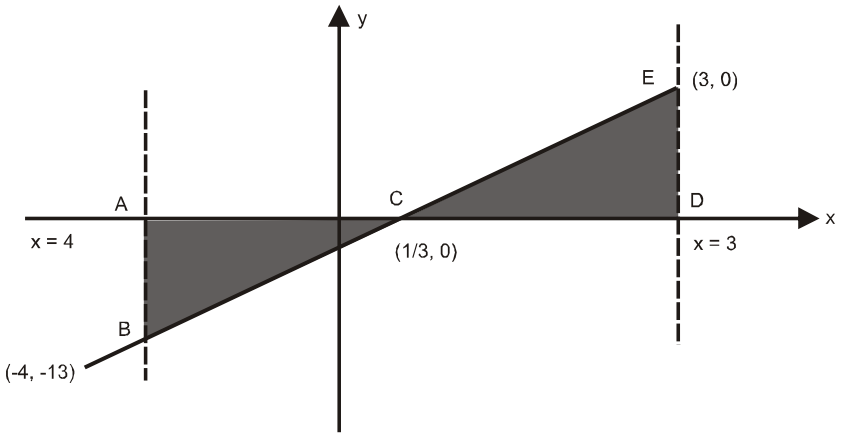


$$\int_a^b f(x)\cdot dx = A_2 - A_1$$

Q.1 $\int_{-4}^{3} (3x - 1)\,.\,dx$

Ans $-\frac{35}{2}$

Sol. $y = 3x - 1$ is a straight line, which lie below the x–axis in $\left[-4, \frac{1}{3}\right)$ and above in $\left(\frac{1}{3}, 3\right]$



Now area of Δ ABC $= \frac{1}{2} \times \frac{13}{3} \times 3 = \frac{169}{6}$ Area of Δ CDE $= \frac{1}{2} \times \frac{5}{3} \times 5 = \frac{32}{3}$

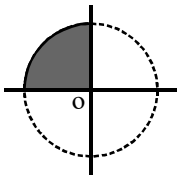
So $\int_{-1}^{4} (2x-3)dx$ $-\frac{169}{6} + \frac{32}{3} = -\frac{105}{6}$ $= -\frac{35}{2}$

Q.2 $\int_{-2}^{0} \sqrt{4-x^2}\, dx$

Ans π

Sol. $y = \sqrt{4-x^2}$, $x \in [-2, \ 0]$

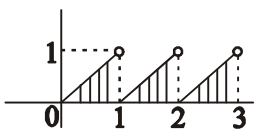
Represents a quarter circle in 2nd quadrant, which is above the x–axis radius of circle is 2.



so $\int_{-2}^{0} \sqrt{4-x^2}dx = \frac{1}{4}\left[\pi(2)^2\right] = \pi$ square unit

Q.3 $\int_{0}^{3} \{x\} \cdot dx$

Sol. $\int_{0}^{3} \{x\}\, dx = \frac{3}{2}$



Q.4 $\int_0^1 \frac{x^3}{\sqrt{1-x^8}}\,dx.$

Ans $\frac{\pi}{8}$

Sol. Let $x^4 = t$, then $4x^3\,dx = dt$

$\therefore\ I = \frac{1}{4}\int_0^1 \frac{dt}{\sqrt{1-t^2}} = \frac{1}{4}[\ \sin^{-1}\ t]_0^1 = \frac{\pi}{8}$

Note :

We have to take proper care of substitution. In the following question, $\tan\frac{x}{2} = t$ is not a valid substitution becasue $\tan\frac{x}{2}$ is discontinuous in $(0, 2\pi)$.

Ex.1 $\int_0^{2\pi} \frac{dx}{2+\sin x}$

$\int_0^{2\pi} \frac{dx}{2+\frac{2\tan x/2}{1+\tan^2 x/2}}$

$\tan\frac{x}{2} = t$

1.2 DEFINITE INTEGRATION WITH THE HELP OF INDEFINITE INTEGRATION

1. $\int_0^2 \left(e^{x^2}\right)x\,dx$

Ans. $\frac{e^4-1}{2}$

Sol. $\int_0^2 e^{x^2}\,x dx$

$\Rightarrow$ Put $x^2 = y$

$= \int_0^4 e^y \frac{dy}{2} = \frac{e^4-1}{2}$

2. $\int_3^8 \frac{\sin\sqrt{x+1}}{\sqrt{x+1}}\,dx$

Ans. 2 [cos2–cos3]

Sol. let $x + 1 = t^2$

3. $\int_0^{1-e^{-2}} \frac{\ell n(1-t)}{1-t}\,dt$

Ans. -2;

Sol. $\int_0^{1-\frac{1}{e^2}} \frac{\ell n(1-t)}{1-t}\,dt$

Put $1-t=y \Rightarrow -dt=dy$

$$-\int_1^{+\frac{1}{e^2}} \frac{\ell ny}{y}\,dy$$

$$=\left(\frac{\ell n^2 y}{2}\right)\Bigg|_{+\frac{1}{e^2}}^{1} = -\frac{1}{2}\cdot 4 = -2$$

4. $\int_0^{\pi/4} \cos 2x\sqrt{4-\sin 2x}\;dx$

Ans. $\frac{8-3\sqrt{3}}{3}$

Sol. $\int_0^{\frac{\pi}{4}} \cos 2x\sqrt{4-\sin 2x}\,dx$

Put $4-\sin 2x=y$

$$=-\int_4^3 \sqrt{y}\,\frac{dy}{2}$$

$$=\frac{1}{2}\int_3^4 \sqrt{y}dy = \frac{1}{2}\left.\frac{y^{\frac{3}{2}}}{\frac{3}{2}}\right|_3^4$$

$$=\frac{1}{3}\left(8-3\sqrt{3}\right)$$

5. $\int_0^1 \frac{\sin^{-1}x}{\sqrt{1-x^2}}\;dx$

Ans. $\frac{\pi^2}{8}$

Sol. $\int_0^1 \frac{\sin^{-1}x}{\sqrt{1-x^2}}\,dx$

$$= \frac{\left(\sin^{-1} x\right)^2}{2}\Bigg|_0^1$$

$$= \frac{\pi^2}{8}$$

6. $\int_0^{\ln 2} \frac{e^x}{e^x + 1} dx$

Ans. $\ln\left(\frac{3}{2}\right)$

Sol. $\int_0^{\ell n 2} \frac{e^x}{e^x + 1} dx$

Put $e^x + 1 = y$

$$= \ell n\left(e^x + 1\right)\Big|_0^{\ell n 2}$$

$$= \ell n \frac{3}{2}$$

7. $\int_0^1 \ln e^{\cos x} \, d(\sin x)$

Ans. $\frac{\pi}{4}$

Sol. $\int_0^1 \cos x d(\sin x)$

$$= \int_0^1 \sqrt{1 - \sin^2 x} d(\sin x)$$

$$= \frac{\sin x}{2}\sqrt{1 - \sin^2 x} + \frac{1}{2}\sin^{-1}(\sin x)\Big|_{\sin x = 0}^{\sin x = 1}$$

$$= \frac{\pi}{4}$$

8. $\int_0^{\pi/2} \frac{dx}{1 + 4\cot^2 x}$

Ans. $\frac{\pi}{6}$

Sol. $\int_0^{\frac{\pi}{2}} \frac{dx}{1+4\cot^2 x}$

$= \int_0^{\frac{\pi}{2}} \frac{\tan^2 x}{\tan^2 x + 4} dx$

$= \int_0^{\frac{\pi}{2}} \frac{\tan^2 x \sec^2 x}{\left(\tan^2 x + 4\right)\sec^2 x} dx$

$= \int_0^{\infty} \frac{y^2}{\left(y^2+4\right)\left(y^2+1\right)} dy$

$= \frac{1}{3}\int_0^{\infty} \left(\frac{4}{y^2+4} - \frac{1}{y^2+1} \right) dy$

$= \frac{1}{3}\frac{4}{2}\tan^{-1}\left(\frac{y}{2}\right) - \frac{1}{3}\tan^{-1} y \Big|_0^{\infty}$

$= \frac{2}{3}\frac{\pi}{2} - \frac{1}{3}\frac{\pi}{2}$

$= \frac{\pi}{6}$

9. $\int_0^{\pi/2} x \sin x \, dx$

Ans. 1

Sol. $\int_0^{\frac{\pi}{2}} x \sin x dx$

$= x\left(-\cos x\right) - (1)\left(-\sin x\right)\Big|_0^{\frac{\pi}{2}}$

$= 1$

10. $\int_1^2 \frac{dx}{(1+x)(2+x)}$

Ans. ln (9/8)

Sol. $\int_1^2 \frac{dx}{\left(1+x\right)\left(2+x\right)}$

$= \int_1^2 \frac{1}{1+x} - \frac{1}{2+x} dx$

$$= \ell n\left(\frac{1+x}{2+x}\right)\Big|_1^2$$

$$= \ell n\frac{3}{4} - \ell n\frac{2}{3}$$

$$= \ell n\frac{9}{8}$$

11. $\int_0^1 x\ \ell n(1+2x)\ dx$

Ans. $\frac{3\ell n3}{8}$

Sol. $\int_0^1 x\ell n(1+2x)dx$

$$= \ell n(1+2x)\cdot\frac{x^2}{2}\Big|_0^1 - \int_0^1 \frac{2}{1+2x}\cdot\frac{x^2}{2}dx$$

$$= \frac{1}{2}\ell n3 - \int_0^1 \frac{x^2}{1+2x}dx = \frac{1}{2}\ell n3 - \frac{1}{4}\int_0^1 2x - 1 + \frac{1}{2x+1}dx$$

$$= \frac{1}{2}\ell n3 - \frac{1}{4}\left(x^2 - x + \frac{\ell n}{2}(2x+1)\right)\Big|_0^1$$

$$= \frac{1}{2}\ell n3 - \frac{1}{4}\left\{\frac{1}{2}\ell n3\right\}$$

$$= \frac{3\ell n3}{8}$$

12. $\int_1^{e^{37}} \frac{\pi\sin(\pi\ell n x)}{x}dx$

Ans. 2

Sol. $\int_1^{e^{37}} \frac{\pi\sin(\pi\ell nx)}{x}dx$

Put $\pi\ell nx = y$

$$= \int_0^{37\pi} \sin y dy = 2$$

13. $\int_0^{\pi/2} \frac{dx}{4+5\sin x}$

Ans. $\frac{1}{3}\ell n2$

Sol. $$\int_0^{\frac{\pi}{2}} \frac{dx}{4+5\sin x} = \int_0^{\frac{\pi}{2}} \frac{dx}{4+\frac{10\tan\frac{x}{2}}{1+\tan^2\frac{x}{2}}}$$

$$= \int_0^{\frac{\pi}{2}} \frac{\sec^2\frac{x}{2}dx}{4\tan^2\frac{x}{2}+10\tan\frac{x}{2}+4}$$

Put $\tan\frac{x}{2} = y$

$$= \int_0^1 \frac{dy}{2y^2+5y+2} = \int_0^1 \frac{dy}{(y+2)(2y+1)}$$

$$= \frac{1}{3}\int_0^1 \frac{2}{2y+1} - \frac{1}{y+2}dy$$

$$= \frac{1}{3}\left(\ell n(2y+1) - \ell n(y+2)\right)\Big|_0^1$$

$$= \frac{1}{3}\ell n2$$

14. $\int_0^{\infty} \frac{dx}{a^2e^x + b^2e^{-x}}$

Ans. $\frac{1}{ab}\cot^{-1}\left(a/b\right)$

Sol. $$\int_0^{\infty} \frac{dx}{a^2e^x + b^2e^{-x}} = \int_0^{\infty} \frac{e^x}{a^2e^{2x}+b^2}dx$$

Put $e^x = y$

$$= \int_1^{\infty} \frac{dy}{a^2y^2+b^2} = \frac{1}{ba}\tan^{-1}\left(\frac{ay}{b}\right)\Big|_1^{\infty}$$

$$= \frac{1}{ab}\left(\frac{\pi}{2} - \tan^{-1}\frac{a}{b}\right)$$

$$= \frac{1}{ab}\cot^{-1}\left(\frac{a}{b}\right)$$

15. $\int_{\alpha}^{\beta} \frac{dx}{\sqrt{(x-\alpha)(\beta-x)}}$ $(\beta > \alpha)$

Ans. π

Sol. $\int_{\alpha}^{\beta} \frac{dx}{\sqrt{(x-\alpha)(\beta-x)}} = \int_{\alpha}^{\beta} \frac{dx}{\sqrt{-\left(x^2 - x(\alpha+\beta)+\alpha\beta\right)}}$

$$= \int_{\alpha}^{\beta} \frac{dx}{\sqrt{\left(\frac{\beta-\alpha}{2}\right)^2 - \left(x - \left(\frac{\alpha+\beta}{2}\right)\right)^2}}$$

$$= \sin^{-1}\left(\frac{x - \left(\frac{\alpha+\beta}{2}\right)}{\left(\frac{\beta-\alpha}{2}\right)} \right)\Bigg|_{\alpha}^{\beta} = \frac{\pi}{2} - \left(-\frac{\pi}{2}\right) = \pi$$

16. $\int_{0}^{1} \sqrt{\frac{x+\sqrt{x^2+1}}{1+x^2}}\, dx$

Ans. $2\left(\sqrt{\sqrt{2}+1}-1\right)$

Sol. Let $I = \int_{0}^{1} \sqrt{\frac{x+\sqrt{x^2+1}}{1+x^2}}\, dx$; $x + \sqrt{x^2+1} = t$

$$\Rightarrow \left(1 + \frac{1 \cdot 2x}{2\sqrt{x^2+1}}\right) dx = dt \Rightarrow \left(\frac{\sqrt{x^2+1}+x}{\sqrt{x^2+1}}\right) dx = dt$$

hence $I = \int_{1}^{\sqrt{2}+1} \frac{dt}{\sqrt{t}} = \int_{1}^{\sqrt{2}+1} t^{-1/2} dt = 2\sqrt{t}\Big]_{1}^{\sqrt{2}+1} = 2\left(\sqrt{\sqrt{2}+1}-1\right)$

17. $\int_{0}^{1/2} \frac{dx}{(1-2x^2)\sqrt{1-x^2}}$

Ans. $\frac{1}{2} ln(2+\sqrt{3})$ or $ln\left(\frac{\sqrt{3}+1}{\sqrt{2}}\right)$

Sol. $\int_0^{\frac{1}{2}} \frac{dx}{\left(1-2x^2\right)\sqrt{1-x^2}}$

Put $x = \sin\theta$

$\Rightarrow \int_0^{\frac{\pi}{6}} \frac{\cos\theta d\theta}{\cos 2\theta \cos\theta} = \int_0^{\frac{\pi}{6}} \sec 2\theta d\theta$

$= \frac{1}{2} \ell n \left|\sec 2\theta + \tan 2\theta\right|_0^{\frac{\pi}{6}}$

$= \frac{1}{2} \ell n \left(2+\sqrt{3}\right)$

18. If $\int_0^1 \frac{2e^{2x} + xe^x + 3e^x + 1}{(e^x+1)^2(e^x+x+1)^2} dx = K - \frac{1}{(e+1)(e+2)}$, where K is a constant. Find the value of K.

Ans $\frac{1}{4}$

Sol Note that $I = \int_0^1 \frac{f'(x)}{f^2(x)} dx$ when $f(x) = (e^x + 1)(e^x + x + 1)$

$\therefore \quad I = -\left.\frac{1}{(e^x+1)(e^x+x+1)}\right]_0^1 = \frac{1}{4} - \frac{1}{(e+1)(e+2)}$; Hence $K = \frac{1}{4}$ Ans.

19. $\int_0^{2008} \left(3x^2 - 8028x + (2007)^2 + \frac{1}{2008}\right) dx$

Ans. 2009

Sol. $\left[x^3 - 4014x^2 + (2007)^2 x + \frac{x}{2008}\right]_0^{2008}$

$= (2008)^3 - 4014(2008)^2 + \left((2007)^2 + \frac{1}{2008}\right) \cdot 2008$

$= (2008)\left[(2008)^2 - (4014)(2008) + (2007)^2\right] + 1$

$= (2008)\left[(2008)^2 - 2(2007)(2008) + (2007)^2\right] + 1 = 2008[(2008 - 2007)^2] + 1$

$= 2009$ **Ans.**

20. $\int_1^e (x+1)e^x \ln x \, dx$

Ans. $e + (e-1)e^e$

Sol. $I = \int_1^e e^x[x\ln x + (\ln x + 1) - 1]\,dx$; now use $\int_1^e e^x\left(f(x)+f'(x)\right) = e^x f(x)$

21. $\int_{1/2}^{2} \left(1 + x - \frac{1}{x}\right) e^{x+\frac{1}{x}}\, dx$

Ans. $\frac{3}{2}e^{5/2}$

Sol. $I = \int_{1/2}^{2} \underbrace{e^{x+\frac{1}{x}}}_{f(x)} + \underbrace{e^{x+\frac{1}{x}}\left(x - \frac{1}{x}\right)}_{x\,f'(x)} dx$

$\therefore \quad I = xe^{x+\frac{1}{x}}\Big]_{1/2}^{2} = 2e^{5/2} - \frac{1}{2}e^{5/2} = \frac{3}{2}e^{5/2}$ Ans.

$f(x) = e^{x+\frac{1}{x}} \Rightarrow \quad f'(x) = e^{x+\frac{1}{x}}\left(1 - \frac{1}{x^2}\right) \quad \Rightarrow \quad x\,f'(x) = e^{x+\frac{1}{x}}\left(1 - \frac{1}{x^2}\right)$

22. $\int_2^3 \frac{\sqrt{(x-2)^2}}{1+\sqrt{(x-2)^3}}\, dx$

Ans. $2 - \frac{2}{3}\ln 2 - \frac{2\pi}{3\sqrt{3}}$

Sol. $I = \int_2^3 \frac{|x-2|}{1+(x-2)^{\frac{3}{2}}}\, dx$

Put $\sqrt{x-2} = y$

$\Rightarrow x - 2 = y^2$

$y = \int_1 \frac{y^2\, 2y\, dy}{1+y^3}$

$\Rightarrow I = 2\int_0^1 \left(1 - \frac{1}{1+y^3}\right) dy$

23. $\int_0^{\pi/4} \frac{\sin\left(x-\frac{\pi}{4}\right)}{\sin 2x + 2(1+\sin x+\cos x)}\,dx$

Ans. $\frac{4-3\sqrt{2}}{4}$

Sol. $I = \frac{1}{\sqrt{2}}\int_0^{\pi/4} \frac{(\sin x-\cos x)}{(1+\sin 2x)+2(\sin x+\cos x)+1}\,dx = \frac{1}{\sqrt{2}}\int_0^{\pi/4} \frac{(\sin x-\cos x)\,dx}{[(\sin x+\cos x)+1]^2}$

put $\sin x + \cos x = t$ and proceed.

24. $\int_2^4 \frac{\sqrt{x^2-4}}{x^4}\,dx$

Ans. $\frac{\sqrt{3}}{32}$

Sol. $\int_2^4 \frac{\sqrt{x^2-4}}{x^4}dx = \int_2^4 \frac{x\sqrt{1-\frac{4}{x^2}}}{x^4}dx$

$= \int_2^4 \frac{\sqrt{1-\frac{4}{x^2}}}{x^3}dx$

Put $1-\frac{4}{x^2} = y$

$= \frac{1}{8}\int_0^{\frac{3}{4}} \sqrt{y}dy = \frac{1}{8}\left.\frac{y^{\frac{3}{2}}}{\frac{3}{2}}\right|_0^{\frac{3}{4}}$

$= \frac{1}{12}\cdot\frac{3\sqrt{3}}{8} = \frac{\sqrt{3}}{32}$

25. $\int_0^{\pi/16} \left(\frac{\sin x+\sin 2x+\sin 3x+.....+\sin 7x}{\cos x+\cos 2x+\cos 3x+.....+\cos 7x}\right)dx$

Ans. $\frac{1}{8}\ln 2$

Sol $\int_0^{\pi/16} \tan 4x\,dx = \frac{1}{4} l\text{n}(\sec 4x)]_0^{\pi/16} = \frac{1}{4} l\text{n}\sqrt{2} = \frac{1}{8} l\text{n}\,2$]

26. $\int_{\pi/4}^{\pi/2} \left(\sqrt{\frac{\sin x}{x}} + \sqrt{\frac{x}{\sin x}} \cos x \right) dx$

Ans. $\sqrt{2\pi} - \sqrt{\frac{\pi}{\sqrt{2}}}$

Sol. $\int_{\pi/4}^{\pi/2} \frac{\sin x - x\cos x}{\sqrt{x \sin x}} dx$ put $x \sin x = t^2$

27. Prove that $\int_a^b \frac{x^{n-1}\left((n-2)x^2 + (n-1)(a+b)x + nab\right)}{(x+a)^2(x+b)^2} dx = \frac{b^{n-1} - a^{n-1}}{2(a+b)}$ $(a > 0, b > 0)$

Ans.

Sol. $\int_a^b \frac{x^{n-1}[(nx^2 + n(a+b)x + nab - (2x^2 + (a+b)x]}{(x+a)^2(x+b)^2} dx$

$$= \int_a^b \frac{x^{n-1}[n(x+a)(x+b) - x(2x+a+b)]}{(x+a)^2(x+b)^2} dx$$

$$\int_a^b \left[\frac{nx^{n-1}}{(x+a)(x+b)} - \frac{x^n(2x+a+b)}{(x+a)^2(x+b)^2} \right] dx = \int_a^b \frac{d}{dx} \left[\frac{x^n}{(x+a)(x+b)} \right] dx$$

$$= \frac{x^n}{(x+a)(x+b)} \Bigg]_a^b = \frac{b^n}{(b+a)2b} - \frac{a^n}{2a(a+b)} = \frac{b^{n-1} - a^{n-1}}{2(a+b)}$$

28. $\int_0^\infty x^{2n+1} e^{-x^2} dx$, $n \in N$

Ans. $\left(\frac{n!}{2} \right)$

Sol. $I = \int_0^\infty (x^2)^n \cdot x\, e^{-x^2} dx$, put $x^2 = t$

29. Assume that f " is continuous and that $f(1) = 3$, $f'(1) = 2$ and $\int_0^1 f(x)\,dx = 5$. Find the value of $\int_0^1 x^2 f''(x)\,dx$.

Ans. 6

Sol. $\int_0^1 x^2 f''(x)$

$= x^2 f'(x)\big|_0^1 - \int_0^1 2xf'(x)\,dx$

$= 2 - \left\{2x\,f(x)\big|_0^1 - \int_0^1 2f(x)\,dx\right\}$

$= 2 - (2f(1)) + 2\int_0^1 f(x)\,dx$

$= 2 - 6 + 10 = 6$

30. $\int_3^5 \frac{x^2\,dx}{\sqrt{(x-3)(5-x)}}$

Ans. $\frac{33\pi}{2}$

Sol. Put $x = 3\cos^2\theta + 5\sin^2\theta$; $dx = 2\sin 2\theta\,d\theta$

$I = 2\int_0^{\pi/2} \frac{(3+2\sin^2\theta)^2 \sin 2\theta}{2\sin\theta\cos\theta} d\theta$ $\quad [x - 3 = 2\sin^2\theta\ ;\ 5 - x = 2\cos^2\theta]$

$= 2\int_0^{\pi/2} (9 + 4\sin^4\theta + 12\sin^2\theta)\,d\theta = 2\left[9\frac{\pi}{2} + 4\frac{3\pi}{16} + 12\frac{\pi}{4}\right] = 2\left[\frac{9\pi}{2} + \frac{3\pi}{4} + 3\pi\right] = \frac{33\pi}{2}$ Ans.

31. $\int_0^{2\pi} [\cos x(1+x) + (1-x)\sin x]\,dx$

Ans 2π

Sol $\int \cos x + x\cos x + \sin x - x\sin x)dx;$

$\int [(\sin x + \cos x) + x(\cos x - \sin x)]dx = \int (f(x) + x f'(x))dx;$

$[x(\sin x + \cos x)]_0^{2\pi} = 2\pi$

32. Let $I = \int_0^{\pi/2} \frac{\cos x}{a\cos x + b\sin x} dx$ and $J = \int_0^{\pi/2} \frac{\sin x}{a\cos x + b\sin x} dx$, where $a > 0$ and $b > 0$.

Compute the values of I and J.

Ans. $I = \frac{1}{a^2+b^2}\left(\frac{a\pi}{2} + b\,ln\left(\frac{b}{a}\right)\right)$; $\quad J = \frac{1}{a^2+b^2}\left(\frac{b\pi}{2} - a\,ln\left(\frac{b}{a}\right)\right)$

Sol. $aI + bJ = \frac{\pi}{2}$ ....(1)

and $\quad bI - aJ = \int_0^{\pi/2} \frac{b\cos x - a\sin x}{a\cos x + b\sin x}dx$

$\therefore \quad bI - aJ = l\text{n}\left[a\cos x + b\sin x\right]_0^{\pi/2} \quad \Rightarrow \quad bI - aJ = l\text{n}\left(\frac{b}{a}\right) \quad$ .....(2)

from (1) and (2)

$a^2I + abJ = \frac{a\pi}{2}$

$b^2I - abJ = b\, l\text{n}(b/a)$

$I = \frac{1}{a^2+b^2}\left(\frac{a\pi}{2} + b\, l\text{n}\left(\frac{b}{a}\right)\right)$ **Ans.**

again $\quad abI + b^2I = \frac{b\pi}{2}$

and $\quad abI - a^2J = a\, l\text{n}\,(b/a)$

subtract ---

$J = \frac{1}{a^2+b^2}\left(\frac{b\pi}{2} - a\, l\text{n}\left(\frac{b}{a}\right)\right)$ **Ans.**

33. Let $5f(x) + 4f\left(\frac{1}{x}\right) = \frac{1}{x} + 3, x > 0$. Then $18\int_1^2 f(x)dx$ is equal to :

Ans. $10\log_e 2 - 6$

Sol. $5f(x) + 4f\left(\frac{1}{x}\right) = \frac{1}{x} + 3$

$x \to \frac{1}{x}$

$5f\left(\frac{1}{x}\right) + 4f(x) = x + 3$

$(1)\times 5 - (2)\times 4$

$\Rightarrow f(x) = \frac{5}{9x} - \frac{4}{9}x + \frac{1}{3}$

$\Rightarrow 18\int_1^2 f(x)dx = 18\left(\frac{5}{9}\ln 2 - \frac{4}{9}\times\frac{3}{2} + \frac{1}{3}\right)$

$= 10\ln 2 - 6$

Important Note : If limits are reciprocal of each other.

Ex.1 $I = \int_{\frac{1}{2}}^{2} \frac{\ln x \cdot dx}{1+x^2}$ ..... (i)

$x = \frac{1}{t} \Rightarrow dx = \frac{-dt}{t^2}$

$I = -\int_{2}^{\frac{1}{2}} \frac{\ln\left(\frac{1}{t}\right)}{1+\frac{1}{t^2}}\left(\frac{dt}{t^2}\right)$

$I = -\int_{\frac{1}{2}}^{2} \frac{\ln t}{1+t^2} \cdot dt$ .....(ii)

(i) + (ii) $2I = 0 \quad I = 0$

Ex.2 Evaluate: $\int_0^{\infty} \frac{l\text{n } x \; dx}{x^2+2x+4}$

Ans. $\frac{\pi l\text{n } 2}{3\sqrt{3}}$

Sol $I = \int_0^{\infty} \frac{l\text{n } x \; dx}{x^2+2x+4}$; (put $x = 2t$; $dx = 2dt$ to make coefficient of x^2 and constant term same)

$= 2\int_0^{\infty} \frac{l\text{n } 2 + l\text{n } t}{4(t^2+t+1)} dt = \frac{l\text{n } 2}{2} \underbrace{\int_0^{\infty} \frac{dt}{t^2+t+1}}_{I_1} + \frac{1}{2} \underbrace{\int_0^{\infty} \frac{l\text{n } t \, dt}{t^2+t+1}}_{I_2 = \text{zero}}$

$I_2 = \int_0^{\infty} \frac{l\text{n } t \, dt}{t^2+t+1}$ (put $t = \frac{1}{y}$; $dt = -\frac{1}{y^2} dy$); $I_2 = \int_{\infty}^{0} \frac{\ln y \cdot}{\left(\frac{1}{y^2}+\frac{1}{y}+1\right)y^2} dy$

$I_2 = \int_{\infty}^{0} \frac{l\text{n } y \, dy}{y^2+y+1} = -\int_0^{\infty} \frac{l\text{n } y \, dy}{y^2+y+1} = -I_2$

now $I_1 = \int_0^{\infty} \frac{dt}{\left(t+\frac{1}{2}\right)^2+\left(\frac{\sqrt{3}}{2}\right)^2} = \frac{2}{\sqrt{3}} \tan^{-1} \frac{(t+(1/2))}{\frac{\sqrt{3}}{2}}\Bigg|_0^{\infty} = \frac{2}{\sqrt{3}}\left[\frac{\pi}{2} - \frac{\pi}{6}\right] = \frac{2\pi}{3\sqrt{3}}$

$\therefore \quad I = \frac{l\text{n } 2}{2} \cdot \frac{2\pi}{3\sqrt{3}} = \frac{\pi l\text{n } 2}{3\sqrt{3}}$ **Ans.**

Important Points

1. If $f(x) \geq 0$ on the interval [a,b], then $\int_a^b f(x)dx \geq 0$

2. If f is a continuous function in (a,b) and $\int_a^b f(x)dx = 0$ then the equation f(x) = 0 has atleast one root lying in (a, b) .

3. $\int_a^b f'(x) \cdot dx = \left(f(x)\right)_a^b$ if f(x) is continuous in (a, b).

Ex. $\int_0^{\pi} \sec^2 x \cdot dx = \tan x)_0^{\pi} = 0$ Wrong

4. $\int_a^a f(x) \cdot dx = 0$ if $x = a$ is in domain of f(x).

5. $\int x \cdot d\left(\ln x\right) = \int x \times \frac{1}{x} \cdot dx$

$$\int_0^1 x \cdot d(\ln x) = \int_1^e x \times \frac{1}{x} \cdot dx$$

6. $\lim_{n\to a} \int_p^q f_n\left(x\right) \cdot dx = \int_p^q \lim_{n\to a} f_n(x) \cdot dx$ if p and q are independent of n.

Eg. $\lim_{n\to\infty} \int_{-a^{\frac{1}{3}}}^{a^{\frac{1}{3}}} \left(1-\frac{t^3}{n}\right)^n t^2 \cdot dt$

Sol $\int_{-a^{\frac{1}{3}}}^{a^{\frac{1}{3}}} \lim_{n\to\infty} \left(1-\frac{t^3}{n}\right)^n t^2 \cdot dt$

put $t^3 = v$

$$\frac{1}{3}\int_{-a}^{a} \lim_{n\to\infty} \left(1-\frac{v}{n}\right)^n \cdot dv$$

$$\frac{1}{3}\int_{-a}^{a} e^{n\times\left(\frac{-v}{n}\right)} \cdot dv$$

$$= \frac{1}{3}\int_{-a}^{a} e^{-v} \cdot dv$$

$$= -\frac{1}{3}\left(e^{-a} - e^{a}\right) = \frac{1}{3}\left(e^{a} - e^{-a}\right)$$

7. If f(x) and g(x) are inverse of each other and f(a) = c, f(b) = d.

then, $\int_a^b f(x)\cdot dx + \int_c^d g(y)\cdot dy = bd - ac$

Proof: $y = f(x)$

$dy = f'(x)\cdot dx$

$$\int_a^b f(x)\cdot dx + \int_a^b \underbrace{g(f(x))}_{x} f'(x) = dx$$

$$\int_a^b \left(f(x) + x(f'(x))\right)\cdot dx = \left(x\left(f(x)\right)\right)_a^b = b(f(b)) - a(f(a)) = bd - ac$$

Que. $\int_1^e \ln x\cdot dx + \int_0^1 e^x\cdot dx = e\times 1 - 1\times 0 = e$

Important results :

1. $\int_0^{\frac{\pi}{2}} \sin x\cdot dx = \int_0^{\frac{\pi}{2}} \cos x\cdot dx = 1$

2. $\int_0^{\frac{\pi}{2}} \sin^2 x\cdot dx = \int_0^{\frac{\pi}{2}} \cos^2 x\cdot dx = \frac{\pi}{4}$

3. $\int_0^{\frac{\pi}{2}} \sin^3 x\cdot dx = \int_0^{\frac{\pi}{2}} \cos^3 x\cdot dx = \frac{2}{3}$

4. $\int_0^{\frac{\pi}{2}} \sin^4 x\cdot dx = \int_0^{\frac{\pi}{2}} \cos^4 x\cdot dx = \frac{3\pi}{16}$

2.1 PROPERTIES OF DEFINITE INTEGRAL - P1

$$\int_a^b f(x)\ dx = \int_a^b f(t)\,dt$$

i.e. definite integral is independent of variable of integration.

2.2 PROPERTIES OF DEFINITE INTEGRAL - P2

$$\int_a^b f(x)\ dx = -\int_b^a f(x)\,dx$$

2.3 PROPERTIES OF DEFINITE INTEGRAL - P3

$$\int_a^b f(x)\ dx = \int_a^c f(x)\,dx + \int_c^b f(x)\ dx$$

Note :

1. c need not lie between a and b
2. This property is helpful in case of non - uniform function/discontinuous function.

Ex.1 If $f(x)=\begin{cases} x+3 & : \ x<3 \\ 3x^2+1 & : \ x\geq 3 \end{cases}$, then find $\int_2^5 f(x)\,dx$.

Ans. $\frac{211}{2}$

Sol. $\int_2^5 f(x)\,dx=\int_2^3 f(x)\,dx+\int_3^5 f(x)\,dx=\int_2^3 (x+3)\,dx+\int_3^5 (3x^2+1)\,dx=\left[\frac{x^2}{2}+3x\right]_2^3+\left[x^3+x\right]_3^5$

$=\frac{9-4}{2}+3\,(3-2)+5^3-3^3+5-3=\frac{211}{2}$

Ex. 2 Evaluate $\int_2^8 |x-5|\,dx$.

Ans 9

Sol. $\int_2^8 |x-5|\,dx=\int_2^5 (-x+5)\,dx+\int_5^8 (x-5)\,dx=9$

Ex. 3 If $f(x)=\begin{cases} x^2, & 0<x<2 \\ 3x-4, & 2\leq x<3 \end{cases}$ then evaluate $\int_0^3 f(x)dx$

Ans 37/6

Sol. $\int_0^3 f(x)dx=\int_0^2 f(x)dx+\int_2^3 f(x)dx=\int_0^2 x^2\,dx+\int_2^3 (3x-4)dx$

$=\left(\frac{x^3}{3}\right)_0^2+\left(\frac{3x^2}{2}-4x\right)_2^3=\frac{8}{3}+\frac{27}{2}-12-6+8=37/6$

Ex. 4 If $f(x)=\begin{cases} 3[x]-5\frac{|x|}{x}, & x\neq 0 \\ 2, & x=0 \end{cases}$ then $\int_{-3/2}^2 f(x)dx$ is equal to ([.] denotes the greatest integer function)

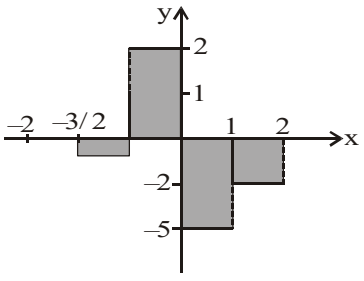
Ans. $-\frac{11}{2}$

Sol. $3[x]-5\frac{|x|}{x}=3[x]-5$, if $x>0$

$=3[x]+5$, if $x<0$

$\Rightarrow \int_{-3/2}^2 f(x)dx=\int_{-3/2}^{-1}(-1)dx+\int_{-1}^{0}(2)dx+\int_0^1(-5)dx+\int_1^2(-2)dx$

$=-1\left(-1+\frac{3}{2}\right)+2(1)+1(-5)+(-2)=-\frac{1}{2}+2-5-2=-\frac{11}{2}$



Questions

1. $\int_0^5 |x-2|\,dx$

Ans. 13/2

Sol. $\int_0^5 |x-2|\,dx=\int_0^2 (2-x)\,dx+\int_2^5 (x-2)\,dx$

$$= 2x - \frac{x^2}{2}\Big|_0^2 + \left(\frac{x^2}{2} - 2x\right)\Big|_2^5$$

$$= 2 + \left(\frac{25}{2} - 4\right) - (2)$$

$$= \frac{13}{2}$$

2. $\int_0^{\pi} \sqrt{\frac{1+\cos 2x}{2}}\ dx$

Ans. 2

Sol. $\int_0^{\pi} \sqrt{\frac{1+\cos 2x}{2}}dx = \int_0^{\pi} \sqrt{\sin^2 x}dx = \int_0^{\pi} |\sin x|\,dx = 2$

3. $\int_0^9 \left[\sqrt{t}\right] dt$

Ans. 13

Sol. $\int_0^9 \left[\sqrt{t}\right] dt = \int_0^1 \left[\sqrt{t}\right] dt + \int_1^4 \left[\sqrt{t}\right] dt + \int_4^9 \left[\sqrt{t}\right] dt$

$= 1\ (4 - 1) + 2(9 - 4)$

$= 13$

4. $\int_0^{3/2} x[x^2]\,dx$

Ans. 3/4

Sol. $\int_0^{\frac{3}{2}} x\left[x^2\right]dx = \int_0^1 x\left[x^2\right]dx + \int_1^{\sqrt{2}} x\left[x^2\right]dx + \int_{\sqrt{2}}^{\frac{3}{2}} x\left[x^2\right]dx$

$$= 0 + \frac{x^2}{2}\Big|_1^{\sqrt{2}} + x^2\Big|_{\sqrt{2}}^{\frac{3}{2}}$$

$$= \frac{1}{2}(2-1) + \left(\frac{9}{4} - 2\right) = \frac{1}{2} + \frac{1}{4} = \frac{3}{4}$$

5. $\int_{-2}^{2} |1 - x^2|\,dx$ [JEE - 89]

Ans. 4

Sol. $\int_{-2}^{2} \left|1 - x^2\right| dx = \int_{-2}^{-1} \left(x^2 - 1\right) dx + \int_{-1}^{1} \left(1 - x^2\right) dx + \int_1^2 \left(x^2 - 1\right) dx$

$= 4$

6. $\int_{-1}^{3}\left[x+\frac{1}{2}\right]dx$

Ans. 4

Sol. $\int_{-1}^{3}\left[x+\frac{1}{2}\right]dx=\int_{-1}^{-\frac{1}{2}}\left[x+\frac{1}{2}\right]dx+\int_{-\frac{1}{2}}^{\frac{1}{2}}\left[x+\frac{1}{2}\right]dx+\int_{\frac{1}{2}}^{\frac{3}{2}}\left[x+\frac{1}{2}\right]dx+\int_{\frac{3}{2}}^{\frac{5}{2}}\left[x+\frac{1}{2}\right]dx+\int_{\frac{5}{2}}^{3}\left[x+\frac{1}{2}\right]dx$

$=-1\left(+\frac{1}{2}\right)+0+1+2+3\cdot\frac{1}{2}$

$=-\frac{1}{2}+3+\frac{3}{2}=4$

7. $\int_{0}^{3}|5x-9|\,dx$

Ans. 117/10

Sol. $\int_{0}^{3}|5x-9|dx=\int_{0}^{\frac{9}{5}}(9-5x)dx+\int_{\frac{9}{5}}^{3}(5x-9)dx$

$=\frac{117}{10}$

8. $\int_{-e}^{-1/e}\left|\ln|x|\right|dx$

Ans. $2-\frac{2}{e}$

Sol. $x=-t;\ -\int_{e}^{1/e}\left|\ln|t|\right|dt=\int_{1/e}^{e}|\ln t|\,dt=-\int_{1/e}^{1}\ln t\,dt+\int_{1}^{e}\ln t\,dt$

9. $\int_{0}^{2\pi}\sqrt{1-\sin x}\;dx$

Ans. $4\sqrt{2}$

Sol. $\int_{0}^{2\pi}\sqrt{1-\sin x}dx=\int_{0}^{2\pi}\left|\cos\frac{x}{2}-\sin\frac{x}{2}\right|dx$

Put $\frac{x}{2}=y$

$$= 2\int_0^{\pi} \left|(\cos y - \sin y)\right| dy = 2\int_0^{\frac{\pi}{4}} (\cos y - \sin y)\, dy + 2\int_{\frac{\pi}{4}}^{\pi} (\sin y - \cos y)\, dy$$

$$= 2(\sin y + \cos y)\Big|_0^{\frac{\pi}{4}} + 2(-\cos y - \sin y)\Big|_{\frac{\pi}{4}}^{\pi}$$

$$= 2\left(\sqrt{2} - 1\right) - 2\left(-1 - \sqrt{2}\right)$$

$$= 4\sqrt{2}$$

10. $\int_0^{2\pi} |\,1 + 2\cos x\,|\, dx$

Ans. $\frac{2\pi}{3} + 4\sqrt{3}$

Sol. $\int_0^{2\pi} \left|1 + 2\cos x\right| dx$

$$= \int_0^{\frac{2\pi}{3}} (1 + 2\cos x)\, dx + \int_{\frac{2\pi}{3}}^{\frac{4\pi}{3}} (-1 - 2\cos x)\, dx + \int_{\frac{4\pi}{3}}^{2\pi} (1 + 2\cos x)\, dx$$

$$= x + 2\sin x\Big|_0^{\frac{2\pi}{3}} - (x + 2\sin x)\Big|_{\frac{2\pi}{3}}^{\frac{4\pi}{3}} + (x + 2\sin x)\Big|_{\frac{4\pi}{3}}^{2\pi}$$

$$= \frac{2\pi}{3} + \sqrt{3} - \left(\frac{4\pi}{3} - \sqrt{3} - \frac{2\pi}{3} - \sqrt{3}\right) + \left(2\pi - \frac{4\pi}{3} + \sqrt{3}\right)$$

$$= \frac{2\pi}{3} + 4\sqrt{3}$$

11. $\int_0^2 [x^2 - x + 1]\, dx$

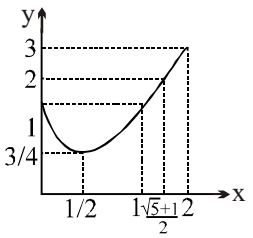
Ans. $\frac{5 - \sqrt{5}}{2}$

Sol: $I = \int_0^1 0\, dx + \int_1^{\frac{\sqrt{5}+1}{2}} 1\, .dx + \int_{\frac{\sqrt{5}+1}{2}}^2 2\, dx$

if $x^2 - x + 1 = 2$

$x^2 - x - 1 = 0$

then $x = \frac{1 \pm \sqrt{5}}{2} = \frac{\sqrt{5}+1}{2}$



12. If $\int_{-a}^{a}(|x|+|x-2|)dx = 22$ $(a > 2)$ and $[x]$ denotes the greatest integer $\le x$, then $\int_{a}^{-a}(x+[x])$ dx is equal to _____. **[JEE(Main)Feb.2021]**

Ans. 3

Sol. $I = \int_{-a}^{a}|x|dx + \int_{-a}^{a}|x-2|dx = 22$

$$2\int_{0}^{a}x\,dx + \int_{-a}^{2}(2-x)dx + \int_{2}^{a}(x-2)dx = 22$$

$$a = 3$$

$$\int_{a}^{-a}(x+[x])dx = \int_{3}^{-3}xdx + \int_{3}^{-3}[x]dx$$

$$= 0+3 = 3$$

13. If $[\cdot]$ represents the greatest integer function, then the value of $\left|\int_{0}^{\sqrt{\frac{\pi}{2}}}\left[\left[x^2\right]-\cos x\right]dx\right|$ is _____.

[JEE(Main)Mar.2021]

Ans. 1

Sol. $$\left|\int_{0}^{\sqrt{\frac{\pi}{2}}}\left[x^2\right]dx + \int_{0}^{\sqrt{\frac{\pi}{2}}}[-\cos x]dx\right|$$

$$= \left|\int_{0}^{1}0\ dx + \int_{1}^{\sqrt{\frac{\pi}{2}}}1\ dx + \int_{0}^{\sqrt{\frac{\pi}{2}}}(-1)\ dx\right|$$

$$= \left|\sqrt{\frac{\pi}{2}} - 1 - \sqrt{\frac{\pi}{2}}\right| = |-1| = 1$$

14. Let the function $f:[0,2]\to\mathbb{R}$ be defined as $f(x) = \begin{cases} e^{\min\{x^2, x-[x]\}}, & x\in[0,1) \\ e^{[x-\log_e x]}, & x\in[1,2] \end{cases}$ where [t] denotes the greatest integer less than or equal to t. Then the value of the integral $\int_0^2 xf(x)dx$ is :

Ans. $2e-\frac{1}{2}$

Sol. $f(x)=\begin{cases} e^{x^2} & x\in(0,1) \\ e & x\in[1,2) \end{cases}$

Now $\int_0^2 xf(x)dx=\int_0^1 xe^{x^2}dx+\int_1^2 x\cdot edx=2e-\frac{1}{2}$

2.4 PROPERTIES OF DEFINITE INTEGRAL - P4 (FOR ODD AND EVEN FUNCTIONS)

$$\int_{-a}^{a} f(x)dx=\int_{0}^{a}[f(x)+f(-x)]dx=\begin{cases} 0 & ;\text{if } f(x) \text{ is an odd function} \\ 2\int_0^a f(x)dx & ;\text{if } f(x) \text{ is an even function} \end{cases}$$

Proof : LHS$=\int_{-a}^{0} f(x)\cdot dx+\int_0^a f(x)\cdot dx$

put $x=-t$

$$\text{LHS}=-\int_a^0 f(-t)\cdot dt+\int_0^a f(t)\cdot dx=\int_0^a f(-t)\cdot dt+\int_0^a f(x)\cdot dx=\int_0^a \big(f(x)+f(-x)\big)\cdot dx$$

Note:

1. This property is applicable for discontinuous functions also,
2. This property is helpful when limits are of opposite signs of each other.

Ex.1 Evaluate : $\int_{-\pi/2}^{\pi/2}\cos^2 x\,dx$.

Ans $\pi/2$

Here I $=2\int_0^{\pi/2}\cos^2 x\,dx \quad \{\because f(-x)=f(x)\}$

$$=\int_0^{\pi/2}(1+\cos 2x)\,dx=\left(x+\frac{\sin 2x}{2}\right)_0^{\pi/2}=\pi/2$$

Ex.2 Evaluate : $\int_{-1}^{1}(x^3+x)dx$

Ans 0

Sol. $\int_{-1}^{1}(x^3+x)dx=0\ [\because\ f(-x)=-f(x)]$

Ex.3 Evaluate : $\int_{-1}^{1}\frac{x^3\sin(1+x^2)}{1+x^2}dx$

Ans 0

Sol. Here $f(x) = \dfrac{x^3 \sin(1+x^2)}{1+x^2}$

& $f(-x) = -\dfrac{x^3 \sin(1+x^2)}{1+x^2}$

$\because f(-x) = -f(x)$

$\therefore I = 0.$

Ex.4 Evaluate : $\int_{-\pi/2}^{\pi/2} \cos^3\theta\,(1+\sin\theta)^2 d\theta$

Ans $\dfrac{8}{5}$

Sol. Here

$I = \int_{-\pi/2}^{\pi/2} (1-\sin^2\theta)(1+\sin\theta)^2 \cos\theta\, d\theta$

Put $\sin\theta = t \Rightarrow \cos\theta\, d\theta = dt$

$\therefore I = \int_{-1}^{1} (1-t^2)(1+t)^2\, dt$

$= \int_{-1}^{1} (1-t^2)(1+2t+t^2)\, dt$

$= \int_{-1}^{1} (1+2t-2t^3-t^4)\, dt$

$= \int_{-1}^{1} (2t-2t^3)\, dt + \int_{-1}^{1} (1-t^4)\, dt$

$= 0 + 2\int_{0}^{1} (1-t^4)\, dt$

$= 2\left[t - \dfrac{t^5}{5}\right]_0^1 = 2\left(1-\dfrac{1}{5}\right) = \dfrac{8}{5}$

Questions

1. $\int_{-1/2}^{1/2} \ln\left(\dfrac{1-x}{1+x}\right) dx$

Ans. 0

Sol. $\int_{-\frac{1}{2}}^{\frac{1}{2}} \ell n\left(\dfrac{1-x}{1+x}\right) dx = \int_{0}^{\frac{1}{2}} \ell n\left(\dfrac{1-x}{1+x}\right) + \ell n\left(\dfrac{1+x}{1-x}\right) dx = 0$

2. $\int_{-3}^{3} \left(\sqrt{e^x} - \sqrt{e^{-x}}\right) dx$

Ans. 0

Sol. $\int_{-3}^{3} \left(\sqrt{e^x} - \sqrt{e^{-x}}\right) dx = \int_{0}^{3} \left(\sqrt{e^x} - \sqrt{e^{-x}}\right) + \left(\sqrt{e^{-x}} - \sqrt{e^x}\right) dx = 0$

3. $\int_{-\pi/4}^{\pi/4} f(x)\,dx$ where $f(x)=\dfrac{x^7-3x^5+3x^3-x+1}{\cos^2 x}$

Ans. 2

Sol. $\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \dfrac{x^7-3x^5+3x^3-x}{\cos^2 x}+\dfrac{1}{\cos^2 x}dx = 2\int_0^{\frac{\pi}{4}} \sec^2 x dx = 2$

4. $\int_{-\pi/2}^{\pi/2} \dfrac{f(x)-f(-x)}{g(x)+g(-x)}dx$

Ans. 0

Sol. $\int_{-\frac{\pi}{2}}^{\frac{\pi}{4}} \dfrac{f(x)-f(-x)}{g(x)+g(-x)}dx = \int_0^{\frac{\pi}{2}} \dfrac{f(x)-f(-x)}{g(x)+g(-x)}+\dfrac{f(-x)-f(x)}{g(x)+g(-x)}dx = 0$

5. $\int_{-a}^{a} x\sqrt{a^2-x^2}\,dx$

Ans. 0

Sol. $\int_{-a}^{a} x\sqrt{a^2-x^2}dx = \int_0^a x\sqrt{a^2-x^2}+(-x)\sqrt{a^2-x^2}dx = 0$

6. $\int_{-1/2}^{1/2} \sec x \ln\dfrac{1-x}{1+x}\,dx$

Ans. 0

Sol. $\int_{-\frac{1}{2}}^{\frac{1}{2}} \sec x \ell n\left(\dfrac{1-x}{1+x}\right)dx = \int_0^{\frac{1}{2}} \sec x \ell n\left(\dfrac{1-x}{1+x}\right)+\sec x \ell n\left(\dfrac{1+x}{1-x}\right)dx$

$= 0$

7. $\int_{-1/2}^{1/2} \left([x]+\ell n\left(\dfrac{1+x}{1-x}\right)\right)dx$ **[JEE 2002]**

Ans. $-\dfrac{1}{2}$

Sol. $\int_{-\frac{1}{2}}^{\frac{1}{2}} [x]+\ell n\left(\dfrac{1+x}{1-x}\right)dx = \int_0^{\frac{1}{2}} [x]+[-x]+\ell n\left(\dfrac{1+x}{1-x}\right)+\ell n\left(\dfrac{1-x}{1+x}\right)dx$

$$= \int_0^{\frac{1}{2}} [x] + [-x]\,dx = -1\left(\frac{1}{2}\right) = -\frac{1}{2}$$

8. $\int_{-2}^{2} |1 - x^2|\,dx$

Ans. 4

Sol. $\int_{-2}^{2} \left|1 - x^2\right| dx = 2\int_0^2 \left|x^2 - 1\right| dx = 2\int_0^1 \left(1 - x^2\right) dx + 2\int_1^2 \left(x^2 - 1\right) dx$

$= 4$

9. $\int_{-1}^{3/2} |x \sin \pi x|\,dx$ **[JEE 82]**

Ans. $\frac{3\pi + 1}{\pi^2}$

Sol. $I = \int_{-1}^{1} \left|(x \sin \pi x)\right| dx + \int_1^{3/2} \left|x \sin \pi x\right| dx$; $I = 2\int_0^1 x \sin \pi x\,dx - \int_1^{3/2} x \sin \pi x\,dx$

10. $\int_{-1}^{3} \left(\tan^{-1} \frac{x}{x^2 + 1} + \tan^{-1} \frac{x^2 + 1}{x} \right) dx$

Ans. π

Sol. $\int_{-1}^{1} f(x)dx + \int_1^3 \left(\tan^{-1} \frac{x}{x^2 + 1} + \cot^{-1} \frac{x}{x^2 + 1} \right) dx = 0 + \frac{\pi}{2} \cdot 2 = \pi$ Ans

2.5 PROPERTIES OF DEFINITE INTEGRAL - P5 (KING PROPERTY)

$$\int_a^b f(x)dx = \int_a^b f(a + b - x)dx$$

Proof : LHS$= \int_a^b f(x) \cdot dx$

Put $x = a + b - t \Rightarrow \quad dx = -dt$

$$\text{LHS} = -\int_b^a f(a + b - t) \cdot dt = \int_a^b f(a + b - t) \cdot dt = \text{RHS}$$

Ex.1 $I = \int_0^{\frac{\pi}{2}} \sin^2 x \cdot dx$ ..... (i)

Apply king

$$I = \int_0^{\frac{\pi}{2}} \sin^2\left(\frac{\pi}{2} - x\right) \cdot dx$$

$$I = \int_0^{\frac{\pi}{2}} \cos^2 x \cdot dx \quad \text{..... (ii)}$$

(i) + (ii)

$$2I = \int_0^{\frac{\pi}{2}} dx = \frac{\pi}{2} \Rightarrow I = \frac{\pi}{4}$$

Note :

1. This property is applicable for discontinuous function also
2. When applying king if D^r remains unchanged then also this property is helpful.

Ex.2 $$I = \int_0^{\frac{\pi}{2}} \frac{\sin^3(x)}{\sin^3 x + \cos^3 x}$$

apply King

$$I = \int_0^{\frac{\pi}{2}} \frac{\cos^3(x)}{\sin^3 x + \cos^3 x}$$

Add.

$$2I = \int_0^{\frac{\pi}{2}} dx \Rightarrow I = \frac{\pi}{4}$$

3. When function is of the form x × f(x) and when on applying king f(x) remains unchanged then also this property is helpful.

Ex.3 $$I = \int_0^{\pi} x \cdot \sin x \cdot dx$$

King

$$I = \int_0^{\pi} (\pi - x) \sin x \cdot dx$$

Add.

$$2I = \pi \int_0^{\pi} \sin x \cdot dx \Rightarrow \quad 2I = \pi \times 2 \quad \Rightarrow I = \pi$$

Questions

1. $I = \int_0^{\pi/2} \sin^2 x \, dx$

Ans. $\frac{\pi}{4}$

Sol. $I = \int_0^{\frac{\pi}{2}} \sin^2 x dx$

$I = \int_0^{\frac{\pi}{2}} \cos^2 x dx$

on adding $\frac{\pi}{2}$

$2I = \int_0^{\frac{\pi}{2}} 1 dx$

$I = \frac{\pi}{4}$

2. $I = \int_0^{\pi/2} \frac{dx}{1+\sqrt{\tan x}}$

Ans. $\frac{\pi}{4}$

Sol. $I = \int_0^{\pi/2} \frac{dx}{1+\sqrt{\tan x}} = \int_0^{\pi/2} \frac{\sqrt{\cos x}}{\sqrt{\cos x}+\sqrt{\sin x}} dx$ .....(i)

then $I = \int_0^{\pi/2} \frac{\sqrt{\cos(\pi/2 - x)dx}}{\sqrt{\cos(\pi/2 - x)}+\sqrt{\sin(\pi/2 - x)}} = \int_0^{\pi/2} \frac{\sqrt{\sin x}}{\sqrt{\sin x}+\sqrt{\cos x}} dx$ .....(ii)

Adding (i) and (ii), we get

$2I = \int_0^{\pi/2} \frac{\sqrt{\sin x}}{\sqrt{\sin x}+\sqrt{\cos x}}.dx + \int_0^{\pi/2} \frac{\sqrt{\cos x}}{\sqrt{\sin x}+\sqrt{\cos x}} dx$

$= \int_0^{\pi/2} \frac{\sqrt{\sin x}+\sqrt{\cos x}}{\sqrt{\sin x}+\sqrt{\cos x}}.dx = \int_0^{\pi/2} 1.dx = [x]_0^{\pi/2} = \frac{\pi}{2} - 0$

$2I = \frac{\pi}{2} \Rightarrow I = \frac{\pi}{4}$

3. $\int_{\pi/6}^{\pi/3} \sin 2x \, ln(\tan x) \, dx$

Ans. 0

Sol. $I = \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \sin 2x \ell n(\tan x)\,dx$

Apply KING and add

$2I = \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \sin 2x \ell n(\tan x) + \sin 2x \ell n(\cot x)\,dx$

$\Rightarrow 2I = 0 \Rightarrow I = 0$

4. $\int_{50}^{100} \frac{\ell n\, x}{\ell n\, x + \ell n(150 - x)}\,dx$

Ans. 25

Sol. $I = \int_{50}^{100} \frac{\ell n x}{\ell n x + \ell n(150 - x)}$

Apply KING and add

$2I = \int_{50}^{100} 1 \cdot dx \Rightarrow I = 25$

5. $\int_{0}^{\pi/2} \frac{\sin^3 x}{\sin x + \cos x}\,dx$

Ans. $\frac{\pi - 1}{4}$

Sol. $I = \int_{0}^{\frac{\pi}{2}} \frac{\sin^3 x}{\sin x + \cos x}\,dx$

Apply KING and add

$2I = \int_{0}^{\frac{\pi}{2}} \frac{\sin^3 x + \cos^3 x}{\sin x + \cos x}\,dx$

$\Rightarrow 2I = \int_{0}^{\frac{\pi}{2}} 1 - \sin x \cos x\,dx$

$= \int_{0}^{\frac{\pi}{2}} \left(1 - \frac{\sin 2x}{2}\right)dx$

$= \frac{\pi - 1}{4}$

6. $\int_2^3 \frac{\sqrt{x}}{\sqrt{5-x}+\sqrt{x}}dx$ **[JEE - 94]**

Ans. $\frac{1}{2}$

Sol. $I=\int_2^3 \frac{\sqrt{x}}{\sqrt{5-x}+\sqrt{x}}dx$

KING and add

$2I=\int_2^3 1\cdot dx$

$I=\frac{1}{2}$

7. $\int_{\pi/8}^{3\pi/8} \ell n\left(\frac{4+3\sin x}{4+3\cos x}\right)dx$

Ans. 0

Sol. $I=\int_{\frac{\pi}{8}}^{\frac{3\pi}{8}} \ell n\left(\frac{4+3\sin x}{4+3\cos x}\right)dx$

KING and add

$2I=\int_{\frac{\pi}{8}}^{\frac{3\pi}{8}} \ell n\left(\frac{4+3\sin x}{4+\cos x}\right)+\ell n\left(\frac{4+3\cos x}{4+3\sin x}\right)dx$

$I=0$

8. $\int_0^{\pi/4} ln\,(1+\tan x)\,dx$

Ans. $\frac{\pi}{8}\,ln\,2$

Sol. $I=\int_0^{\frac{\pi}{4}} \ell n\left(1+\tan x\right)dx$

KING and add

$2I=\int_0^{\frac{\pi}{4}} \ell n\left(1+\tan x\right)+\ell n\left(1+\frac{1-\tan x}{1+\tan x}\right)dx$

$2I=\int_0^{\frac{\pi}{4}} \ell n\left(1+\tan x\right)+\ell n\left(\frac{2}{1+\tan x}\right)dx$

$$2I = \int_0^{\frac{\pi}{4}} \ell n2 \Rightarrow I = \frac{\pi}{8}\ell n2$$

9. $\int_{-\pi/2}^{\pi/2} \left(\frac{1}{(2007)^x + 1}\right) \cdot \frac{\sin^{2008} x}{\sin^{2008} x + \cos^{2008} x} dx$

Ans. $\frac{\pi}{4}$

Sol. $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \left(\frac{1}{(2007)^x + 1}\right) \frac{(\sin x)^{2008}}{(\sin x)^{2008} + (\cos x)^{2008}} dx$

KING and add

$$2I = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{(\sin x)^{2008}}{(\sin x)^{2008} + (\cos x)^{2008}} \left(\frac{1}{(2007)^x + 1} + \frac{1}{(2007)^{-x} + 1}\right) dx$$

$$2I = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{(\sin x)^{2008}}{(\sin x)^{2008} + (\cos x)^{2008}} dx$$

$$\Rightarrow 2I = 2\int_0^{\frac{\pi}{2}} \frac{(\sin x)^{2008}}{(\sin x)^{2008} + (\cos x)^{2008}} dx$$

KING/add.

$$I = \frac{\pi}{4}$$

10. $\int_{-\pi/4}^{\pi/4} \frac{\tan^2 x}{1 + e^x} dx$

Ans. $\left(1 - \frac{\pi}{4}\right)$

Sol. Apply king and add

11. $\int_0^{\pi} \frac{dx}{1 + 2^{\tan x}}$

Ans. $\pi/2$

Sol. $I = \int_0^{\pi} \frac{dx}{1 + 2^{\tan x}}$

KING and add

$$2I = \int_0^{\pi} \frac{dx}{1+2^{\tan x}} + \int_0^{\pi} \frac{dx}{1+2^{-\tan x}} dx$$

$$I = \frac{\pi}{2}$$

12. $\int_0^{\pi/2} \frac{\sqrt{\cot x}}{\sqrt{\cot x}+\sqrt{\tan x}} dx$ **[JEE - 83]**

Ans. $\frac{\pi}{4}$

Sol. $I = \int_0^{\frac{\pi}{2}} \frac{\sqrt{\cot x}}{\sqrt{\tan x}+\sqrt{\cot x}} dx$

KING/add

$$2I = \int_0^{\frac{\pi}{2}} 1 \cdot dx \Rightarrow I = \frac{\pi}{4}$$

13. $\int_0^1 \frac{ln(1+x)}{1+x^2} dx$

Ans. $\frac{\pi}{8} \cdot ln\, 2$

Sol. $I = \int_0^1 \frac{\ell n(1+x)}{1+x^2} dx$

Put $x = \tan\theta$

$$I = \int_0^{\frac{\pi}{4}} \frac{\ell n(1+\tan\theta)\sec^2\theta d\theta}{\sec^2\theta}$$

$$I = \int_0^{\frac{\pi}{4}} \ell n(1+\tan\theta) d\theta$$

$$= \frac{\pi}{8}\ell n2$$

14. $\int_2^3 \frac{x^2\ dx}{2x^2 - 10x + 25}$

Ans. $\frac{1}{2}$

Sol. $\int_2^3 \frac{x^2}{(x-5)^2+x^2}dx$ Apply king and add

15. $\int_0^1 \ln\left(\frac{1}{x}-1\right)dx$

Ans. 0

Sol. $I = \int_0^1 \log\left(\frac{1-x}{x}\right)dx$ ...(1)

$\Rightarrow I = \int_0^1 \log\left[\frac{1-(1-x)}{1-x}\right]dx$ (by P – 4)

$= \int_0^1 \log\left(\frac{x}{1-x}\right)dx = -\int_0^1 \log\left(\frac{1-x}{x}\right)dx$

$= -I$

$\Rightarrow 2I = 0 \quad \Rightarrow I = 0$

16. $\int_0^{\pi} \frac{\sin 8x}{\sin x}dx$

Ans. 0

Sol. $I = \int_0^{\pi} \frac{\sin 8x}{\sin x}dx$

KING/add

$2I = \int_0^{\pi} \frac{\sin 8x}{\sin x} - \frac{\sin 8x}{\sin x}dx$

$\Rightarrow I = 0$

17. $\int_{-\pi/2}^{\pi/2} \sqrt{\cos x - \cos^3 x}\,dx$

Ans. $\frac{4}{3}$

Sol. $I = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sqrt{\cos x - \cos^3 x}dx$

$\Rightarrow I = 2\int_0^{\frac{\pi}{2}} \sqrt{\cos x}\,|\sin x|\,dx$

$$\Rightarrow I = 2\int_0^{\frac{\pi}{2}} \sqrt{\cos x}\,\sin x\,dx$$

$$\Rightarrow I = 2(-1)\frac{(\cos x)^{\frac{3}{2}}}{\frac{3}{2}}\Bigg|_0^{\frac{\pi}{2}}$$

$$= -\frac{4}{3}(0-1) = \frac{4}{3}$$

18. $\displaystyle\int_0^{\pi/2n} \frac{dx}{1+\tan^n(nx)}$

Ans. $\dfrac{\pi}{4n}$

Sol. $\displaystyle\int_0^{\frac{\pi}{2n}} \frac{dx}{1+\tan^n(nx)}dx$

Put $nx = y$

$$\Rightarrow I = \int_0^{\frac{\pi}{2}} \frac{dx}{1+(\tan y)^n}\frac{dy}{n}$$

$$\Rightarrow I = \int_0^{\frac{\pi}{2}} \frac{(\cos y)^n}{(\sin y)^n + (\cos y)^n}\frac{dy}{n}$$

Apply KING/add

$$2I = \int_0^{\frac{\pi}{2}} 1\cdot dy \Rightarrow I = \frac{\pi}{4n}$$

19. $\displaystyle\int_0^1 \tan^{-1}\left(\frac{2x-1}{1+x-x^2}\right)dx$

Ans. 0

Sol. $\displaystyle I = \int_0^1 \tan^{-1}\left(\frac{2x-1}{1+x-x^2}\right)dx$

$$= \int_0^1 \tan^{-1}\left(\frac{x-(1-x)}{1+x(1-x)}\right)dx$$

$$I = \int_0^1 \tan^{-1} x\,dx - \int_0^1 \tan^{-1}(1-x)\,dx$$

$$I = \int_0^1 \tan^{-1} x - \tan^{-1}(1-x)\,dx$$

KING and add. on $\tan^{-1}(1-x)$

$I = 0$

20. $\int_0^{\pi/2} \frac{\sin 8x \cdot \ell n(\cot x)}{\cos 2x}\,dx$

Ans. 0

Sol. $I = \int_0^{\frac{\pi}{2}} \frac{\sin 8x \ell n(\cot x)}{\cos 2x}\,dx$

KING and add

$$2I = \int_0^{\frac{\pi}{2}} \frac{\sin 8x}{\cos 2x} \ell n(\cot x) + \frac{(-\sin 8x)}{(-\cos 2x)} \ell n(\tan x)\,dx$$

$\Rightarrow I = 0$

21. $\int_{\pi/2}^{3\pi/2} [2\sin x]\,dx$ is **[JEE 99]**

Ans. $-\frac{\pi}{2}$

Sol. $I = \int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} [2\sin x]dx$

KING/add

$$2I = \int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} [2\sin x]` + [-2\sin x]dx$$

$2I = (-1)(\pi) \Rightarrow I = -\frac{\pi}{2}$

22. $\int_0^{\pi} \frac{e^{\cos x}}{e^{\cos x} + e^{-\cos x}}\,dx$

Ans. $\frac{\pi}{2}$

Sol. $I = \int_0^{\pi} \frac{e^{\cos x}}{e^{\cos x} + e^{-\cos x}}\,dx$

KING/add

$$2I = \int_0^{\pi} \frac{e^{\cos x}}{e^{\cos x} + e^{-\cos x}} + \frac{e^{-\cos x}}{e^{\cos x} + e^{-\cos x}} dx$$

$$I = \frac{\pi}{2}$$

23. $\int_{\pi/4}^{3\pi/4} \frac{x \sin x}{1 + \sin x} \, dx$

Ans. $\pi[\frac{\pi}{4} - (\sqrt{2} - 1)]$

Sol. $I = \int_{\frac{\pi}{4}}^{\frac{3\pi}{4}} \frac{x \sin x}{1 + \sin x} dx$

KING/add

$$2I = \int_{\frac{\pi}{4}}^{\frac{3\pi}{4}} \frac{x \sin x}{1 + x \sin x} + \frac{(\pi - x)\sin x}{1 + \sin x} dx$$

$$\Rightarrow 2I = \pi \int_{\frac{\pi}{4}}^{\frac{3\pi}{4}} \left(1 - \frac{1}{1 + \sin x}\right) dx$$

$$\Rightarrow 2I = \pi \int_{\frac{\pi}{4}}^{\frac{3\pi}{4}} \left(1 - \frac{(1 - \sin x)}{\cos^2 x}\right) dx$$

$$\Rightarrow 2I = \pi \int_{\frac{\pi}{4}}^{\frac{3\pi}{4}} \left(1 - \sec^2 x + \sec x \tan x\right) dx$$

$$\Rightarrow 2I = \pi \left(x - \tan x + \sec x\right)\Big|_{\frac{\pi}{4}}^{\frac{3\pi}{4}}$$ Now break into half angle

$$2I = \pi \left(x + \tan\left(\frac{\pi}{4} - \frac{x}{2}\right)\right)\Big|_{\frac{\pi}{4}}^{\frac{3\pi}{4}}$$

$$I = \pi\left(\frac{\pi}{4} + 1 - \sqrt{2}\right)$$

24. $\int_0^2 \frac{dx}{\left(17+8x-4x^2\right)\left(e^{6(1-x)}+1\right)}$

Ans. $\frac{1}{4\sqrt{2}}\, ln\left|\frac{2+\sqrt{21}}{2-\sqrt{21}}\right|$

Sol. Let $I=\int_0^2 \frac{dx}{(17+8x-4x^2)[e^{6(1-x)}+1]}$

Also $I=\int_0^2 \frac{dx}{(17+8x-4x^2)[e^{-6(1-x)}+1]}$ $\left[\because \int_0^a f(x)dx=\int_0^a f(a-x)dx\right]$

Adding, we get

$$2I=\int_0^2 \frac{1}{17+8x-4x^2}\left(\frac{1}{e^{6(1-x)}+1}+\frac{1}{e^{-6(1-x)}+1}\right)dx$$

$$=\int_0^2 \frac{1}{17+8x-4x^2}dx=-\frac{1}{4}\int_0^2 \frac{dx}{x^2-2x-17/4}$$

$$=-\frac{1}{4}\int_0^2 \frac{dx}{(x-1)^2-21/4}=-\frac{1}{4}\times\frac{1}{2\times\frac{\sqrt{21}}{2}}\left[\log\left|\frac{x-1-\frac{\sqrt{21}}{2}}{x-1+\frac{\sqrt{21}}{2}}\right|\right]_0^2$$

$$=-\frac{1}{4\sqrt{21}}\left[\log\left|\frac{2x-2-\sqrt{21}}{2x-2+\sqrt{21}}\right|\right]_0^2 \quad \Rightarrow \quad I=-\frac{1}{8\sqrt{21}}\left[\log\left|\frac{2-\sqrt{21}}{2+\sqrt{21}}\right|-\log\left|\frac{2+\sqrt{21}}{\sqrt{21}-2}\right|\right]$$

$$=-\frac{1}{4\sqrt{21}}\left[\log\left|\frac{\sqrt{21}-2}{2+\sqrt{21}}\right|\right]$$

25. $\int_0^{\pi/2} \sqrt{\sin 2\theta}\,\sin\theta\, d\theta$

Ans. $\frac{\pi}{4}$

Sol. $I=\int_0^{\frac{\pi}{2}} \sqrt{\sin 2\theta}\,\sin\theta d\theta$

KING/add

$$2I=\int_0^{\frac{\pi}{2}} \sqrt{\sin 2\theta}\left(\sin\theta+\cos\theta\right)d\theta$$

$$2I=\int_0^{\frac{\pi}{2}} \sqrt{1-(s-c)^2}\left(\sin\theta+\cos\theta\right)d\theta$$

Let $\sin\theta-\cos\theta=y$

$$2I = \int_{-1}^{1} \sqrt{1-y^2}\,dy$$

$$\Rightarrow I = \int_{0}^{1} \sqrt{1-y^2}\,dy = \frac{\pi}{4}$$

26. $\int_{0}^{\pi/2} \frac{x \sin x \cos x}{\sin^4 x + \cos^4 x}\,dx$

Ans. $\frac{\pi^2}{16}$

Sol. $I = \int_{0}^{\frac{\pi}{2}} \frac{x \sin x \cos x}{(\sin x)^4 + (\cos x)^4}\,dx$

KING/add

$$2I = \frac{\pi}{2} \int_{0}^{\frac{\pi}{2}} \frac{\frac{\sin 2x}{2}\,dx}{1 - 2\sin^2 x \cos^2 x}$$

$$\Rightarrow I = \frac{\pi}{8} \int_{0}^{\frac{\pi}{2}} \frac{\sin 2x\,dx}{1 - \frac{\sin^2 2x}{2}}$$

$$\Rightarrow I = \frac{\pi}{4} \int_{0}^{\frac{\pi}{2}} \frac{\sin 2x}{1 + \cos^2 2x}\,dx$$

$$I = -\frac{\pi}{4} \cdot \tan^{-1}\left(\frac{\cos 2x}{1}\right) \cdot \frac{1}{2}\Bigg|_{0}^{\frac{\pi}{2}}$$

$$I = -\frac{\pi}{8}\left(-\frac{\pi}{4} - \frac{\pi}{4}\right) = \frac{\pi^2}{16}$$

27. $\int_{0}^{\pi/4} \frac{x\,dx}{1 + \cos 2x + \sin 2x}$

Ans. $\frac{\pi}{16} \ln 2$

Sol. $I = \int_{0}^{\frac{\pi}{4}} \frac{x\,dx}{1 + \cos 2x + \sin 2x}$

KING/add

$$2I = \frac{\pi}{4}\int_0^{\frac{\pi}{4}} \frac{dx}{1+\frac{1-\tan^2 x}{1+\tan^2 x}+\frac{2\tan x}{1+\tan^2 x}}$$

$$I = \frac{\pi}{8}\int_0^{\frac{\pi}{4}} \frac{\sec^2 x dx}{2(1+\tan x)} = \frac{\pi}{16}\ell n\left|(1+\tan x)\right|_0^{\frac{\pi}{4}}$$

$$= \frac{\pi}{16}\ell n 2$$

28. $$\int_{-1/\sqrt{3}}^{1/\sqrt{3}} \frac{x^4}{1-x^4}\cos^{-1}\left(\frac{2x}{1+x^2}\right)dx$$

Ans. $$\frac{\pi^2}{12}-\frac{\pi}{\sqrt{3}}+\frac{\pi}{4}ln\left(2+\sqrt{3}\right)]$$

Sol. Use King and $\cos^{-1}(-x) = \pi - \cos^{-1}x$ and add & use P-5

29. Prove that: $\int_0^{\pi/2} f(\sin 2x)\sin x\,dx = \sqrt{2}\int_0^{\pi/4} f(\cos 2x)\cos x\,dx$

Ans.

Sol. $$I = \int_0^{\frac{\pi}{2}} f(\sin 2x)\sin x dx$$

KING/add

$$2I = \int_0^{\frac{\pi}{2}} f(\sin 2x)(\sin x+\cos x)dx$$

$$2I = \sqrt{2}\int_0^{\frac{\pi}{2}} f(\sin 2x)\cos\left(x-\frac{\pi}{4}\right)dx$$

Put $x = \frac{\pi}{4}+y$

$$= \sqrt{2}\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} f(\cos 2y)\cos y dy$$

Using even/odd property

$$I = \sqrt{2}\int_0^{\frac{\pi}{4}} f(\cos 2y)\cos y dy$$

30. $\int_{a/2}^{\sqrt{3}a/2} \frac{dx}{x+\sqrt{a^2-x^2}}\ (a>0)$

Ans. $\frac{\pi}{12}$

Sol. $I=\int_{\frac{a}{2}}^{\frac{\sqrt{3}a}{2}} \frac{dx}{x+\sqrt{a^2-x^2}}$

Put $x = a\sin\theta$

$$I=\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{a\cos\theta d\theta}{a(\sin\theta+\cos\theta)}$$

KING and add

$$2I=\left(\frac{\pi}{3}-\frac{\pi}{6}\right)\Rightarrow I=\frac{\pi}{12}$$

31. $\int_0^{\pi/2} \frac{\sin^2 x}{\sin x+\cos x}dx$

Ans. $\frac{1}{\sqrt{2}}ln(\sqrt{2}+1)$

Sol. $I=\int_0^{\frac{\pi}{2}} \frac{\sin^2 x}{\sin x+\cos x}dx$

KING/add

$$2I=\int_0^{\frac{\pi}{2}} \frac{dx}{\sin x+\cos x}$$

$$2I=\frac{1}{\sqrt{2}}\int_0^{\frac{\pi}{2}} \sec\left(x-\frac{\pi}{4}\right)dx$$

$$I=\frac{1}{2\sqrt{2}}\ell n\left|\sec\left(x-\frac{\pi}{4}\right)+\tan\left(x-\frac{\pi}{4}\right)\right|_0^{\frac{\pi}{2}}$$

$$I=\frac{1}{\sqrt{2}}\left(\ell n\left(\sqrt{2}+1\right)\right)$$

32. $\int_0^{\pi/2} \frac{\sin x - \cos x}{1+\sin x \cos x} dx$

Ans. 0

Sol. $I = \int_0^{\frac{\pi}{2}} \frac{\sin x - \cos x}{1+\sin x \cos x} dx$

KING and add

$2I = 0 \Rightarrow I = 0$

33. $\int_0^{\pi/2} \frac{\sin^2 x}{1+\sin x \cos x} dx$

Ans. $\frac{\pi}{3\sqrt{3}}$

Sol. KING and add

$$2I = \int_0^{\frac{\pi}{2}} \frac{1}{1+\sin x \cos x} dx$$

$$2I = \int_0^{\frac{\pi}{2}} \frac{1}{1+\frac{\tan x}{1+\tan^2 x}} dx$$

$$2I = \int_0^{\frac{\pi}{2}} \frac{\sec^2 x dx}{1+\tan^2 x+\tan x}$$

Put $\tan x = y$

$$I = \frac{1}{2}\int_0^{\infty} \frac{dy}{\left(y+\frac{1}{2}\right)^2+\left(\frac{\sqrt{3}}{2}\right)^2}$$

$$\Rightarrow I = \frac{1}{2}\cdot\frac{1}{\frac{\sqrt{3}}{2}} \tan^{-1}\left(\frac{y+\frac{1}{2}}{\frac{\sqrt{3}}{2}}\right)\Bigg|_0^{\infty}$$

$$= \frac{1}{\sqrt{3}}\left(\frac{\pi}{2}-\frac{\pi}{6}\right) = \frac{\pi}{3\sqrt{3}}$$

34. $\int_0^1 \cot^{-1}(1 - x + x^2)\, dx$

Ans. $\frac{\pi}{2} - l\text{n}\,2$

Sol. $I = \int_0^1 \tan^{-1}\left(\frac{1}{1-x+x^2}\right)dx = \int_0^1 \tan^{-1}\left(\frac{x+(1-x)}{1-x(1-x)}\right)dx$

$= \int_0^1 [\tan^{-1}x + \tan^{-1}(1-x)]\ dx = \int_0^1 \tan^{-1}x\ dx + \int_0^1 \tan^{-1}(1-x)\ dx$

$= 2\int_0^1 \tan^{-1}x\ dx = 2\left[x\ \tan^{-1}x - \frac{1}{2}\log(1+x^2)\right]_0^1 = 2\frac{\pi}{4} - \log 2 = \frac{\pi}{2} - \log 2$

35. $\int_{-\pi}^{\pi} (\cos px - \sin qx)^2\,dx$, p, q ∈ I

Ans. 2π

Sol. $I = \int_{-\pi}^{\pi} (\cos px - \sin 2x)^2\,dx$

$I = \int_0^{\pi} (\cos px - \sin 2x)^2 + (\cos px + \sin 2x)^2\,dx$

$I = 2\int_0^{\pi} \cos^2 px + \sin^2 2x\,dx$

$I = \int_0^{\pi} (1 + \cos 2px + 1 - \cos 2qx)\,dx$

$I = 2\pi$ as p, q ∈ Integer

36. Prove that $\int_0^{\pi} x\ f(\sin x)\ dx = \frac{\pi}{2}\int_0^{\pi} f(\sin x)dx$. **[JEE 82]**

Ans.

Sol. $I = \int_0^{\pi} xf(\sin x)\,dx$

KING/add

$2I = \int_0^{\pi} xf(\sin x) + (\pi - x)f(\sin x)\,dx$

$I = \frac{\pi}{2}\int_0^{\pi} f(\sin x)\,dx$

37. $\int_{-2}^{2} \left(x^3 f(x) + x \cdot f''(x) + 2\right)dx$, where $f(x)$ is an even differentiable function

Ans. 8

Sol. $I = \int_{-2}^{2} \left(x^3 f(x) + x f''(x) + 2\right)$

Given $f(x) = f(-x) \rightarrow f'(x) = f'(-x)$

Apply KING and add

$$2I = \int_{-2}^{2} x^3 f(x) + x f''(x) + 2 + (-x)^3 f(x) + (-x) f''(x) + 2 dx$$

$$I = \frac{16}{2} = 8$$

38. Let f(x) be a differentiable function defined on [0, 2] such that $f'(x) = f'(2-x)$ for all $\forall$ x Î (0, 2), f(0) = 1 and $f(2) = e^2$. Then the value of $\int_0^2 f(x)\,dx$ is : **[JEE(Main)Feb.2021]**

माना [0, 2] में f(x) एक अवकलनीय फलन है, जिसके लिए $f'(x) = f'(2-x)$ $\forall$ $x \in (0, 2)$, f(0) = 1 तथा $f(2) = e^2$. हैं। तो $\int_0^2 f(x)$ का मान है।

Ans. $1 + e^2$

Sol. $f'(x) = f'(2-x)$

$f'(x) - f'(2-x) = 0$

Integrate w.r.t. x

$f(x) + f(2-x) = C$

$\because$ $f(0) = 1$, $f(2) = e^2 \Rightarrow C = e^2 + 1$

$$I = \int_0^2 f(x)\,dx$$

$$I = \int_0^2 f(2-x)\,dx \qquad \left(\because \int_a^b f(x)\,dx = \int_a^b f(a+b-x)\,dx\right)$$

Add

$$2I = \int_0^2 (f(x) + f(2-x))dx$$

$$2I = \int_0^2 (e^2 + 1)\,dx$$

$\Rightarrow I = e^2 + 1$

2.6 PROPERTIES OF DEFINITE INTEGRAL - P6 (QUEEN PROPERTY)

$$\int_0^{2a} f(x)dx = \int_0^{a} f(x)dx + \int_0^{a} f(2a-x)dx = \begin{cases} 2\int_0^{a} f(x)dx \ ; \ \text{if} \ f(2a-x) = f(x) \\ 0 \quad ; \ \text{if} \ f(2a-x) = -f(x) \end{cases}$$

Proof:

$$\text{LHS} = \int_0^{a} f(x)\cdot dx + \int_a^{2a} f(x)\cdot dx$$

Put $x = 2a - t$

Ex.1 $I = \int_0^{\pi} \sin^4 x \cdot dx$ Ex.2 $I = \int_0^{4\pi} x \sin^4 x \cos^6 x \cdot dx$

Sol. Ex.1

$$I = \int_0^{\pi} (\sin x)^4 \, dx$$

$$\Rightarrow I = 2\int_0^{\frac{\pi}{2}} (\sin x)^4 \, dx$$

$$= 2\cdot\frac{3\pi}{16} = \frac{3\pi}{8}$$

Sol. Ex. 2

$$I = \int_0^{4\pi} x(\sin x)^4 (\cos x)^6 \, dx$$

Apply queen

$$I = \int_0^{2\pi} xs^4c^6 + (4\pi - x)s^4c^6 dx$$

$$I = \int_0^{2\pi} 4\pi(s^4c^6)\, dx$$

Apply queen again

$$I = 8\pi\int_0^{\pi} (\sin x)^4 (\cos x)^6 \, dx$$

Apply queen again

$$I = 16\pi\int_0^{\frac{\pi}{2}} (\sin x)^4 (\cos x)^6 \, dx$$

Apply KING/add

$$2I = 16\pi\int_0^{\frac{\pi}{2}} s^4c^6 + c^4s^6 dx$$

$$\Rightarrow I = 8\pi\int_0^{\frac{\pi}{2}} s^4c^4 dx$$

$$\Rightarrow I = \frac{8\pi}{16}\int_0^{\frac{\pi}{2}} (\sin 2x)^4 \, dx$$

Put $2x = y$

$$\Rightarrow I = \frac{\pi}{2}\int_0^{\pi} (\sin y)^4 \frac{dy}{2}$$

$$I = \frac{\pi}{4}\int_0^{\pi} (\sin y)^4$$

Apply queen

$$I = \frac{\pi}{2}\int_0^{\frac{\pi}{2}} (\sin y)^4 = \frac{\pi}{2}\cdot\frac{3\pi}{16} = \frac{3\pi^2}{32}$$

Note :

1. This property is helpful when we need to reduce the upper limit.
2. To apply this property lower limit should be zero.

Questions

1. $\int_0^{2\pi} \sin x \, dx$

Ans. 0

Sol. $I = \int_0^{2\pi} (\sin x) dx = \int_0^{\pi} \sin x + \sin(2\pi - x) dx = 0$

2. $\int_0^{2\pi} \sin^5(x)\cos^4(x) dx$

Ans. 0

Sol. $I = \int_0^{2\pi} (\sin x)^5 (\cos x)^4 \, dx = \int_0^{\pi} (\sin x)^5 (\cos x)^4 + (-\sin x)^5 (\cos x)^4 \, dx$

$= 0$

3. $\int_0^{\pi} \frac{dx}{1 + 2\sin^2 x}$

Ans. $\frac{\pi}{\sqrt{3}}$

Sol. Let $f(x) = \frac{1}{1+2\sin^2 x} \Rightarrow f(\pi - x) = f(x) \Rightarrow \int_0^{\pi} \frac{dx}{1+2\sin^2 x}$

$= 2\int_0^{\frac{\pi}{2}} \frac{dx}{1+2\sin^2 x} = 2\int_0^{\frac{\pi}{2}} \frac{\sec^2 x \, dx}{1+\tan^2 x + 2\tan^2 x} = 2\int_0^{\frac{\pi}{2}} \frac{\sec^2 x \, dx}{1+3\tan^2 x} = \frac{2}{\sqrt{3}} \left[\tan^{-1}\left(\sqrt{3}\tan x\right)\right]_0^{\frac{\pi}{2}}$

$\because \tan\frac{\pi}{2}$ is undefined, we take limit

$= \frac{2}{\sqrt{3}} \left[\lim_{x \to \frac{\pi}{2}^-} \tan^{-1}\left(\sqrt{3}\tan x\right) - \tan^{-1}\left(\sqrt{3}\tan 0\right) \right] = \frac{2}{\sqrt{3}} \frac{\pi}{2} = \frac{\pi}{\sqrt{3}}$

4. $\int_0^{\pi} \frac{\sin 3x}{\sin 4x} dx$

Ans. 0

Sol. $I = \int_0^{\pi} \frac{\sin 3x}{\sin 4x} dx = \int_0^{\frac{\pi}{2}} \frac{\sin 3x}{\sin 4x} + \frac{\sin(3\pi - 3x)}{\sin(4\pi - 4x)} dx = 0$

5. $\int_0^{2\pi} \sin^4 x \, dx$

Ans. $\frac{3\pi}{4}$

Sol. $I = \int_0^{2\pi} (\sin x)^4 \, dx = \int_0^{\pi} 2(\sin x)^4 \, dx = 4\int_0^{\frac{\pi}{2}} (\sin x)^4 \, dx = \frac{3\pi}{4}$

6. $\int_0^{2\pi} \cos^5 x \, dx$

Ans. 0

Sol. $I = \int_0^{2\pi} (\cos x)^5 \, dx = 2\int_0^{\pi} (\cos x)^5 \, dx$

Apply KING and add

$$2I = 2\int_0^{\pi} (\cos x)^5 + (\cos x)^5 \, dx = 0$$

7. $\int_0^{2\pi} \frac{x(\sin x)^{2n}}{(\sin x)^{2n} + (\cos x)^{2n}} dx$, $n \in N$

Ans. π^2

Sol. $I = \int_0^{2\pi} \frac{x(\sin x)^{2n}}{(\sin x)^{2n} + (\cos x)^{2n}} dx$

Queen

$$I = \int_0^{\pi} \frac{x(\sin x)^{2n}}{(\sin x)^{2n} + (\cos x)^{2n}} + \frac{(2\pi - x)(\sin x)^n}{(\sin x)^{2n} + (\cos x)^{2n}} dx$$

$$I = 2\pi \int_0^{\pi} \frac{(s)^{2n}}{s^{2n} + c^{2n}} dx$$

Applyn Queen

$$I = 4\pi \int \frac{(\sin x)^{2n}}{(\sin x)^{2n} + (\cos x)^{2n}} dx$$

Apply KING

$$I = 2\pi \cdot \frac{\pi}{2} = \pi^2$$

8. $\int_0^{\pi} \sin^3 x \cos^3 x \, dx$

Ans. 0

Sol. $I = \int_0^{\pi} (\sin x)^3 (\cos x)^3 \, dx$

Apply Queen

$$I = \int_0^{\frac{\pi}{2}} (\sin x)^3 (\cos x)^3 + (\sin x)^3 (-\cos x)^3 \, dx = 0$$

9. $\int_0^{\pi} x\left(\sin^2(\sin x) + \cos^2(\cos x)\right) dx$

Ans. $\frac{\pi^2}{2}$

Sol. $I = \int_0^{\pi} x\left(\sin^2(\sin x) + \cos^2(\cos x)\right)dx$; $I = \int_0^{\pi} (\pi - x)\left(\sin^2(\sin x) + \cos^2(\cos x)\right)dx$

$2I = \pi \int_0^{\pi} \left(\sin^2(\sin x) + \cos^2(\cos x)\right)dx$;

$2I = 2\pi \int_0^{\pi/2} \left(\sin^2(\sin x) + \cos^2(\cos x)\right)dx$ (Using Queen)

also $I = \pi \int_0^{\pi/2} \left(\sin^2(\cos x) + \cos^2(\sin x)\right)dx$;

use king and add, $2I = \pi \int_0^{\pi/2} 2\, dx \Rightarrow I = \pi \int_0^{\pi/2} dx = \frac{\pi^2}{2}$ Ans.

10. $\int_0^{\pi} x\left(\sin(\cos^2 x)\cos(\sin^2 x)\right)dx$

Ans. $\frac{\pi^2}{4}(\sin 1)$

Sol. $I = \int_0^{\pi} x\left(\sin(\cos^2 x)\cos(\sin^2 x)\right)dx$; $I = \int_0^{\pi} (\pi - x)\sin(\cos^2 x)\cos(\sin^2 x)\,dx$

$2I = \pi \int_0^{\pi} \sin(\cos^2 x)\cos(\sin^2 x)\,dx$; $2I = 2\pi \int_0^{\pi/2} \sin(\cos^2 x)\cos(\sin^2 x)\,dx$

$I = \pi \int_0^{\pi/2} \sin(\cos^2 x)\cos(\sin^2 x)\,dx$ ...(1);

$I = \pi \int_0^{\pi/2} \sin(\sin^2 x)\cos(\cos^2 x)dx$ ..(2) (King)

add (1) and (2) (sin A cos B + cos A sin B)

$I = \frac{\pi}{2} \int_0^{\pi/2} \sin(\cos^2 x + \sin^2 x)dx$ $2I = \pi \int_0^{\pi/2} (\sin 1)dx \Rightarrow I = \frac{\pi^2}{4}(\sin 1)$

11. $\int_0^{2\pi} x \sin^4 x \cos^6 x \, dx$

Ans. $\frac{3\pi^2}{128}$

Sol. Ex 2 on Page 18

12. $\int_0^{\pi/2} \ln(\sin x)dx = -\frac{\pi}{2} \ln 2$

Ans. $-\frac{\pi}{2} \ln 2$

Sol. Let $I = \int_0^{\frac{\pi}{2}} \ell n \sin x \, dx$ .....(i)

$\Rightarrow I = \int_0^{\frac{\pi}{2}} \ell n\left(\sin\left(\frac{\pi}{2} - x\right)\right) dx$ (by property P – 5)

$I = \int_0^{\frac{\pi}{2}} \ell n (\cos x) \, dx$ .....(ii)

Adding (i) and (ii)

$2I = \int_0^{\frac{\pi}{2}} \ell n (\sin x \cdot \cos x) \, dx = \int_0^{\frac{\pi}{2}} \ell n\left(\frac{\sin 2x}{2}\right) dx$

$2I = \int_0^{\frac{\pi}{2}} \ell n (\sin 2x) \, dx - \int_0^{\frac{\pi}{2}} \ell n 2 \, dx$

$2I = I_1 - \frac{\pi}{2} \ell n 2$

where $I_1 = \int_0^{\frac{\pi}{2}} \ell n (\sin 2x) \, dx$

put $2x = t$ $\Rightarrow$ $dx = \frac{1}{2} dt$

L . L : $x = 0$ $\Rightarrow$ $t = 0$

U . L : $x = \frac{\pi}{2}$ $\Rightarrow$ $t = \pi$

$\Rightarrow I_1 = \int_0^{\pi} \ell n (\sin t) \cdot \frac{1}{2} dt = \frac{1}{2} \times 2 \int_0^{\frac{\pi}{2}} \ell n (\sin t) \, dt$ (by using property P – 6)

$\Rightarrow I_1 = I$

$\therefore$ (iii) gives $I = -\frac{\pi}{2} \ell n 2$

13. $\int_0^1 \ln \sin\left(\frac{\pi x}{2}\right) dx$

Ans. $-l\text{n}\,2$

Sol. $I = \int_0^1 \ell n\left(\sin\frac{\pi x}{2}\right)dx$

Put $\frac{\pi x}{2} = y$

$I = \frac{2}{\pi}\int_0^{\pi} \ell n(\sin y)\,dy = -\ell n2$

14. $\int_0^{\pi} x \ln(\sin x)dx$

Ans. $-\frac{\pi^2}{2}\cdot l\text{n}2$

Sol. $I = \int_0^{\pi} x\ell n(\sin x)\,dx$

Queen Apply

$I = \int_0^{\frac{\pi}{2}} x\ell n(\sin x) + (\pi - x)\ell n(\sin x)\,dx$

$= \pi\left(-\frac{\pi}{2}\ell n2\right) = -\frac{\pi^2}{2}\ell n2$

15. $\int_0^{\pi} \ln(1-\cos x)dx$

Ans. $-\pi\, l\text{n}\,2$

Sol. $I = \int_0^{\pi} \ell n(1-\cos x)\,dx$

Apply Queen

$I = \int_0^{\frac{\pi}{2}} \ell n(1-\cos x) + \ell n(1+\cos x)\,dx$

$I = 2\int_0^{\frac{\pi}{2}} \ell n(\sin x)\,dx = -\pi\ell n2$

16. $\int_0^1 \frac{\sin^{-1} x}{x} dx$

Ans. $\frac{\pi}{2} \cdot l\text{n}2$

Sol. $I = \int_0^1 \frac{\sin^{-1} x}{x} dx$

Put $\sin^{-1}x = \theta$

$$I = \int_0^{\frac{\pi}{2}} \frac{\theta \cos\theta d\theta}{\sin\theta} = \int_0^{\frac{\pi}{2}} \theta \cot\theta d\theta$$

$$\Rightarrow I = \theta \ell n|\sin\theta|_0^{\frac{\pi}{2}} - \int_0^{\frac{\pi}{2}} \ell n|\sin\theta| d\theta$$

$$\Rightarrow I = 0 - \lim_{\theta\to 0^+} (\ell n \sin\theta)\theta + \frac{\pi}{2}\ell n2$$

$$\Rightarrow I = \frac{\pi}{2}\ell n2$$

17. $\int_0^{\pi/2} (2\cos^2 x) l\text{n}(\sin 2x) dx$

Ans. $-\frac{\pi}{2} l\text{n}\,2$

Sol. $I = \int_0^{\frac{\pi}{4}} 2\cos^2 x \ell n(\sin 2x) dx$

Apply Queen

$$I = 2\int_0^{\frac{\pi}{2}} \cos^2 x \ell n(\sin 2x) + \sin^2 x \ell n(\sin 2x) dx$$

$$I = 2\int_0^{\frac{\pi}{4}} \ell n(\sin 2x) dx$$

Put $2x = y$

$$I = 2\int_0^{\frac{\pi}{2}} \ell n(\sin y)\frac{dy}{2} = -\frac{\pi}{2}\ell n2$$

18. $\int_0^{\infty}\left(\frac{\tan^{-1}x}{x}\right)^3 dx$

Ans. $\frac{3\pi}{2}ln\,2-\frac{\pi^3}{16}$

Sol. put $x=\tan\theta$

$$=\int_0^{\pi/2}\frac{\theta^3}{\tan^3\theta}\cdot\sec^2\theta\,d\theta=\int_0^{\pi/2}\underbrace{\frac{\sec^2\theta}{\tan^3\theta}}_{II}\cdot\underbrace{\theta^3}_{I}d\theta=-\frac{\theta^3}{2\tan^2\theta}\Big|_0^{\pi/2}+\frac{3}{2}\int_0^{\pi/2}\frac{\theta^2}{\tan^2\theta}d\theta$$

$$=-\frac{\pi^3}{16}+\frac{3}{2}\int_0^{\pi/2}\theta^2(\text{cosec}^2\theta-1)\,d\theta=\frac{3}{2}\left[\int_0^{\pi/2}\underbrace{(\theta^2}_{I}\cdot\underbrace{\text{cosec}^2\theta)}_{II}d\theta-\frac{\pi^3}{24}\right]$$

$$=\frac{3}{2}\left[-\theta^2\cot\theta\Big|_0^{\pi/2}+2\int_0^{\pi/2}(\theta\cot\theta)d\theta-\frac{\pi^3}{24}\right]=\frac{3}{2}\left[(0)+2\int_0^{\pi/2}(\theta\cot\theta)d\theta-\frac{\pi^3}{24}\right]$$

2.7 PROPERTIES OF DEFINITE INTEGRAL - P7 (FOR PERIODIC FUNCTIONS)

If f(x) is a periodic function with period T, then

(i) $\int_0^{nT}f(x)\,dx=n\int_0^{T}f(x)dx,\ n\in z$ (ii) $\int_a^{a+nT}f(x)\,dx=n\int_0^{T}f(x)dx,\ n\in z,\ a\in R$

(iii) $\int_{mT}^{nT}f(x)\,dx=(n-m)\int_0^{T}f(x)dx,\ m,n\in z$ (iv) $\int_{nT}^{a+nT}f(x)\,dx=\int_0^{a}f(x)dx,\ n\in z,\ a\in R$

Ex.1 $\int_0^{16\pi/3}|\sin x|\,dx$

Ans 21/2

Sol. $\int_0^{16\pi/3}|\sin x|\,dx=\int_0^{5\pi}|\sin x|\,dx+\int_{5\pi}^{5\pi+\pi/3}|\sin x|\,dx=5\int_0^{\pi}|\sin x|\,dx+\int_0^{\pi/3}|\sin x|\,dx$

$$=5[-\cos x]_0^{\pi}+[-\cos x]_0^{\pi/3}=10+\left(-\frac{1}{2}+1\right)=\frac{21}{2}$$

Ex.2 $\int_0^{4\pi}|\cos x|\,dx$

Ans 8

Sol. Note that |cos x| is a periodic function with period π. Hence the given integral.

$$I=4\int_0^{\pi}|\cos x|\,dx=4\left[\int_0^{\frac{\pi}{2}}\cos x dx-\int_{\frac{\pi}{2}}^{\pi}\cos x dx\right]=4\left[[\sin x]_0^{\pi/2}-[\sin x]_{\pi/2}^{\pi}\right]=4[1+1]=8$$

Questions

1. $\int_{\pi/3}^{13\pi/3} |\sin x|\,dx$

Ans. 8

Sol. $I = \int_{\frac{\pi}{3}}^{\frac{13\pi}{3}} |\sin x|\,dx = \int_{0}^{4\pi} |\sin x|\,dx = 4\int_{0}^{\pi} |\sin x|\,dx = 8$

2. $\int_{50}^{100} \{x\}\,dx$

Ans. 25

Sol. $I = \int_{50}^{100} \{x\}\,dx = 50\int_{0}^{1} \{x\}\,dx = 50\int_{0}^{1} x\,dx = 25$

3. $\int_{0}^{13\pi/3} |\sin x|\,dx$

Ans. 17/2

Sol. $\int_{0}^{\frac{13\pi}{3}} |\sin x|\,dx = \int_{0}^{4\pi} |\sin x|\,dx + \int_{0}^{\frac{\pi}{3}} |\sin x|\,dx$

$= 8 + (-\cos x)\Big|_0^{\frac{\pi}{3}}$

$= 8 - \left(\frac{1}{2} - 1\right) = 8 + \frac{1}{2} = \frac{17}{2}$

4. $\int_{0}^{200\pi} \sqrt{1+\cos x}\ dx$

Ans. $400\sqrt{2}$

Sol. $\int_{0}^{200\pi} \sqrt{1+\cos x}\,dx = \sqrt{2}\int_{0}^{200\pi} \left|\cos\frac{x}{2}\right| dx$

Put $\frac{x}{2} = y$

$= 2\sqrt{2}\int_{0}^{100\pi} |\cos y|\,dy$

$$\Rightarrow I = 200\sqrt{2}\int_0^{\pi}|\cos y|\,dy = 400\sqrt{2}$$

5. $\int_0^{2n\pi}\left(|\sin x| - \left[\left|\frac{\sin x}{2}\right|\right]\right)dx$ [.] denotes greatest integer function, $n \in N$.

Ans. 4n

Sol. $I = \int_0^{2n\pi}|\sin x| - \left[\frac{|\sin x|}{2}\right]dx$

$$\frac{|\sin x|}{2} \in \left[0, \frac{1}{2}\right]$$

$$\Rightarrow I = \int_0^{2n\pi}|\sin x|\,dx = 2n\int_0^{\pi}|\sin x|\,dx = 4n$$

6. $\int_0^{1000} e^{x-[x]}\,dx$

Ans. $1000(e-1)$

Sol. $I = \int_0^{1000} e^{\{x\}}dx = 1000\int_0^1 e^{\{x\}}dx = 1000\int_0^1 e^x dx$

$= 1000(e-1)$

7. $\int_0^{2000\pi}\frac{dx}{1+e^{\sin x}}$

Ans. 1000π

Sol. $I = \int_0^{2000\pi}\frac{dx}{1+e^{\sin x}} = 1000\int_0^{2\pi}\frac{dx}{1+e^{\sin x}}$

Apply KING and add

$$2I = 1000\int_0^{2\pi}1\cdot dx \Rightarrow I = 1000\pi$$

8. $\int_0^{n\pi+v}|\cos x|\,dx$ where $\frac{\pi}{2} < v < \pi$ & $n \in N$

Ans. $2n + 2 - \sin v$

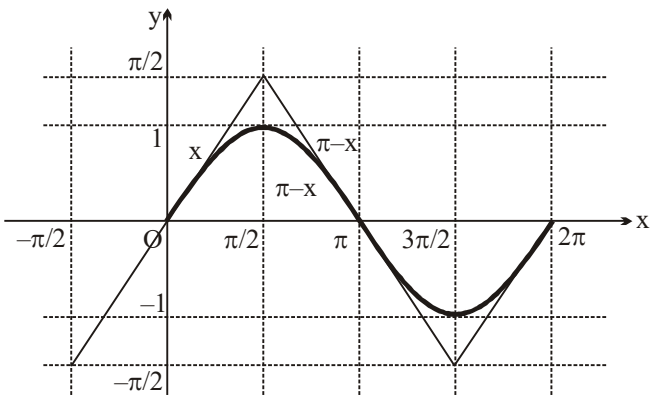
Sol. $I = \int_0^{n\pi}|\cos x|\,dx + \int_{n\pi}^{n\pi+v}|\cos x|\,dx = n\int_0^{\pi}|\cos x|\,dx + \int_0^{v}|\cos y|\,dy$ where $n\pi + y = x$

9. The value of the definite integral $\int_0^{2n\pi} \max.\left(\sin x, \sin^{-1}(\sin x)\right)dx$ equals (where $n \in I$)

(A) $\frac{n(\pi^2-4)}{2}$ (B) $\frac{n(\pi^2-4)}{4}$ (C) $\frac{n(\pi^2-8)}{4}$ (D) $\frac{n(\pi^2-2)}{4}$

Ans. C

Sol. $I = \int_0^{2n\pi} \max.\left(\sin x, \sin^{-1}(\sin x)\right)dx$



$$= n\left[\int_0^{\pi/2} x\,dx + \int_{\pi/2}^{\pi} (\pi - x)\,dx + \int_{\pi}^{2\pi} (\sin x)\,dx\right]$$

$$= n\left[\frac{\pi^2}{8} + \frac{\pi^2}{2} - \frac{1}{2}\left(\pi^2 - \frac{\pi^2}{4}\right) - 2\right]$$

$$= n\left[\frac{\pi^2}{8} + \frac{\pi^2}{2} - \frac{3\pi^2}{8} - 2\right] = \frac{n(\pi^2-8)}{4}$$ Ans.

10. Let $f : R \to R$ be a continuous function such that $f(x) + f(x + 1) = 2$, for all $x \in R$. If $I_1 = \int_0^8 f(x)\,dx$ and

$I_2 = \int_{-1}^{3} f(x)\,dx$, then the value of $I_1 + 2I_2$ is equal to _____. **[JEE(Main)Mar.2021]**

Ans. 16

Sol. $f(x) + f(x + 1) = 2$

$x \to x + 1$

$f(x + 1) + f(x + 2) = 2$

$\Rightarrow f(x) = f(x + 2)$

Period of $f(x)$ is 2.

$$I = \int_0^2 f(x)\,dx = \int_0^1 f(x)\,dx + \int_1^2 f(x)\,dx$$

$x = t + 1$

$$I = \int_0^1 f(x)\,dx + \int_0^1 f(t+1)\,dt$$

$I = 2$

$I_1 = \int_0^8 f(x)\,dx = 4\int_0^2 f(x)\,dx = 8$ ($\because$ f(x) is periodic with period 2)

$I_2 = \int_{-1}^3 f(x)\,dx = 2\int_0^2 f(x)\,dx = 4$ ($\because$ f(x) is periodic with period 2)

$I_1 + 2I_2 = 8 + 8 = 16$

3. DIFFERENTIATION OF INTEGRATION - LEIBNITZ RULE

Indefinite Integration

$I = \int f(x)\cdot dx$

$\frac{dI}{dx} = f(x)$;

Definite Integration

Leibnitz Rule: If $F(x) = \int_{g(x)}^{h(x)} f(t)\,dt$, then

$\frac{dF(x)}{dx} = h'(x)\,f(h(x)) - g'(x)\,f(g(x))$ f(x) is continuous.

Let $P(t) = \int f(t)\,dt$ $\Rightarrow$ $F(x) = \int_{g(x)}^{h(x)} f(t)\,dt = P(h(x)) - P(g(x))$

$\Rightarrow$ $\frac{dF(x)}{dx} = P'(h(x))\,h'(x) - P'(g(x))\,g'(x) = f(h(x))\,h'(x) - f(g(x))\,g'(x)$

Ex.1 $f(x) = \int_1^x te^{t^3}\cdot dt$ find f'(x)

Sol. $f'(x) = xe^{x^3}$

Ex.2 $f(x) = \int_0^x e^{t+x}\cdot dt$ find f'(x) Ex.3 $f(x) = \int_0^x \frac{dt}{t^2+x^2}$ find f'(x)

Sol. **Ex. 2** $f(x) = \int_0^x e^{t+x}dt = e^x\int_0^x e^t dt$

$\Rightarrow f(x) = e^x(e^x - 1) \Rightarrow f'(x) = 2e^{2x} - e^x$

Ex. 3

$f(x) = \int_0^x \frac{dt}{t^2+x^2} = \frac{1}{x}\tan^{-1}\left(\frac{t}{x}\right)\Big|_0^x$

$$\Rightarrow f(x)=\frac{\pi}{4x}\Rightarrow f'(x)=-\frac{\pi}{4x^2}$$

Questions

1. $f(x)=\int_{x^2}^{x^3} e^{\sin(t)}\cdot dt$ find f'(x)

Sol. $f(x)=\int_{x^2}^{x^3} e^{\sin(t)}dt\Rightarrow f'(x)=e^{\sin x^3}\left(3x^2\right)-e^{\sin x^2}\cdot 2x$

2. $I=\int_{2}^{\sin x} e^{t^2}\,dt$ Find $\frac{dI}{dx}$

Sol. $\frac{dI}{dx}=e^{(\sin x)^2}\cdot\cos x$

3. Let $G(x)=\int_{2}^{x^2}\frac{dt}{1+\sqrt{t}}$ $(x>0)$. Find G ' (9).

Ans $\frac{9}{5}$

Sol: $G'(x)=\frac{1}{1+x}\cdot 2x=\frac{2x}{1+x}$; $G''(x)=\frac{2}{(1+x)^2}\Rightarrow G''(1)=\frac{1}{2}$; $G'(9)=\frac{18}{10}=\frac{9}{5}$

4. If $x\in[0,\pi/2]$ Prove that $f(x)=\int_{0}^{\sin^2 x}\sin^{-1}\sqrt{t}\,dt+\int_{0}^{\cos^2 x}\cos^{-1}\sqrt{t}\,dt=\frac{\pi}{4}$

Sol. $f'(x)=\sin^{-1}|\sin x|\sin 2x+\cos^{-1}|\cos x|(-\sin 2x)$

as $x\in\left[0,\frac{\pi}{2}\right]$

$\Rightarrow f(x)=\text{constant}=f\left(\frac{\pi}{4}\right)$

$\Rightarrow f(x)=\int_{0}^{\frac{1}{2}}\sin^{-1}\sqrt{t}dt+\int_{0}^{\frac{1}{2}}\cos^{-1}\sqrt{t}dt=\frac{\pi}{4}$

5. Prove that $\int_{1/e}^{\tan x}\frac{t\,dt}{1+t^2}+\int_{1/e}^{\cot x}\frac{dt}{t\left(1+t^2\right)}=1$

Sol. $f(x)=\int_{\frac{1}{e}}^{\tan x}\frac{tdt}{1+t^2}dt+\int_{\frac{1}{e}}^{\cot x}\frac{dt}{t\left(1+t^2\right)}$

$$f'(x)=\frac{\tan x}{1+\tan^2 x}\sec^2 x+\frac{1}{\cot x\left(1+\cot^2 x\right)}\left(-\operatorname{cosec}^2 x\right)$$

$$\Rightarrow f(x)=\text{constant}=f\left(\frac{\pi}{4}\right)$$

$$f\left(\frac{\pi}{4}\right)=\int_{\frac{1}{e}}^{1}\frac{t}{1+t^2}+\frac{1}{t\left(1+t^2\right)}dt=\int_{\frac{1}{e}}^{1}\frac{dt}{t}$$

$$=\ell n|t|\Big|_{\frac{1}{e}}^{1}=1$$

6. $\frac{d}{dx}\int_{\sqrt{x}}^{x^2}\sqrt{\sin t}\,(t-x)dt$;

Sol. $I=\int_{\sqrt{x}}^{x^2}\sqrt{\sin t}\,t\,dt-x\int_{\sqrt{x}}^{x^2}\sqrt{\sin t}\,dt$

$$\frac{dI}{dx}=\sqrt{\sin x^2}\cdot x^2\cdot(2x)-\sqrt{\sin\sqrt{x}}\sqrt{x}\cdot\frac{1}{2\sqrt{x}}-x\left(\sqrt{\sin x^2}\cdot 2x-\sqrt{\sin\sqrt{x}}\cdot\frac{1}{2\sqrt{x}}\right)-\int_{\sqrt{x}}^{x^2}\sqrt{\sin t}\,dt$$

$$\frac{dI}{dx}=\sqrt{\sin\sqrt{x}}\left(-\frac{1}{2}+\frac{\sqrt{x}}{2}\right)-\int_{\sqrt{x}}^{x^2}\sqrt{\sin t}\,dt$$

7. The function $f(x)=\int_{-1}^{x}t\left(e^t-1\right)(t-1)(t-2)^3(t-3)^5\,dt$ has a local minimum at x = **[JEE - 99]**

(A) 0 (B) 1 (C) 2 (D) 3

Sol. $f'(x)=x(e^x-1)(x-1)(x-2)^3(x-3)^5$

$\Rightarrow$ Local min at x = 1,3.

Ans B, D.

8. $\frac{d}{dx}\int_{x}^{x^2}\frac{dt}{(x^2+t^2)}$ $(x>0)$

Sol. $I=\int_{x}^{x^2}\frac{dt}{x^2+t^2}=\frac{1}{x}\tan^{-1}\left(\frac{t}{x}\right)\Big|_{x}^{x^2}$

$$I=\frac{1}{x}\left(\tan^{-1}(x)-\frac{\pi}{4}\right)$$

$$\frac{dI}{dx}=\frac{1}{x}\left(\frac{1}{1+x^2}\right)+\left(\tan^{-1}x-\frac{\pi}{4}\right)\left(-\frac{1}{x^2}\right)$$

9. Let $f:R\to R$ be a differentiable function and f(1) = 4. Then the value of $\lim\limits_{x\to1}\int\limits_4^{f(x)}\frac{2t}{x-1}dt$ is

(A) 8 f′(1) (B) 4 f′(1) (C) 2f′(1) (D) f′(1) **[JEE - 90]**

Ans A

Sol. $$\lim_{x\to1}\frac{\int\limits_4^{f(x)}2t}{x-1}=\lim_{x\to1}\frac{2f(x)-f'(x)}{1}=8f'(1)$$

10. $f(x)=\int\limits_{e^{2x}}^{e^{3x}}\frac{t}{\ell n\,t}dt$ $x>0$. Find derivative of f(x) w.r.t. $\ell n\,x$ when $x=\ell n\,2$

Ans 48

Sol. $$f'(x)=\frac{e^{3x}}{\ell ne^{3x}}\cdot3e^{3x}-\frac{e^{2x}}{\ell ne^{2x}}\cdot2e^{2x}$$

$$=\frac{e^{6x}-e^{4x}}{x}$$

$$g(x)=\ell nx\Rightarrow g'(x)=\frac{1}{x}$$

$$\frac{df(x)}{dg(x)}=e^{6x}-e^{4x}$$

Given $x=\ell n2$

$\Rightarrow e^x=2$

$= 64 - 16 = 48$

11. $x=\int\limits_0^y\frac{dt}{\sqrt{1+4t^2}}$, If $\frac{d^2y}{dx^2}=ky$, find k.

Ans. 4

Sol. $$\frac{dx}{dy}=\frac{1}{\sqrt{1+4y^2}}\Rightarrow\frac{dy}{dx}=\sqrt{1+4y^2}$$

$$\Rightarrow\frac{d^2y}{dx^2}=\frac{1}{2\sqrt{1+4y^2}}\cdot8y\left(\frac{dy}{dx}\right)$$

$$\Rightarrow\frac{d^2y}{dx^2}=4y\cdot(1)\Rightarrow k=4$$

12. If $x = \int_1^{t^2} z\, l\text{n} z\, dz$ and $y = \int_{t^2}^{1} z^2\, l\text{n} z\, dz$ find $\frac{dy}{dx}$

Ans. $-t^2$

Sol. $\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = -\frac{t^4\left(\ell n t^2\right)(2t)}{t^2\left(\ell n t^2\right)(2t)} = -t^2$

13. If $y = \int_0^{z^2} \frac{dx}{1+x^3}$ find $\frac{d^2y}{dz^2}$ at $z = 1$

Ans. -2

Sol. $\frac{dy}{dz} = \frac{1}{1+z^6}\cdot 2z$

$\frac{d^2y}{dz^2} = \frac{\left(1+z^6\right)2-(2z)\left(6z^5\right)}{\left(1+z^6\right)^2}$

when $z = 1$

$\frac{d^2y}{dz^2} = \frac{4-12}{4} = -2$

14. $\underset{x\to 0}{\text{Limit}} \frac{\int_0^{x^2} \cos t^2\ dt}{x \sin x}$ **[JEE 97]**

Ans. 1

Sol. Apply Lopital Rule

$= \lim_{x\to 0} \frac{\left(\cos x^4\right)(2x)}{2x} = 1$

15. $f(x) = \int_0^x \frac{\sin^2 \frac{t}{2}\, dt}{t}$ then find $\underset{x\to 0}{Limit}\ \frac{f'(x)}{x}$

Ans. $\frac{1}{4}$

Sol. $f'(x) = \frac{\sin^2\left(\frac{x}{2}\right)}{x}$

$$\lim_{x\to 0}\frac{\sin^2\left(\frac{x}{2}\right)}{x^2}=\frac{1}{4}$$

16. $\underset{x\to 0}{\text{Limit}}\ \dfrac{\int_0^x x\,e^{t^2}\,dt}{1-e^{x^2}}$

Ans. –1

Sol. $\lim_{x\to 0}\dfrac{x\int_0^x e^{t^2}dt}{1-e^{x^2}}=\lim_{n\to 0}\dfrac{xe^{x^2}+\int_0^x e^{t^2}dt}{-2xe^{x^2}}$

$$=-\frac{1}{2}\lim_{n\to 0}\frac{xe^{x^2}\cdot 2x+e^{x^2}+e^{x^2}}{1}=-1$$

17. $\underset{x\to 0}{\text{Lim}}\ \dfrac{1}{x^3}\int_0^x\dfrac{t^2}{t^4+1}dt$

Ans. $\dfrac{1}{3}$

Sol. Using L'Hospital's Rule $\underset{x\to 0}{\text{Lim}}\ \dfrac{x^2}{(x^4+1)}\cdot\dfrac{1}{3x^2}=\dfrac{1}{3}$

18. Evaluate $\underset{x\to\infty}{\text{Lim}}\ x\int_0^x\left(e^{t^2-x^2}\right)dt$

Ans. 1/2

Sol. $I=\underset{x\to\infty}{\text{Lim}}\ xe^{-x^2}\int_0^x e^{t^2}dt=\underset{x\to\infty}{\text{Lim}}\ \dfrac{x\cdot\int_0^x e^{t^2}dt}{e^{x^2}}$; now use L'Hospital rule twice to get 1/2]

19. $\int_0^x f(t)dt = x\cos(\pi x)$, for $x > 0$, $f(4)$ is equal to

Ans. 1

Sol. $\int_0^x f(t)\,dt = x\cos(\pi x)$

differentiating w.r.t. x

$\Rightarrow f(x) = \cos(\pi x) + x(\pi)(-\sin \pi x)$

$\Rightarrow f(4) = 1$

20. $\lim\limits_{x\to 0} \dfrac{1}{x}\int\limits_0^x (1-\tan 2t)^{1/t}\, dt$

Ans. e^{-2}

Sol $= \lim\limits_{x\to 0} (1-\tan 2x)^{1/x} = e^{\lim\limits_{x\to 0} -\frac{\tan 2x}{x}} = e^{-2}$ Ans.

(using L'hospital's Rule)

21. Given a function g, continuous everywhere such that $g(1) = 5$ and $\int\limits_0^1 g(t)\, dt = 2$.

If $f(x) = \dfrac{1}{2}\int\limits_0^x (x-t)^2 g(t)\, dt$, then compute the value of $f'''(1) - f''(1)$.

Ans 3

Sol. $g(1) = 5$ and $\int\limits_0^1 g(t)\, dt = 2$

$$2f(x) = \int\limits_0^x (x^2 - 2xt + t^2) g(t)\, dt = x^2\int\limits_0^x g(t)\, dt - 2x\int\limits_0^x t\, g(t)\, dt + \int\limits_0^x t^2 g(t)\, dt$$

Differentiating

$$2f'(x) = x^2 g(x) + \int\limits_0^x g(t)dt \cdot 2x - 2\left[x^2 g(x) + \left(\int\limits_0^x t\, g(t)\, dt\right)\right] + x^2 g(x)$$

$$2f'(x) = 2x\int\limits_0^x g(t)dt - 2\int\limits_0^x t\, g(t)\, dt$$

$$f''(x) = x\, g(x) + \int\limits_0^x g(t)dt - x\, g(x) = \int\limits_0^x g(t)dt$$

hence $f''(1) = \int\limits_0^1 g(t)\, dt = 2$ (given)

also $f'''(x) = g(x) \Rightarrow f'''(1) = g(1) = 5$

$\therefore$ $f'''(1) - f''(1) = 5 - 2 = 3$ Ans.

22. Let $f(x)$ is a derivable function satisfying $f(x) = \int\limits_0^x e^t \sin(x-t)\, dt$ and $g(x) = f''(x) - f(x)$. Find the range of $g(x)$.

Ans. $\left[-\sqrt{2}, \sqrt{2}\right]$

Sol. $f(x) = \int_0^x e^t \sin(x-t)\,dt$ **[12th, 09-11-2008, P-2]**

$= \int_0^x e^{x-t} \sin(t)\,dt$ (using King)

$f(x) = e^x \int_0^x e^{-t} \sin t\,dt$

$f'(x) = e^x \cdot e^{-x} \sin x + \left(\int_0^x e^{-t} \sin t\,dt\right) e^x$

$f'(x) = \sin x + f(x)$ ....(1)

$f''(x) = \cos x + f'(x) = \cos x + \sin x + f(x)$ [using (1)]

$f''(x) - f(x) = \sin x + \cos x$ ....(2)

$\Rightarrow \quad g(x) = \sin x + \cos x \quad \Rightarrow \quad g(x) \in \left[-\sqrt{2}, \sqrt{2}\right]$

23. Let $f(x)$ be a continuous function such that $f(x) > 0$ for all $x \geq 0$ and $\left(f(x)\right)^{101} = 1 + \int_0^x f(t)\,dt$. The value of $\left(f(101)\right)^{100}$ is

Ans. 101

Sol. Given $\left(f(x)\right)^{101} = 1 + \int_0^x f(t)\,dt$

differentiating

$101 \cdot \left(f(x)\right)^{100} \cdot f'(x) = f(x)$

$\therefore \quad 101 \cdot \left(f(x)\right)^{99} \cdot f'(x) = 1 \quad (\text{as } f(x) > 0)$

integrating $\dfrac{(101)\left(f(x)\right)^{100}}{100} = x + C$

but $f(0) = 1 \quad \Rightarrow \quad C = \dfrac{101}{100}$

$\therefore \quad \dfrac{101}{100}\left(f(x)\right)^{100} = x + \dfrac{101}{100}$

put $x = 101$

$\dfrac{101}{100}\left(f(101)\right)^{100} = 101 + \dfrac{101}{100} = \dfrac{(101)(101)}{100}$

$\therefore \quad (f(101))^{100} = 101$ Ans.

4. DETERMINATION OF FUNCTION

1. f(x) is a continuous function such that $f(x) = 1 + \int_0^x f(t)\,dt$. Find f(x).

Ans. $f(x) = e^x$

Sol. Differentiating w.r.t. x

$$f'(x) = f(x) \Rightarrow \frac{f'(x)}{f(x)} = 1$$

$\Rightarrow \ell n|f(x)| = x + C$

$\Rightarrow f(0) = 1 \Rightarrow C = 0$

$f(x) = \pm e^x$ but $f(0) = 1$

$\Rightarrow f(x) = e^x$

2. Find a function f, continuous for all x (and not zero every where), such that $f^2(x) = \int_0^x \frac{f(t)\sin t}{2+\cos t}\,dt$

Ans. $\frac{1}{2} ln \frac{3}{2+\cos x}$

Sol. $f^2(x) = \int_0^x \frac{f(t)\sin t}{2+\cos t}\,dt$

(note that $f^2(x)$, being an integral function of a continuous function is continuous and differentiable)

$$2 f(x) f'(x) = \frac{f(x).\sin x}{2+\cos x}$$

Integrating $2 f(x) = c - ln (2 + \cos x)$

$x = 0 ; f(0) = 0 \Rightarrow c = ln3$

$$f(x) = \frac{1}{2} ln \frac{3}{2+\cos x}$$

3. If $f(x) = x + \int_0^{\pi/2} \sin(x+y).f(y)\,dy$ where x and y are independent variables, find f(x).

Ans. $f(x) = x - \frac{8(\pi-1)}{\pi^2-4}\sin x - \frac{2(\pi^2-2\pi+4)}{\pi^2-4}\cos x, x \in R$

Sol. $f(x) = x + \int_0^{\pi/2} (\sin x \cos y + \cos x \sin y)\, f(y)\,dy$

$$= x + \sin x \underbrace{\int_0^{\pi/2} \cos y\ f(y)\ dy}_{A} + \cos x \underbrace{\int_0^{\pi/2} \sin y\ f(y)\ dy}_{B}$$

$f(x) = x + A\sin x + B\cos x$

$$A = \int_0^{\pi/2} \cos y\ f(y)\ dy \quad \text{or} \quad \int_0^{\pi/2} \cos y\ (y + A\sin y + B\cos y)\ dy$$

$$= \int_0^{\pi/2} y\cos y\ dy + A\int_0^{\pi/2} \sin y\cos y\ dy + B\int_0^{\pi/2} \cos^2 y\ dy$$

$$= y\sin y\Big]_0^{\pi/2} - \int_0^{\pi/2} \sin y\ dy + \frac{A}{2}\sin^2 y\Big]_0^{\pi/2} + B\frac{\pi}{4}$$

$$A = \left(\frac{\pi}{2} - 1\right) + \frac{A}{2} + \frac{B\pi}{4} \Rightarrow \frac{A}{2} = \frac{\pi}{2} - 1 + \frac{B\pi}{4} \Rightarrow 2A - B\pi = 2\pi - 4 \qquad ....(1)$$

|||ly $\quad B = \int_0^{\pi/2} \sin y\ f(y)\,dy$

4. Find the function f(x) satisfying, the relation $6 + \int_a^x \frac{f(t)dt}{t^2} = 2\sqrt{x}$. Hence deduce the value of 'a'.

Ans $f(x) = x^{3/2}$ and $a = 9$

Sol. Differentiating both sides of the given expression, we get

$$\frac{f(x)}{x^2} = 2\cdot\frac{1}{2\sqrt{x}} \quad \text{or} \quad f(x) = x^{3/2}$$

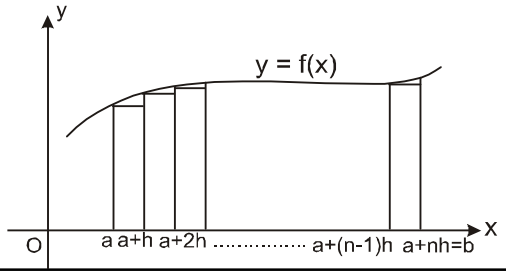
Substituting this value in the given relation, we get,

$$\therefore \quad 6 + \int_a^x \frac{dt}{\sqrt{t}} = 2\sqrt{x}\ ; \qquad 6 + 2\sqrt{t}\Big]_a^x = 2\sqrt{x}\ ; 6 + 2[\sqrt{x} - \sqrt{a}\,] = 2\sqrt{x}$$

$$2\sqrt{a} = 6 \quad \Rightarrow \quad a = 9; \quad \text{Hence } \boxed{f(x) = x^{3/2}} \text{ and } \boxed{a = 9} \text{ Ans.}$$

5.1 DEFINITE INTEGRAL AS A LIMIT OF SUM

Let f(x) be a continuous real valued function defined on the closed interval [a, b] which is divided into n parts as shown in figure.



The point of division on x-axis are a, a + h, a + 2h .....a + (n – 1)h, a + nh, where $\frac{b-a}{n} = h$.

Let S_n denotes the area of these n rectangles.

Then, $S_n = hf(a) + hf(a+h) + hf(a+2h) + ...+hf(a+(n-1)h)$

Clearly, S_n is area very close to the area of the region bounded by curve y = f(x), x–axis and the ordinates x = a, x = b.

Hence $\int_a^b f(x)\,dx = \underset{n\to\infty}{Lt}\ S_n$

$$\int_a^b f(x)\,dx = \underset{n\to\infty}{Lt}\sum_{r=0}^{n-1} h\,f(a+rh) = \underset{n\to\infty}{Lt}\sum_{r=0}^{n-1}\left(\frac{b-a}{n}\right) f\left(a+\frac{(b-a)r}{n}\right)$$

$$\text{If } a = 0, b = 1,\ \int_0^1 f(x)\,dx = \underset{n\to\infty}{Lt}\sum_{r=0}^{n-1}\frac{1}{n} f\left(\frac{r}{n}\right)$$

For finding sum of an infinite series with the help of definite integration, following formula is used-

$$\lim_{n\to\infty}\sum_{r=0}^{n-1} f\left(\frac{r}{n}\right).\frac{1}{n} = \int_0^1 f(x)\ dx.$$

The following method is used to solve the questions on summation of series.

(i) After writing $(r-1)^{th}$ or r^{th} term of the series, express it in the form $\frac{1}{n} f\left(\frac{r}{n}\right)$. Therefore the given series will take the form $\lim_{n\to\infty}\sum_{r=0}^{n-1} f\left(\frac{r}{n}\right).\frac{1}{n}$

(ii) Now writing $\int$ in place of $\left(\lim_{n\to\infty}\sum\ \right)$, x in place of $\left(\frac{r}{n}\right)$ and dx in place of $\frac{1}{n}$, we get the integral $\int f(x)\ dx$ in place of above series.

(iii) The lower limit of this integral = $\lim_{n\to\infty}\left(\frac{r}{n}\right)_{r=0}$ where r = 0 is taken corresponding to first term of the series and upper limit = $\lim_{n\to\infty}\left(\frac{r}{n}\right)_{r=n-1}$ where r = n – 1 is taken corresponding to the last term.

Questions

1. $\lim_{n\to\infty}\frac{1}{n}\left(e^{1/n}+e^{2/n}+e^{3/n}+....e^{n/n}\right)$

Ans. e–1

Sol. $\lim_{n\to\infty}\frac{1}{n}\sum_{r=1}^{n} e^{\frac{r}{n}} = \int_0^1 e^x dx = e-1$

2. $\lim_{n\to\infty}\frac{1}{\sqrt{n}}+\frac{1}{\sqrt{2n}}+\frac{1}{\sqrt{3n}}+\frac{1}{\sqrt{4n}}+.....\frac{1}{2n}$

Ans. 4

Sol. $\lim_{n\to\infty}\sum_{r=1}^{4n}\frac{1}{\sqrt{rn}} = \lim_{n\to\infty}\frac{1}{n}\sum_{r=1}^{4n}\frac{1}{\sqrt{\frac{r}{n}}}$

$$= \int_0^4 \frac{1}{\sqrt{x}}\,dx = 4$$

3. $\underset{n\to\infty}{\text{Limit}}\ \dfrac{\sqrt{1}+\sqrt{2}+\sqrt{3}+.....+2\sqrt{n}}{n\sqrt{n}}$

Ans. 16/3

Sol. $\lim\limits_{n\to\infty}\sum\limits_{r=1}^{4n}\dfrac{\sqrt{r}}{n\sqrt{n}} = \int_0^4 \sqrt{x}dx = \dfrac{2}{3}x^{\frac{3}{2}}\Big|_0^4 = \dfrac{2}{3}\cdot 8 = \dfrac{16}{3}$

4. $\underset{n\to\infty}{\text{Limit}}\ \dfrac{1}{n}+\dfrac{1}{n+1}+\dfrac{1}{n+2}+.....+\dfrac{1}{4n}$

Ans. $l\,n\,4$

Sol. $\lim\limits_{n\to\infty}\sum\limits_{r=0}^{3n}\dfrac{1}{n+r} = \lim\limits_{n\to\infty}\dfrac{1}{n}\sum\limits_{r=0}^{3n}\dfrac{1}{1+\frac{r}{n}}$

$$= \int_0^3 \frac{dx}{1+x} = \ell n(1+x)\Big|_0^3 = \ell n 4$$

5. $\underset{n\to\infty}{\text{Limit}}\ \left[\dfrac{n+1}{n^2+1^2}+\dfrac{n+2}{n^2+2^2}+\dfrac{n+3}{n^2+3^2}+.....+\dfrac{3}{5n}\right]$

Ans. $\tan^{-1}2+\dfrac{1}{2}\ l\,n\,5$

Sol. $T_r = \dfrac{n+r}{n^2+r^2} = \dfrac{1}{n}\dfrac{1+\frac{r}{n}}{1+\left(\frac{r}{n}\right)^2}$; $S = \lim\limits_{n\to\infty}\dfrac{1}{n}\sum\limits_{r=1}^{2n}\dfrac{1+\frac{r}{n}}{1+\left(\frac{r}{n}\right)^2}$

$$= \int_0^2 \frac{1+x}{1+x^2}\,dx$$

6. $\underset{n\to\infty}{\text{Limit}}\ \dfrac{n^2}{(n^2+1)^{3/2}}+\dfrac{n^2}{(n^2+2^2)^{3/2}}+.....+\dfrac{n^2}{\left[n^2+(n-1)^2\right]^{3/2}}$

Ans. $\dfrac{1}{\sqrt{2}}$

Sol. $$\lim_{n\to\infty}\sum_{r=1}^{n-1}\frac{n^2}{\left(n^2+r^2\right)^{\frac{3}{2}}}=\lim_{n\to\infty}\frac{1}{n}\sum_{r=1}^{n-1}\frac{1}{\left(1+\left(\frac{r}{n}\right)^2\right)^{\frac{3}{2}}}$$

$$=\int\frac{dx}{\left(1+x^2\right)^{\frac{3}{2}}}$$

Put $x=\tan\theta$

$$I=\int_0^{\frac{\pi}{4}}\frac{\sec^2\theta d\theta}{\sec^3\theta}$$

$$=\int_0^{\frac{\pi}{4}}\cos\theta d\theta=\sin\theta\Big|_0^{\frac{\pi}{4}}=\frac{1}{\sqrt{2}}$$

7. $$\lim_{n\to\infty}\frac{n}{(n+1)\sqrt{(2n+1)}}+\frac{n}{(n+2)\sqrt{2(2n+2)}}+\frac{n}{(n+3)\sqrt{3(2n+3)}}$$ ..... up to n terms

Ans. $\frac{\pi}{3}$

Sol. $$\int_0^1\frac{dx}{(1+x)\sqrt{x^2+2x}}=\frac{\pi}{3}\ ;T_r=\frac{n}{(n+r)\sqrt{r(2n+r)}}$$

8. $$\lim_{n\to\infty}\frac{1^2}{n^3+1^3}+\frac{2^2}{n^3+2^3}+.....+\frac{n^2}{2n^3}$$

Ans. $\frac{1}{3}\ell n\,2$

Sol. $$\lim_{n\to\infty}\sum_{r=1}^{n}\frac{r^2}{n^3+r^3}=\lim_{n\to\infty}\frac{1}{n}\sum\frac{\left(\frac{r}{n}\right)^2}{1+\left(\frac{r}{n}\right)^3}$$

$$=\int_0^1\frac{x^2dx}{1+x^3}=\frac{1}{3}\ell n\left(1+x^3\right)\Big|_0^1=\frac{\ell n2}{3}$$

9. $$\lim_{n\to\infty}\sum_{r=1}^{r=n-1}\frac{1}{n}\sqrt{\frac{n+r}{n-r}}$$

Ans. $\frac{\pi}{2}+1$

Sol. $\lim\limits_{n\to\infty}\sum\limits_{r=1}^{n-1}\frac{1}{n}\sqrt{\frac{1+\frac{r}{n}}{1-\frac{r}{n}}}=\int\limits_0^1\sqrt{\frac{1+x}{1-x}}dx$

Put $x=\cos\theta$

$I=-\int\limits_{\frac{\pi}{2}}^{0}\cot\frac{\theta}{2}\sin\theta d\theta$

$I=\int\limits_0^{\frac{\pi}{2}}\frac{\cos\frac{\theta}{2}}{\sin\frac{\theta}{2}}\cdot 2\sin\frac{\theta}{2}\cos\frac{\theta}{2}d\theta$

$I=\int\limits_0^{\frac{\pi}{2}}(1+\cos\theta)d\theta=\theta+\sin\theta\Big|_0^{\frac{\pi}{2}}$

$=\frac{\pi}{2}+1$

10. $\underset{n\to\infty}{\text{Lim}}\sum\limits_{i=1}^{n}\frac{2}{n}\left[\left(1+\frac{2i}{n}\right)^2+1\right]$

Ans. $\frac{32}{3}$

Sol. $\underset{n\to\infty}{\text{Lim}}\sum\limits_{i=1}^{n}\frac{2}{n}\left[\left(1+\frac{2i}{n}\right)^2+1\right]=2\int\limits_0^1\left((1+2x)^2+1\right)dx$

$=2\left[\frac{(1+2x)^3}{2\cdot 3}+x\right]_0^1=2\left[\left(\frac{27}{6}+1\right)-\frac{1}{6}\right]=2\left[\frac{27}{6}+\frac{5}{6}\right]=\frac{32}{3}$ **Ans.**

11. $\underset{n\to\infty}{\text{Lim}}\frac{1}{n}\sum\limits_{r=1}^{2n}\frac{r}{\sqrt{n^2+r^2}}$ equals : **[JEE - 97]**

Ans $-1+\sqrt{5}$

Sol. $\lim\limits_{n\to\infty}\frac{1}{n}\sum\limits_{r=1}^{2n}\frac{r}{\sqrt{r^2+n^2}}=\lim\limits_{n\to\infty}\frac{1}{n}\sum\limits_{r=1}^{2n}\frac{\frac{r}{n}}{\sqrt{\left(\frac{r}{n}\right)^2+1}}$

$=\int\limits_0^2\frac{x}{\sqrt{1+x^2}}dx$

$=\sqrt{1+x^2}\Big|_0^2=\sqrt{5}-1$

12. $\underset{n\to\infty}{\text{Lim}}\left[\frac{1}{n}+\frac{n^2}{(n+1)^3}+\frac{n^2}{(n+2)^3}+.....+\frac{1}{27n}\right]$

Ans. $\frac{4}{9}$

Sol. $\lim\limits_{n\to\infty}\sum\limits_{r=0}^{2n}\frac{n^2}{(n+r)^3}=\lim\limits_{n\to\infty}\frac{1}{n}\sum\limits_{r=0}^{2n}\frac{1}{\left(1+\frac{r}{n}\right)^3}$

$=\int\limits_0^2\frac{dx}{(1+x)^3}=\frac{(1+x)^{-2}}{-2}\Big|_0^2=-\frac{1}{2}\left(\frac{1}{9}-1\right)=\frac{4}{9}$

13. $\underset{n\to\infty}{\text{Lim}}\left(\tan^{-1}\frac{1}{n}\right)\left(\sum\limits_{k=1}^{n}\frac{1}{1+\tan(k/n)}\right)$

Ans. $\frac{1+\ell n(\sin 1+\cos 1)}{2}$

Sol. $\underset{n\to\infty}{\text{Lim}}\left(n\tan^{-1}\frac{1}{n}\right)\cdot\underset{n\to\infty}{\text{Lim}}\frac{1}{n}\sum\limits_{k=1}^{n}\frac{1}{1+\tan(k/n)}=\int\limits_0^1\frac{dx}{1+\tan x}=\frac{1}{2}[1+\ell n(\sin 1+\cos 1)]$

14. $\underset{n\to\infty}{\text{Limit}}\ \frac{\left[(n+1)(n+2).....(n+n)\right]^{1/n}}{n}$

Ans. $\frac{4}{e}$

Sol. $\ell=\lim\limits_{n\to\infty}\left(\frac{n+1}{n}.\frac{n+2}{n}.....\frac{n+n}{n}\right)^{\frac{1}{n}}$

$$\Rightarrow \ell n\ell = \lim_{n\to\infty}\frac{1}{n}\sum_{r=1}^{n}\ell n\left(1+\frac{r}{n}\right)$$

$$\Rightarrow \ell n\ell = \int_0^1 \ell n\left(1+x\right)dx = \left(1+x\right)\ell n\left(1+x\right)-\left(1+x\right)\Big|_0^1$$

$= (2\ell n\, 2) - 2 + 1$

$= \ell n4 - \ell ne$

$$\Rightarrow \ell = \frac{4}{e}$$

15. $\underset{n\to\infty}{\text{Limit}}\ \left[\left(1+\frac{1}{n^2}\right)^{\frac{2}{n^2}}\cdot\left(1+\frac{2^2}{n^2}\right)^{\frac{4}{n^2}}\cdot\left(1+\frac{3^2}{n^2}\right)^{\frac{6}{n^2}}\ .....\left(1+\frac{n^2}{n^2}\right)^{\frac{2n}{n^2}}\right]$

Ans. $\frac{4}{e}$

Sol. $\ell = \lim_{n\to\infty}\prod_{r=1}^{n}\left(1+\frac{r^2}{n^2}\right)^{\frac{2r}{n^2}}$

$$\Rightarrow \ell n\ell = \lim_{n\to\infty}\frac{2}{n}\sum_{r=1}^{n}\frac{r}{n}\ell n\left(1+\frac{r^2}{n^2}\right)$$

$$\Rightarrow \ell n\ell = 2\int_0^1 x\ell n\left(1+x^2\right)dx$$

$$\Rightarrow \ell n\ell = \left(1+x^2\right)\ell n\left(1+x^2\right)-\left(1+x^2\right)\Big|_0^1$$

$\Rightarrow \ell n\ell = (2\ell n2 - 2) + 1$

$$\Rightarrow \ell = \frac{4}{e}$$

16. $\underset{n\to\infty}{\text{Lim}}\left({}^{2n}C_n\right)^{1/n}$.

Ans. 4

Sol. Let $P = \underset{n\to\infty}{\text{Lim}}\left(\frac{(2n)!}{n!n!}\right)^{1/n}$; $P = \lim_{n\to\infty}\left[\frac{n!(n+1)(n+2).....(n+n)}{n!n!}\right]^{1/n}$;

$$P = \lim_{n\to\infty}\left(\frac{n+1}{1}\cdot\frac{n+2}{2}.....\frac{n+n}{n}\right)^{1/n}$$

$$ln\, P = \lim_{n\to\infty}\frac{1}{n}\left(\ln\frac{n+1}{1}+\ln\frac{n+2}{2}+.....+\ln\frac{n+n}{n}\right);$$

$$\therefore\ T_r = \frac{1}{n}\left(l\text{n}\,\frac{n+r}{r}\right) = \frac{1}{n}\ln\left(1+\frac{1}{r/n}\right)$$

$$S_n = \lim_{n\to\infty}\frac{1}{n}\sum_{r=1}^{n}\ln\left(1+\frac{1}{r/n}\right)$$

$$l\text{n}\,P = \int_0^1 l\text{n}\left(1+\frac{1}{x}\right)dx = \int_0^1 \left(l\text{n}(1+x) - l\text{n}\,x\right)dx$$

$$= (1+x)\ln(1+x) - (1+x) - [x\,l\text{n}\,x - x]$$

$$= \left[(1+x)\,l\text{n}(1+x) - 1 - x\,l\text{n}\,x\right]_0^1 = (2\,l\text{n}\,2 - 1 - 0) - (0 - 1)$$

$l\text{n}\,P = l\text{n}\,4 \quad \Rightarrow \quad \boxed{P = 4}$ Ans.

17. $\lim\limits_{n\to\infty} n^{-\frac{1}{2}\left(1+\frac{1}{n}\right)}\cdot(1^1\cdot 2^2\cdot 3^3.....n^n)^{\frac{1}{n^2}}$

Ans. $\dfrac{1}{\sqrt[4]{e}}$

Sol. Let $L = \lim\limits_{n\to\infty} n^{-\frac{1}{2}\left(1+\frac{1}{n}\right)}\cdot(1^1\cdot 2^2\cdot 3^3.....n^n)^{\frac{1}{n^2}}$

$$l\text{n}\,L = \lim_{n\to\infty} -\frac{1}{2}\left(\frac{n+1}{n}\right)l\text{n}\,n + \frac{1}{n^2}\sum_{k=1}^{n} k\,l\text{n}\,k$$

$$= \lim_{n\to\infty} -\frac{1}{2}\left(\frac{n+1}{n}\right)l\text{n}\,n + \frac{1}{n^2}\sum_{k=1}^{n}\left(k\,l\text{n}\,k - k\,l\text{n}\,n + k\,l\text{n}\,n\right)$$

$$= \lim_{n\to\infty} -\frac{1}{2}\left(\frac{n+1}{n}\right)l\text{n}\,n + \frac{1}{n^2}\sum_{k=1}^{n} k\,l\text{n}\frac{k}{n} + \frac{l\text{n}\,n}{n^2}\sum_{k=1}^{n} k$$

$$= \lim_{n\to\infty} -\frac{1}{2}\left(\frac{n+1}{n}\right)l\text{n}\,n + \frac{1}{n}\sum_{k=1}^{n}\frac{k}{n}\,l\text{n}\frac{k}{n} + \frac{l\text{n}\,n}{n^2}\cdot\frac{n(n+1)}{2}$$

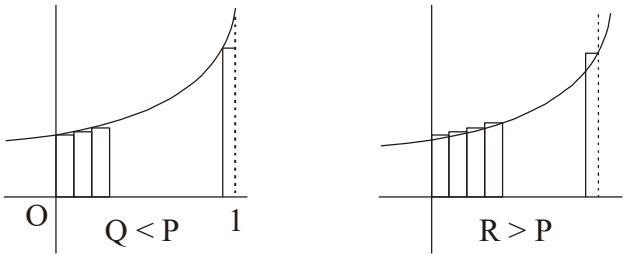
$$= -\frac{1}{2}\left(\frac{n+1}{n}\right)l\text{n}\,n + \int_0^1 x\,l\text{n}\,x\,dx + \frac{1}{2}\left(\frac{n+1}{n}\right)l\text{n}\,n$$

$$= \int_0^1 \underset{II}{x}\underbrace{\ln x}_{I}\,dx = -\frac{1}{4}; \qquad \therefore \quad L = e^{-\frac{1}{4}}$$ **Ans.]**

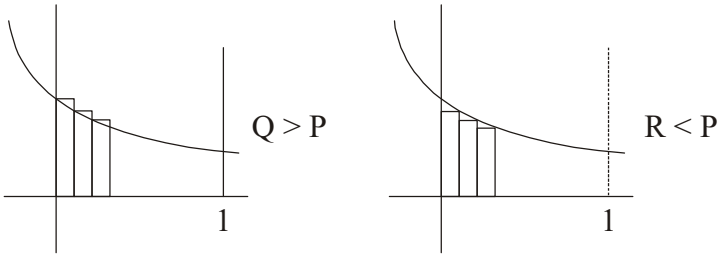
5.2 IMPORTANT NOTE ON LIMIT OF SUM

$$P = \lim_{n\to\infty}\frac{1}{n}\sum_{r=0}^{n-1} f\left(\frac{r}{n}\right) = \int_0^1 f(x)\cdot dx;\ Q = \frac{1}{n}\sum_{r=0}^{n-1} f\left(\frac{r}{n}\right);\ R = \frac{1}{n}\sum_{r=1}^{n} f\left(\frac{r}{n}\right)$$

Case I : f is increasing



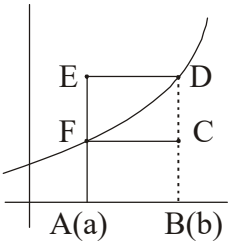
Case II : f is decreasing



6. DEFINITE INTEGRATION BY ESTIMATION

6.1 PROFILE 1

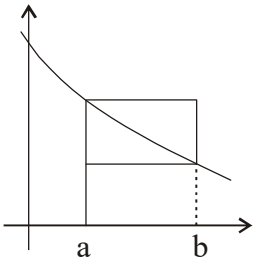
Case I : f is increasing



$$\text{Area of } \square ABCF < \int_a^b f(x)\cdot dx < \text{Area of } \square ABDE$$

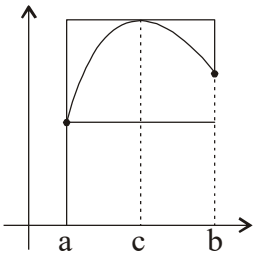
$$(b-a)\,f(a) < \int_a^b f(x)\cdot dx < (b-a)\,f(b)$$

Case II : f is decreasing



$$(b-a)\,f(b) < \int_a^b f(x)\cdot dx < (b-a)\,f(a)$$

Case III : f is non - monotonic



$$(b-a)\,f(a) < \int_a^b f(x)\cdot dx < (b-a)\,f(c)$$

Questions

1. Prove that $\frac{\pi}{128} < \int_{\pi/4}^{\pi/2} (\sin x)^{10}\,dx < \frac{\pi}{4}$,

Sol. $\frac{\pi}{128} < \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} (\sin x)^{10} < \frac{\pi}{4}$

Let $f(x) = (\sin x)^{10}$

In $x \in \left(\frac{\pi}{4}, \frac{\pi}{2}\right)$

$$\Rightarrow \left(\frac{1}{\sqrt{2}}\right)^{10} < (\sin x)^{10} < 1$$

$$\Rightarrow \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \frac{1}{32}\,dx < I < \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} 1\,dx$$

2. Prove that $1 < \int_0^{\pi/2} \frac{\sin x}{x}\,dx < \frac{\pi}{2}$;

Sol. Let $f(x)=\frac{\sin x}{x}$

$\Rightarrow f'(x)=\frac{x\cos x-\sin x}{x^2}=\frac{\cos x}{x^2}(x-\tan x)<0$ in $\left(0,\frac{\pi}{2}\right)$

$\Rightarrow$ f(x) is strictly decreasing in $\left(0,\frac{\pi}{2}\right)$

$f\left(\frac{\pi}{2}\right)<f(x)<\lim_{x\to 0}\frac{\sin x}{x}$

$\frac{2}{\pi}<f(x)<1$

3. Prove that $\frac{e-1}{3}<\int_1^e\frac{dx}{2+\ell nx}<\frac{e-1}{2}$;

Sol. Let f(x) = 2 + ℓnx in (1,e)
f(x) > 0 and f(x) is increasing in (1, e)

$\Rightarrow \frac{1}{2+\ell nx}$ is decreasing in (1, e)

$\frac{1}{3}<\frac{1}{2+\ell nx}<\frac{1}{2}$

$\Rightarrow \int_1^e\frac{1}{3}dx<\int_1^e\frac{dx}{2+\ell nx}<\int_1^e\frac{1}{2}dx$

4. Prove that $e^2-1<\int_{\frac{1}{e}}^{e}\frac{e^x}{x}\cdot dx<e^{e-1}\left(e-\frac{1}{e}\right)$

Sol. Let $f(x)=\frac{e^x}{x}$ $\quad x\in\left(\frac{1}{e},e\right)$

$f'(x)=\frac{xe^x-e^x}{x}=\frac{e^x}{x}(x-1)$

$\Rightarrow$ f(x) is decreasing in $\left(\frac{1}{e},1\right)$ and increasing in (1, e)

$\Rightarrow f(1)=e,\ f(e)=\frac{e^e}{e}-e^{e-1}, f\left(\frac{1}{e}\right)=e\cdot e^{\frac{1}{e}}=e^{1+\frac{1}{e}}$

$\Rightarrow \int_{\frac{1}{e}}^{e}edx<\int_{\frac{1}{e}}^{e}\frac{e^x}{x}dx<\int_{\frac{1}{e}}^{e}e^{e-1}dx$

$$\Rightarrow e\left(e-\frac{1}{e}\right)<\int_{\frac{1}{e}}^{e}\frac{e^x}{x}dx<e^{e-1}\left(e-\frac{1}{e}\right)$$

5. Prove that $4 \le \int_1^3 \sqrt{3+x^3}\,dx \le 2\sqrt{30}$

Sol. : Since the function $f(x) = \sqrt{3+x^2}$ increases monotonically on the interval [1, 3], m = 2, M = $\sqrt{30}$, b – a =2.

Hence, $2.2 \le \int_1^3 \sqrt{3+x^3}\,dx \le 2\sqrt{30} \Rightarrow 4 \le \int_1^3 \sqrt{3+x^3}\,dx \le 2\sqrt{30}$

6. Prove that $\frac{\pi}{\pi^3+10\pi+5} < \int_0^{\pi} \frac{dx}{x^3+10x+9\ \sin x+5} < \frac{\pi}{5}$.

Sol. : Let $f(x) = x^3 + 10x + 9\sin x + 5$

$f'(x) = 3x^2 + 10 + 9\cos x > 0 \ \forall\ x \in R$

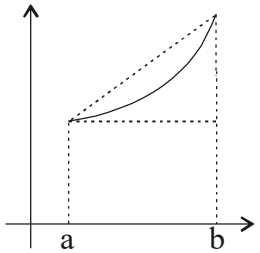
$\Rightarrow$ f(x) is entirely increasing $\Rightarrow \frac{1}{f(x)}$ is decreasing in $(0, \pi)$

$\Rightarrow$ Absolute maximum of f(x) in $[0, \pi]$ is $\frac{1}{5}$ and absolute minimum is $\frac{1}{\pi^3+10\pi+5}$

so $\frac{\pi}{\pi^3+10\pi+5} < \int_0^{\pi} \frac{dx}{x^3+10x+9\sin x+5} < \frac{\pi}{5}$

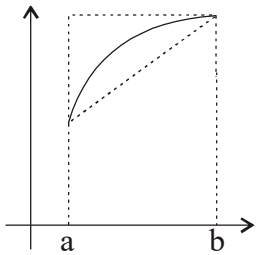
6.2 PROFILE 2 : USE OF CONCAVITY

Case I: f is concave up



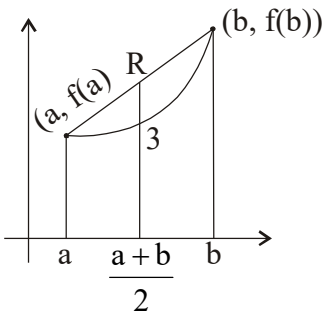
$$(b-a)\,f(a) < \int_a^b f(x)\cdot dx < \frac{1}{2}\,(b-a)\,(f(a)+f(b))$$

Case II : f is concave down



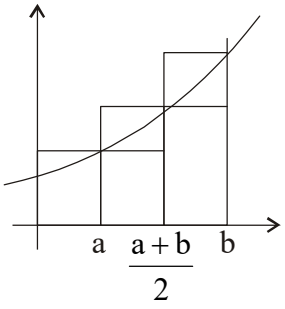
$$\frac{1}{2}(b-a)\big(f(a)+f(b)\big) < \int_a^b f(x)\cdot dx < (b-a)f(b)$$

Concave up :



$R\left(\frac{a+b}{2}, \frac{f(a)+f(b)}{2}\right)$; $S\left(\frac{a+b}{2}, f\left(\frac{a+b}{2}\right)\right)$ So $\quad f\left(\frac{a+b}{2}\right) < \frac{f(a)+f(b)}{2}$

6.3 PROFILE 3 : MORE ACCURATE APPROXIMATION



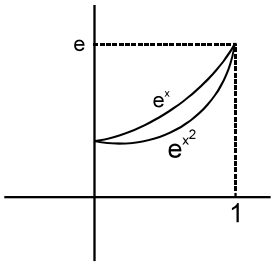
$$\left(\frac{b-a}{2}\right)\left(f(a)+f\left(\frac{a+b}{2}\right)\right) < \int_a^b f(x)\cdot dx < \left(\frac{b-a}{2}\right)\left(f(b)+f\left(\frac{a+b}{2}\right)\right)$$

6.4 PROFILE 4 : ASSUMPTION OF FUNCTION(S)

If $f_1(x) \le f(x) \le f_2(x)\ \forall x \in [a,\ b]$, then $\int_a^b f_1(x)dx \le \int_a^b f(x)dx \le \int_a^b f_2(x)dx$.

Q.1 Estimate the value of $\int_0^1 e^{x^2}dx$ by using $\int_0^1 e^x dx$.

Sol. For $x \in (0, 1)$, $e^{x^2} < e^x$



$\Rightarrow \qquad 1 \times 1 < \int_0^1 e^{x^2}dx < \int_0^1 e^x dx$

$$1 < \int_0^1 e^{x^2}dx < e - 1$$

Q.2 Let $g(x) = \int_0^x f(t)dt$, where f is such that $\frac{1}{2} \le f(t) \le 1$ for $t \in (0,1]$ and $0 \le f(t) \le \frac{1}{2}$ for $t \in (1, 2]$. Then g(2) satisfies the inequality : **[JEE - 2000]**

(A) $-\frac{3}{2} \le g(2) < \frac{1}{2}$ (B) $\frac{1}{2} \le g(2) \le \frac{3}{2}$ (C) $\frac{3}{2} < g(2) \le \frac{5}{2}$ (D) $2 < g(2) < 4$

Ans. B

Sol. $g(x)=\int_0^x f(t)\,dt \Rightarrow g'(x)=f(x)$

$$g(2)=\int_0^2 f(t)\,dt=\int_0^1 f(t)\,dt+\int_1^2 f(t)\,dt$$

But

$$\frac{1}{2}\le\int_0^1 f(t)\,dt\le 1 \text{ and } 0\le\int_1^2 f(t)\,dt\le\frac{1}{2}$$

$$\Rightarrow \frac{1}{2}\le g(2)\le\frac{3}{2}$$

Q.3 Prove that $\frac{1}{2}<\int_0^1\frac{dx}{\sqrt{4-x^2+x^5}}<\frac{\pi}{6}$; .

Sol. in (0, 1) $\sqrt{4-x^2+x^5}>\sqrt{4-x^2}$

$$\therefore \quad \int_0^1\frac{1}{\sqrt{4-x^2+x^5}}<\int_0^1\frac{1}{\sqrt{4-x^2}}=\sin^{-1}\frac{x}{2}\Big|_0^1=\sin^{-1}\frac{1}{2}=\frac{\pi}{6}$$

Also $\int_0^1\frac{1}{\sqrt{4-(x^2-x^5)}}<\frac{1}{\sqrt{4}}=\int_0^1\frac{1}{2}dx$

Q.4 Prove that $1\le\int_0^{\pi/2}\sqrt{1-\sin^3 x}\,dx\le\frac{1}{2}\left(\sqrt{2}+\ln(1+\sqrt{2})\right)$

Sol. $\sin^4x\le\sin^3x\le\sin^2x \Rightarrow -\sin^2x\le-\sin^3x\le-\sin^4x \Rightarrow 1-\sin^2x\le1-\sin^3x\le1-\sin^4x$

so, we get $\sqrt{1-\sin^2 x}\le\sqrt{1-\sin^3 x}\le\sqrt{1-\sin^4 x}$

$$\int_0^{\pi/2}\sqrt{1-\sin^2 x}\,dx=\int_0^{\pi/2}\cos x\,dx=[\sin x]_0^{\pi/2}=1$$

$$\int_0^{\pi/2}\sqrt{1-\sin^4 x}\,dx=\int_0^{\pi/2}\sqrt{(1+\sin^2 x)\cos^2 x}\,dx=\int_0^{\pi/2}\cos x\sqrt{1+\sin^2 x}\,dx$$

$$=\int_0^1\sqrt{1+t^2}\,dt=\left[\frac{1}{2}\left(t\sqrt{t^2+1}+\ln\left(t+\sqrt{t^2+1}\right)\right)\right]_0^1=\frac{1}{2}\left(\sqrt{2}+\ln\left(1+\sqrt{2}\right)\right)$$

$$\therefore \quad 1 \le \int_0^{\pi/2} \sqrt{1-\sin^3 x}\, dx \le \frac{1}{2}\left(\sqrt{2} + l\text{n}(1+\sqrt{2})\right) \ ]$$

Note: $\left| \int_a^b f(x)dx \right| \le \int_a^b |f(x)|dx$.

Q.5 Prove that $\left| \int_{10}^{19} \frac{\sin x}{1+x^8} dx \right| < 10^{-7}$

Ans.

Sol. To find $I = \left| \int_{10}^{19} \frac{\sin x}{1+x^8} dx \right| \le \int_{10}^{19} \left| \frac{\sin x}{1+x^8} \right| dx$ ..... (i)

Since $|\sin x| \le 1$ for $x \ge 10$

The inequality $\left| \frac{\sin x}{1+x^8} \right| \le \frac{1}{|1+x^8|}$ ..... (ii)

also, $10 \le x \le 19$

$\Rightarrow 1 + x^8 > 10^8$

$\Rightarrow \frac{1}{1+x^8} < \frac{1}{10^8}$ or $\frac{1}{|1+x^8|} < 10^{-8}$ ..... (iii)

from (ii) and (iii) ;

$$\left| \frac{\sin x}{1+x^8} \right| < 10^{-8}$$

$$\left| \int_{10}^{19} \frac{\sin x}{1+x^8} dx \right| < \int_{10}^{19} 10^{-8} dx$$

$$\therefore \left| \int_{10}^{19} \frac{\sin x}{1+x^8} dx \right| < (19-10).10^{-8} < 10^{-7}$$

Q.6 If $f(x)$ is integrable function such that $|f(x) - f(y)| \le |x^2 - y^2|$, $\forall\, x, y \in [a,b]$ then prove that

$$\left| \int_a^b \frac{f(x)-f(a)}{x+a} dx \right| \le \frac{(a-b)^2}{2} .$$

Sol. : Given, $\left| \int_a^b \frac{f(x)-f(a)}{x+a} dx \right| \le \int_a^b \left| \frac{f(x)-f(a)}{x+a} \right| dx$

$$\le \int_a^b \left| \frac{x^2-a^2}{x+a} \right| dx = \int_a^b |x-a|\, dx = \int_a^b (x-a)dx = \frac{(a-b)^2}{2}$$

Q.7 Given that f satisfies $|f(u)-f(v)| \le |u-v|$ for u and v in [a, b] then prove that

(i)f is continuous in [a, b] and (ii) $\left|\int_a^b f(x)\,dx-(b-a)f(a)\right| \le \frac{(b-a)^2}{2}$

Sol. (i) put $u = x+h$ and $v = x$

(ii) L.H.S. $= \left|\int_a^b f(x)\,dx-\int_a^b f(a)\,dx\right| = \left|\int_a^b f(x)-f(a)\,dx\right| \le \int_a^b |f(x)-f(a)|\,dx \le \int_a^b |x-a|\,dx$

Hence $\int_a^b (x-a)\,dx = \frac{(b-a)^2}{2}$

8. Let $f: R \to R$ be defined as $f(x) = e^{-x}\sin x$. If $F: [0, 1] \to R$ is differential function such that $F(x) = \int_0^x f(t)\,dt$, then prove that $\frac{330}{360} \le \int_0^1 (F'(x)+f(x))e^x\,dx \le \frac{331}{360}$

Sol. $f(x) = e^{-x}\sin^x$

$F(x) = \int_0^x f(t)\,dt$

$F'(x) = f(x)$

$I = \int_0^1 \left(f(x)+f(x)\right)e^x\,dx$

$I = \int_0^1 2f(x)e^x\,dx$

$I = \int_0^1 2\left(e^{-x}\sin x\right)e^x\,dx$

$I = 2\int_0^1 2\sin x\,dx$

Now by approximation

$x-\frac{x^3}{3!} \le \sin x \le x-\frac{x^3}{3!}+\frac{x^5}{5!}$

$\int_0^1\left(x-\frac{x^3}{3!}\right) \le \int_0^1 \sin x\,dx \le \int_0^1\left(x-\frac{x^3}{3!}+\frac{x^5}{5!}\right)dx$

$\frac{11}{24} \le \int_0^1 \sin x\,dx \le \frac{331}{720}$

$$\frac{11}{12} \le 2\int_0^1 \sin x\ dx \le \frac{331}{360}$$

$$\frac{330}{360} \le 2\int_0^1 \sin x\ dx \le \frac{331}{360}$$

7. WALLI'S THEOREM

$$\int_0^{\pi/2} \sin^n x.\cos^m x\,dx = \frac{[(n-1)(n-3)(n-5)....1 \text{ or } 2][(m-1)(m-3)....1 \text{ or } 2]}{(m+n)(m+n-2)(m+n-4)....1 \text{ or } 2}K$$

Where $K = \begin{cases} \frac{\pi}{2} & \text{if both m and n are even } (m, n \in N) \\ 1 & \text{otherwise} \end{cases}$

Ex.1 $\int_{-\pi/2}^{\pi/2} \sin^4 x\ \cos^6 x\ dx =$

Ans $\frac{3\pi}{256}$

Sol. : $I = \int_{-\pi/2}^{\pi/2} \sin^4 x\ \cos^6 x\ dx = 2\int_0^{\pi/2} \sin^4 x\ \cos^6 x.dx = 2\frac{(3.1)(5.3.1)}{10.8.6.4.2}.\frac{\pi}{2} = \frac{3\pi}{256}$

Ex.2 Evaluate $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sin^2 x \cos^2 x(\sin x + \cos x)\,dx$.

Ans 4/15

Sol. Given integral $= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sin^3 x \cos^2 x\,dx + \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sin^2 x \cos^3 x\,dx$

$= 0 + 2\int_0^{\frac{\pi}{2}} \sin^2 x \cos^3 x\,dx$ ($\because$ $\sin^3x \cos^2x$ is odd and $\sin^2x \cos^3x$ is even)

$= 2.\ \frac{1.2}{5.3.1} = \frac{4}{15}$

Ex.3 Evaluate $\int_0^{\pi} x\sin^5 x\cos^6 x\,dx$.

Ans $\frac{8\pi}{693}$

Sol. Let $I = \int_0^{\pi} x\sin^5 x\cos^6 x\,dx$

$I = \int_0^{\pi} (\pi - x)\sin^5(\pi - x)\cos^6(\pi - x)\,dx = \pi\int_0^{\pi} \sin^5 .\cos^6 x\,dx - \int_0^{\pi} x\sin^5 x .\cos^6 x\,dx$

$\Rightarrow 2I = \pi \cdot 2\int_0^{\frac{\pi}{2}} \sin^5 x \cdot \cos^6 x\, dx$

$I = \pi \dfrac{4.2.5.3.1}{11.9.7.5.3.1}$; $I = \dfrac{8\pi}{693}$

Ex.4 Evaluate $\int_0^1 x^3(1-x)^5 dx$.

Ans $\dfrac{1}{504}$

Sol. Put $x = \sin^2\theta$ $\Rightarrow$ $dx = 2\sin\theta\cos\theta\, d\theta$

L . L : $x = 0$ $\Rightarrow$ $\theta = 0$

U.L. : $x = 1$ $\Rightarrow$ $\theta = \dfrac{\pi}{2}$

$\therefore \int_0^1 x^3(1-x)^5 dx = \int_0^{\frac{\pi}{2}} \sin^6\theta(\cos^2\theta)^5\, 2 \cdot \sin\theta \cdot \cos\theta\, d\theta = 2 \cdot \int_0^{\frac{\pi}{2}} \sin^7\theta\cos^{11}\theta\, d\theta$

$= 2 \cdot \dfrac{6.4.2.10.8.6.4.2}{18.16.14.12.10.8.6.4.2} = \dfrac{1}{504}$

Questions

1. $\int_0^{\pi/2} \sin^6(x)\cos^7(x)\, dx$

Sol. $I = \int_0^{\frac{\pi}{2}} (\sin x)^6 (\cos x)^7\, dx = \dfrac{5\cdot3\cdot1\cdot6\cdot4\cdot2\cdot1}{13\cdot11\cdot9\cdot7\cdot5\cdot3\cdot1} = \dfrac{48}{9009}$

2. $\int_0^{\pi/2} \sin^{10} x\, dx$

Sol. $I = \int_0^{\frac{\pi}{2}} (\sin x)^{10}\, dx = \dfrac{9\cdot7\cdot5\cdot3\cdot1}{10\cdot8\cdot6\cdot4\cdot2}\cdot\dfrac{\pi}{2} = \dfrac{63\pi}{512}$

3. $\int_0^{\pi} \sin^8(x)\cos^7(x)\, dx$

Sol. $I = \int_0^{\pi} (\sin x)^8 (\cos x)^7\, dx$

Apply KING and add

$\Rightarrow I = 0$

4. $\int_0^{2\pi} x \sin^6 x \cos^4 x\, dx$

Sol. $I = \int_0^{2\pi} x(\sin x)^6(\cos x)^4\,dx$

Apply Queen

$I = \int_0^{\pi} x(\sin x)^6(\cos x)^4 + (2\pi - x)(\sin x)^6(\cos)^4\,dx$

$= 2\pi\int_0^{\pi}(\sin x)^6(\cos x)^4\,dx$

Apply Queen

$I = 4\pi\int_0^{\frac{\pi}{2}}(\sin x)^6(\cos x)^4\,dx = 4\pi\cdot\frac{3\pi}{512} = \frac{3\pi^2}{128}$

5. $\int_0^{3\pi/2}\cos^4 3x\cdot\sin^2 6x\,dx$

Sol. $I = 4\int_0^{\frac{3\pi}{2}}(\cos 3x)^6(\sin 3x)^2\,dx$

Put $3x = y$

$I = \frac{4}{3}\int_0^{\frac{9\pi}{2}}(\cos y)^6(\sin y)^2\,dy = \frac{4}{3}\times 9\int_0^{\frac{\pi}{2}}(\cos y)^6(\sin y)^2\,dy$

$= \frac{12\cdot 5\cdot 3\cdot 1\cdot 1}{3\cdot 6\cdot 4\cdot 2}\cdot\frac{\pi}{2} = \frac{15\pi}{64}$

6. $I = \int_0^1 x^{70}(1-x)^{30}\,dx$. Find ${}^{100}C_{30}I$.

Ans $\frac{1}{101}$

Sol. Put $x = \sin^2\theta$

$I = 2\int_0^{\pi/2}\sin^{141}\theta\cdot\cos^{61}\theta\,d\theta$; $I = \frac{2(140\cdot 138.....2)\ (60\cdot 58.....2)}{202\cdot 200.....\cdot 2}$

$I = \frac{2\cdot 2^{70}(70\cdot 69.....1)\ 2^{30}(30\cdot 29.....1)}{2^{101}(101\cdot 100.....1)} = \frac{70!\ 30!}{101\times 100!} = \frac{1}{101}\cdot\frac{1}{{}^{100}C_{30}}$

$$\therefore \qquad I \cdot {}^{100}C_{30} = \frac{1}{101} \textbf{ Ans.}$$

8. INTEGRATION BY REDUCTION

Q.1 $I_n = \int (\ln x)^n \cdot dx \qquad n \in N$

Sol. $I_n = \int (\ell nx)^n \, dx = \int 1 \cdot (\ell nx)^n \, dx$

$\Rightarrow I_n = (\ell nx)^n \, x - \int n \cdot (\ell nx)^{n-1} \cdot \frac{1}{x} \cdot x dx$

$\Rightarrow I_n = x(\ell nx)^n - n\, I_{n-1}$

Q.2 $I_n = \int_1^e (\ell n \, x)^n \, dx$ Prove that $I_n = e - nI_{n-1}$

Sol. $I_n = x(\ell nx)^n \Big|_1^e - nI_{n-1}$

$\Rightarrow I_n = e - nI_{n-1}$

Q.3 $I_n = \int_0^1 e^x (x-1)^n \, dx$ Prove that $I_n + nI_{n-1} = (-1)^{n-1}$

Sol. $I_n = \int_0^1 e^x (x-1)^n \, dx$

$\Rightarrow I_n = (x-1)^n e^x \Big|_0^1 - \int_0^1 n(x-1)^{n-1} e^x dx$

$\Rightarrow I_n = (-1)^{n+1} - nI_{n-1}$

Q.4 $I_n = \int_0^{\pi/2} (\sin x)^n \, dx$ Prove that $nI_n = (n-1)I_{n-2}$

Ans.

Sol. $I_n = \int_0^{\frac{\pi}{2}} \sin^n x \, dx$

$I_n = \left[-\sin^{n-1} x \cos x \right]_0^{\frac{\pi}{2}} + \int_0^{\frac{\pi}{2}} (n-1) \sin^{n-2} x \, . \cos^2 x \, dx$

$= (n-1) \int_0^{\frac{\pi}{2}} \sin^{n-2} x \, . (1 - \sin^2 x) \, dx$

$$= (n-1)\int_0^{\frac{\pi}{2}} \sin^{n-2} x\,dx - (n-1)\int_0^{\frac{\pi}{2}} \sin^{n} x\,dx$$

$$I_n + (n-1)\,I_n = (n-1)\,I_{n-2}$$

$$I_n = \left(\frac{n-1}{n}\right) I_{n-2}$$

Q.5 $I_n = \int_0^{\pi/4} (\tan\theta)^n\,d\theta$ Prove that $I_n + I_{n-2} = \frac{1}{n-1}$

Ans.

Sol. $I_n = \int_0^{\frac{\pi}{4}} (\tan x)^{n-2}\,.\tan^2 x\,dx = \int_0^{\frac{\pi}{4}} (\tan x)^{n-2}(\sec^2 x - 1)\,dx$

$$= \int_0^{\frac{\pi}{4}} (\tan x)^{n-2}\sec^2 x\,dx - \int_0^{\frac{\pi}{4}} (\tan x)^{n-2}\,dx = \left[\frac{(\tan x)^{n-1}}{n-1}\right]_0^{\frac{\pi}{4}} - I_{n-2}$$

$$I_n = \frac{1}{n-1} - I_{n-2} \qquad \therefore \qquad I_n + I_{n-2} = \frac{1}{n-1}$$

Q.6 If $u_n = \int_0^{\pi/2} x(\sin x)^n\,dx$, n>0, then prove that $u_n = \frac{n-1}{n}u_{n-2} + \frac{1}{n^2}$

Ans.

Sol. $= \int_0^{\pi/2} \underset{\text{I}}{x}\,(\sin x)^{n-1}\,.\,\underset{\text{II}}{\sin x}\;dx$

$$= -x\,(\sin x)^{n-1}\,.\,\cos x\Big]_0^{\pi/2} + \int_0^{\pi/2}\Big[(n-1)\,x\,(\sin x)^{n-2}\,.\cos x + (\sin x)^{n-1}\Big]\cos x\,dx$$

$$= 0 + (n-1)\int_0^{\pi/2} x\,(\sin x)^{n-2}(1-\sin^2 x)\,dx + \int_0^{\pi/2}(\sin x)^{n-1}.\cos x\,dx$$

$$\text{or } u_n = (n-1)\,u_{n-2} - (n-1)\,u_n + \frac{1}{n}$$

$$n\,u_n = (n-1)\,u_{n-2} + \frac{1}{n} \qquad \Rightarrow \qquad u_n = \frac{n-1}{n}u_{n-2} + \frac{1}{n^2}$$

Q.7 If $u_n = \int_0^1 x^n \tan^{-1} x\,dx$ then prove that $(n+1)\,u_n + (n-1)\,u_{n-2} = \frac{\pi}{2} - \frac{1}{n}$

Sol. $u_n = \int_0^1 \underset{\text{II}}{x^n}\;\underset{\text{I}}{\tan^{-1} x}\,dx = \frac{\tan^{-1} x\cdot x^{n+1}}{n+1}\Big|_0^1 - \frac{1}{n+1}\int_0^1 \frac{x^{n+1}}{1+x^2}\,dx$

$$= \frac{\pi}{4(n+1)} - \frac{1}{n+1}\int_0^1 \frac{x^{n-1}[1+x^2-1]}{1+x^2}dx$$

$$(n+1)u_n = \frac{\pi}{4} - \left[\int_0^1 x^{n-1}dx - \underset{I}{\int_0^1} \underset{II}{x^{n-1}\cdot\frac{1}{1+x^2}}dx\right]$$

$$= \frac{\pi}{4} - \frac{1}{n} + x^{n-1}\tan^{-1}x\Big|_0^1 - \int_0^1 (n-1)x^{n-2}\cdot\tan^{-1}x\,dx$$

$$(n+1)\,u_n = \frac{\pi}{2} - \frac{1}{n} - (n-1)u_{n-2}\ ; \quad (n+1)\,u_n + (n-1)u_{n-2} = \frac{\pi}{2} - \frac{1}{n}$$

Q.8 $I_n = \int_0^{\pi} \frac{\sin nx}{\sin x}dx, \quad n \in N$ Prove that $I_n = I_{n-2}$

Sol. Consider $\sin n\,x - \sin(n-2)x = 2\cos(n-1)\,x \sin x$

$$\therefore \quad \frac{\sin nx}{\sin x} = \frac{\sin(n-2)x}{\sin x} + 2\cos(n-1)\,x,$$

hence $\int \frac{\sin nx}{\sin x}dx = \int 2\cos(n-1)\,x\,dx + I_{n-2}$

$$= \frac{2}{n-1}\sin(n-1)\,x + I_{n-2}$$

$$I_n = \int_0^{\pi} \frac{\sin nx}{\sin x}dx = 2\int_0^{\pi} \cos(n-1)x\,dx + I_{n-2} = \frac{2}{n-1}\sin(n-1)x\Big|_0^{\pi} + I_{n-2}$$

If n is even $I_n = I_{n-2} = I_{n-4} = \ldots\ldots = I_0 = 0$

If n is odd $\therefore\ I_n = I_{n-2} = I_{n-4} = \ldots\ldots = I_1 = \pi$

again if $I_n = \int_0^{\pi/2} \frac{\sin nx}{\sin x}dx$; $\therefore \quad I_n = \frac{2}{n-1}\sin(n-1)x\Big|_0^{\pi/2}$

If n is odd $I_n = I_{n-2} = I_{n-4} = \ldots\ldots = I_1 = \frac{\pi}{2}$

If n is even $\int_0^{\pi/2} \frac{\sin 2kx}{\sin x}dx$

$2\sin x\cos x = \sin 2x$

$2\sin x\cos 3x = \sin 4x - \sin 2x$

$2\sin x\cos 5x = \sin 6x - \sin 4x$

$\vdots$

$2\sin x\cos(2k-1)x = \sin 2kx - \sin(2k-2)x$

hence $\dfrac{\sin 2kx}{\sin x} = 2(\cos x + \cos 3x + \cos 5x + ... + \cos(2k-1)x)$

9. DIFFERENTIATING AND INTEGRATING SERIES [O]

1 Find the sum of the series

$$\frac{x^2}{1.2} - \frac{x^3}{2.3} + \frac{x^4}{3.4} - .....+ (-1)^{n+1}\frac{x^{n+1}}{n(n+1)} + ... \quad |x| < 1$$

Ans. $(x+1)\,\ell n\,(1+x) - x$

Sol. $\ell n(1+x) = x - \dfrac{x^2}{2} + \dfrac{x^3}{3} .....|x| < 1$

$$\Rightarrow \int_0^x \ell n(1+x)\,dx = \int\left(x - \frac{x^2}{2} + \frac{x^3}{3} - .....\right)$$

$$\Rightarrow (x+1)\,\ell n(1+x) - (1+x)\Big|_0^x = \frac{x^2}{1\cdot 2} - \frac{x^3}{2\cdot 3} + \frac{x^4}{3\cdot 4} .....$$

$$\Rightarrow (x+1)\,\ell n(1+x) - (1+x) - \{1\} = \frac{x^2}{1\cdot 2} - \frac{x^3}{2\cdot 3} + ...$$

$\Rightarrow (x+1)\,\ell n(1+x) - x$

2 If $|x| < 1$ then find the sum of the series

$$f(x) = \frac{1}{1+x} + \frac{2x}{1+x^2} + \frac{4x^3}{1+x^4} + \frac{8x^7}{1+x^8} + ...\infty .$$

Ans. $\dfrac{1}{1-x}$

Sol. Let $f(x) = \dfrac{1}{1+x} + \dfrac{2x}{1+x^2} + \dfrac{4x^3}{1+x^4} + ...$

Consider

$g(x) = \ell n(1+x)\,(1+x^2)\,(1+x^4)\,.....$

$$\Rightarrow g(x) = \ell n\left(\frac{1}{1-x}\right)$$

$$\Rightarrow \ell n\left(\frac{1}{1-x}\right) = \ell n(1+x) + \ell n(1+x^2) + \ell n(1+x^4) + ...$$

differentiating w.r.t. x

$$\Rightarrow -\frac{1}{(1-x)}(-1) = \frac{1}{1+x} + \frac{2x}{1+x^2} + ...\infty$$

$$\Rightarrow f(x)=\frac{1}{1-x}$$

3 Let $f(x)=\frac{1}{2}\tan\frac{x}{2}+\frac{1}{2^2}\tan\frac{x}{2^2}+\frac{1}{2^3}\tan\frac{x}{2^3}+.....\infty$ then prove that $f(x)=\frac{1}{x}-\cot x$

Ans.

Sol. differentiate both side we get ; $f(x)=\frac{d}{dx}\left(\ln x-ln(\sin x)\right)=\frac{1}{x}\cdot\cot x$]

[Note : $\underset{n\to\infty}{\text{Limit}}\cos\frac{x}{2}.\cos\frac{x}{2^2}.\cos\frac{x}{2^3}.....\cos\frac{x}{2^n}=\frac{\sin x}{x}$

4 Prove that $\sum_{n=0}^{\infty}\left(\frac{1}{3n+1}-\frac{1}{3n+2}\right)=\frac{\pi}{3\sqrt{3}}$

Sol. Since the terms do not cancel, hence to proceed differently

$$\sum_{n=0}^{\infty}\int_0^1(t^{3n}-t^{3n+1})\,dt=\sum_{n=0}^{\infty}\int_0^1(t^{3n}(1-t)\,dt=\int_0^1\left((1-t)\sum_{n=0}^{\infty}t^{3n}\right)dt$$

$$=\int_0^1(1-t)\cdot\{1+t^3+t^6+t^8+.....\infty\}\,dt$$

$$=\int_0^1\frac{(1-t)}{1-t^3}dt=\int_0^1\frac{dt}{1+t^2+t}=\int_0^1\frac{dt}{\left(t+\frac{1}{2}\right)^2+\left(\frac{\sqrt{3}}{2}\right)^2}$$

$$=\frac{2}{\sqrt{3}}\tan^{-1}\frac{(t+(1/2))^2}{\sqrt{3}}\Bigg]_0^1=\frac{2}{\sqrt{3}}\tan^{-1}\sqrt{3}-\tan^{-1}\left(\frac{1}{\sqrt{3}}\right)=\frac{2}{\sqrt{3}}\left[\frac{\pi}{3}-\frac{\pi}{6}\right]=\frac{\pi}{3\sqrt{3}}\text{ Ans.}$$

10 EVALUATING INTEGRALS DEPENDENT ON A PARAMETER (NOT IN IIT JEE SYLLABUS)

Differentiate I w.r.t. the parameter within the sign of integrals taking variable of the integrand as constant. Now evaluate the integral so obtained as usual as a function of the parameter and then integrate the result to get I. Constant of integration to be computed by giving some arbitrary values to the parameter so that I is zero.

Explain by taking $I=\int_0^{\pi/2}\sin(x+c)\,dx$

EXAMPLES:

1(a). $I=\int_0^1\frac{\tan^{-1}ax}{x\sqrt{1-x^2}}\,dx$ [Ans: $\frac{\pi}{2}\left\{\ell n\left(a+\sqrt{a^2+1}\right)\right\}$]

[Sol. $I = \int_0^1 \frac{\tan^{-1} a x}{x\sqrt{1-x^2}}\,dx$; $\frac{dI}{da} = \int_0^1 \frac{\cancel{x}\,dx}{\cancel{x}\sqrt{1-x^2}\,(1+a^2x^2)}$

(put $x = \sin\theta$) $= \int_0^{\pi/2} \frac{d\theta}{1+a^2\sin^2\theta}$ or $\int_0^{\pi/2} \frac{\sec^2\theta\,d\theta}{a^2\tan^2\theta+1+\tan^2\theta}$

(put $\tan\theta = t$) $= \int_0^{\infty} \frac{dt}{(1+a^2)t^2+1}$ or $\frac{1}{1+a^2}\int_0^{\infty} \frac{dt}{t^2+\left(\frac{1}{\sqrt{1+a^2}}\right)^2}$

$= \frac{1}{1+a^2}\sqrt{1+a^2}\left[\tan^{-1} t\sqrt{1+a^2}\right]_0^{\infty}$ or $\frac{1}{\sqrt{1+a^2}}\frac{\pi}{2}$

$dI = \frac{\pi\,da}{2\sqrt{1+a^2}}$; $I = \frac{\pi}{2}\left[ln\left(a+\sqrt{1+a^2}\right)\right]+C$; $a = 0$; $I = 0 \Rightarrow C = 0$

$I = \frac{\pi}{2} ln\left(a+\sqrt{1+a^2}\right)$]

1.(b) Evaluate $\int_0^{\infty} \frac{\tan^{-1} ax - \tan^{-1} x}{x}\,dx$ where a is a parameter.

[Sol. Let $I = \int_0^{\infty} \frac{\tan^{-1} ax - \tan^{-1} x}{x}\,dx$

$\frac{dI}{da} = \int_0^{\infty} \frac{1\cdot x}{(1+a^2x^2)x}\,dx = \int_0^{\infty} \frac{dx}{1+a^2x^2} = \frac{1}{a^2}\int_0^{\infty} \frac{dx}{x^2+\frac{1}{a^2}} = \frac{1}{a^2}\,a\tan^{-1} x\Big]_0^{\infty} = \frac{\pi}{2a}$

$\int dI = \frac{\pi}{2}\int\frac{da}{a}$; $I = \frac{\pi}{2}\,ln\,a + C$

when $a = 1$, $I = 0 \Rightarrow C = 0$

Hence $I = \frac{\pi}{2}\,ln\,a$ Ans.]

2. $\int_0^1 \frac{\ell n\,(1-a^2x^2)}{x^2\sqrt{(1-x^2)}}\,dx$ $(a^2 < 1)$ [Ans: $\pi\left(\sqrt{1-a^2}-1\right)$]

[Sol. $I = \int_0^1 \frac{\ell n\,(1-a^2x^2)}{x^2\sqrt{(1-x^2)}}\,dx \quad |a|<1\,; \qquad \frac{dI}{da} = \int_0^1 \frac{-2ax^2}{(1-a^2x^2)}\,\frac{dx}{x^2\sqrt{1-x^2}}$

(put $x = \sin\theta$)

$$= \int_0^{\pi/2} \frac{2a\,d\theta}{a^2\sin^2\theta - 1} \quad \text{or} \quad \int_0^{\pi/2} \frac{2a\,\sec^2\theta d\theta}{a^2\tan^2\theta - (1+\tan^2\theta)} \quad \text{or} \quad -\int_0^{\pi/2} \frac{2a\,\sec^2\theta d\theta}{(1-a^2)\tan^2\theta + 1}$$

(put $\tan\theta = t$) $= \int_0^{\infty} \frac{2a\,dt}{(1-a^2)t^2+1}$ or $-\frac{2a}{(1-a^2)}\int_0^{\infty} \frac{dt}{t^2+\left(\frac{1}{\sqrt{1-a^2}}\right)^2}$

$$\frac{dI}{da} = -\frac{2a}{\sqrt{1-a^2}}\left[\tan^{-1} t\sqrt{1-a^2}\right]_0^{\infty} \quad \text{or} \quad -\frac{\pi.2a}{2\sqrt{1-a^2}} \quad \text{or} \quad -\frac{\pi a}{\sqrt{1-a^2}}$$

$I = \pi\sqrt{1-a^2} + C$

if $a = 0$; $I = 0 \Rightarrow C = -\pi$

$\therefore \quad I = \pi\left[\sqrt{1-a^2} - 1\right]$]

3. $\int_0^{\infty} \frac{\tan^{-1}(ax)}{x(1+x^2)}\,dx$ [Ans: $\frac{\pi}{2} l\mathrm{n}(a+1)$]

Sol. $I = \int_0^{\infty} \frac{\tan^{-1}(ax)\,dx}{x(1+x^2)}$ $I = 0$ when $a = 0$

Differentiating w.r.t. a

$$\frac{dI}{da} = \int_0^{\infty} \frac{1}{(1+a^2x^2)} \cdot \frac{x}{x(1+x^2)}\,dx$$

$$\frac{dI}{da} = \int_0^{\infty} \frac{1}{(1+a^2x^2)} \cdot \frac{1}{(1+x^2)}\,dx$$

$$\frac{dI}{da} = \int_0^{\infty} \left(\frac{a^2}{1+a^2x^2} - \frac{1}{1+x^2}\right)dx \cdot \frac{1}{a^2-1}$$

$$\frac{dI}{da} = \left(\frac{a^2}{a}\cdot\tan^{-1}(ax)\Big|_0^{\infty} - \tan^{-1}(x)\Big|_0^{\infty}\right)\frac{1}{(a^2-1)}$$

$$\frac{dI}{da}=\left(a\frac{\pi}{2}-\frac{\pi}{2}\right)\cdot\frac{1}{a^2-1}\Rightarrow\frac{dI}{da}=\frac{\pi}{2}\cdot\frac{1}{a+1}$$

$$\Rightarrow I=\frac{\pi}{2}\ell n(a+1)+c$$

when $a=0\Rightarrow I=0\Rightarrow c=0$

4. $\int_0^1 \frac{\ell n\,(1-a^2x^2)}{\sqrt{1-x^2}}\,dx\ \ (a^2<1)$

Sol. $I=\int_0^1\frac{\ell n(1-a^2x^2)}{\sqrt{1-x^2}}dx\quad(a^2<1)$

when $a=0\Rightarrow I=0$

Put $x=\sin\theta$

$$I(a)=\int_0^{\frac{\pi}{2}}\ell n(1-a^2\sin^2\theta)\,d\theta$$

$$\frac{dI}{da}=\int_0^{\frac{\pi}{2}}\frac{-2a\sin^2\theta}{1-a^2\sin^2\theta}d\theta$$

$$\frac{dI}{da}=-2a\int_{\frac{\pi}{2}}\frac{\sin^2\theta d\theta}{1-a^2\sin^2\theta}0$$

$$=-2a\int_0^{\frac{\pi}{2}}\frac{\sin^2\theta d\theta}{1-a^2(1-\cos^2\theta)}$$

$$=-2a\int_0^{\frac{\pi}{2}}\frac{\sin^2\theta d\theta}{(1-a^2)+a^2\cos^2\theta}$$

Multiply divide by $\sec^4\theta$

$$=-2a\int_0^{\frac{\pi}{2}}\frac{\tan^2\theta\sec^2\theta d\theta}{(1-a^2)\sec^4\theta+a^2\sec^2\theta}$$

$$=-2a\int_0^{\frac{\pi}{2}}\frac{\tan^2\theta\sec^2\theta d\theta}{(1-a^2)(1+\tan^2\theta)^2+a^2(1+\tan^2\theta)}$$

Put $\tan\theta=y$

$$\frac{dI}{da}=-2a\int_0^{\infty}\frac{y^2dy}{\left(1+y^2\right)\left(a^2+\left(1-a^2\right)\left(1+y^2\right)\right)}$$

$$=-2a\int_0^{\infty}\frac{y^2dy}{\left(1+y^2\right)\left(1+y^2-a^2y^2\right)}=-2a\int_0^{\infty}\frac{y^2dy}{\left(1+y^2\right)\left(1+y^2\left(1-a^2\right)\right)}$$

$$\frac{dI}{da}=\frac{-2a}{-a^2}\int_0^{\infty}\frac{1}{1+y^2}-\frac{1}{1+y^2\left(1-a^2\right)}dy$$

$$\frac{dI}{da}=\frac{2}{a}\tan^{-1}(y)-\frac{1}{\sqrt{1-a^2}}\tan^{-1}\left(\frac{y\sqrt{1-a^2}}{1}\right)\Bigg|_0^{\infty}$$

$$\frac{dI}{da}=\frac{2}{a}\left(\frac{\pi}{2}-\frac{\pi}{2}\cdot\frac{1}{\sqrt{1-a^2}}\right)$$

$$\frac{dI}{da}=\pi\left(\frac{1}{a}-\frac{1}{a\sqrt{1-a^2}}\right)$$

$$I=\pi\left(\ell na-\ell n\left|\frac{1}{a}-\frac{\sqrt{1-a^2}}{a}\right|\right)+c$$

$$I=\pi\ell n\left(\frac{a^2}{1-\sqrt{1-a^2}}\right)+c$$

$$I=\pi\ell n\left(\frac{a^2\left(1+\sqrt{1-a^2}\right)}{a^2}\right)+c$$

$$I=\pi\ell n\left(1+\sqrt{1-a^2}\right)+c$$

when $a=0\Rightarrow I=0\Rightarrow \pi\ell n2+c=0$

$$I=\pi\ell n\left(\frac{1+\sqrt{1-a^2}}{2}\right)$$

5. $\int_0^{\pi}\frac{\ell n\,(1+a\,\cos x)}{\cos x}\,dx\;\;(|a|<1)=\pi\sin^{-1}a$

Sol. $I=\int_0^{\pi}\frac{\ell n(1+a\cos x)}{\cos x}dx$

when $a=0\Rightarrow I=0$

$$\frac{dI}{da}=\int_0^{\pi}\frac{\cos x}{\cos x(1+a\cos x)}dx$$

$$\Rightarrow \frac{dI}{da}=\int_0^{\pi}\frac{dx}{1+a\cos x}$$

$$\Rightarrow \frac{dI}{da}=\int_0^{\frac{\pi}{2}}\frac{1}{1+a\cos x}+\frac{1}{1-a\cos x}dx$$

$$=2\int_0^{\frac{\pi}{2}}\frac{\sec^2 x dx}{(1-a^2)+\tan^2 x}$$

Put $\tan x = y$

$$\frac{dI}{da}=2\int_0^{\infty}\frac{dy}{(1-a^2)+y^2}=\frac{2}{\sqrt{1-a^2}}\tan^{-1}\left(\frac{y}{\sqrt{1-a^2}}\right)\Bigg|_0^{\infty}$$

$$=\frac{2}{\sqrt{1-a^2}}\cdot\frac{\pi}{2}$$

$$\Rightarrow \frac{dI}{da}=\frac{\pi}{\sqrt{1-a^2}} \Rightarrow I=\pi\sin^{-1}a+C$$

when $a=0 \Rightarrow I=0 \Rightarrow c=0$

6. $\int_0^{\pi/2} ln\left(\frac{1+a\sin x}{1-a\sin x}\right)\frac{dx}{\sin x}$ $(|a|<1)$ [Ans: $\pi\sin^{-1}a$]

[Sol. $I=\int_0^{\pi/2} ln\left(\frac{1+a\sin x}{1-a\sin x}\right)\frac{dx}{\sin x}$ $(|a|<1)$

$$\frac{dI}{da}=\int_0^{\pi/2}\frac{2\sin x}{(1-a^2\sin^2 x)}\cdot\frac{dx}{\sin x}$$

$$=\int_0^{\pi/2}\frac{2\sec^2 x}{(1+\tan^2 x-a^2\tan^2 x}dx \quad \text{or} \quad \int_0^{\pi/2}\frac{2\sec^2 x}{(1-a^2)\tan^2 x+1}dx$$

(put $\tan x = t$) $\int_0^{\infty}\frac{2\,dt}{(1-a^2)t^2+1}$ or $\frac{2}{(1-a^2)}\int_0^{\infty}\frac{dt}{t^2+\left(\frac{1}{\sqrt{1-a^2}}\right)^2}$

$$= \frac{2}{\sqrt{1-a^2}}\left[\tan^{-1}\left(t\sqrt{1-a^2}\right)\right]_0^{\infty} = \frac{\pi}{\sqrt{1-a^2}}$$

$dI = \dfrac{\pi\, da}{\sqrt{1-a^2}}$; $I = \pi \sin^{-1}a + 0$; $a = 0$, $I = 0$]

7. $\displaystyle\int_0^{\pi/2} \ln(a^2\cos^2 x + b^2\sin^2 x)\, dx$ [Ans. $\pi \ln\dfrac{a+b}{2}$]

[Sol. $I = \displaystyle\int_0^{\pi/2} \ln(a^2\cos^2 x + b^2\sin^2 x)\, dx$ $(a > 0 , b > 0)$

$$= \int_0^{\pi/2} \ln\left\{a^2(\cos^2 x + k^2\sin^2 x)\right\} dx \quad \text{or} \quad \pi \ln a + \underbrace{\int_0^{\pi/2} \ln(\cos^2 x + k^2\sin^2 x)\, dx}_{I_1}$$

$$I_1 = \int_0^{\pi/2} \ln(\cos^2 x + k^2\sin^2 x)\, dx$$

$$\frac{dI_1}{dk} = \int_0^{\pi/2} \frac{2k\sin^2 x}{\cos^2 x + k^2\sin^2 x}\, dx \quad \text{or} \quad 2k\int_0^{\pi/2} \frac{\tan^2 x\ \sec^2 x}{(1+k^2\tan^2 x)(1+\tan^2 x)}\, dx$$

(put $\tan x = t$) $= 2k\displaystyle\int_0^{\infty} \frac{t^2}{(1+k^2t^2)(1+t^2)}\, dt$ or $\dfrac{2k}{k^2-1}\displaystyle\int_0^{\infty} \frac{(1+k^2t^2)-(1+t^2)}{(1+k^2t^2)(1+t^2)}\, dt$

$$= \frac{2k}{k^2-1}\left[\int_0^{\infty}\frac{dt}{1+t^2} - \int_0^{\infty}\frac{dt}{1+k^2t^2}\right] \quad \text{or} \quad \frac{2k}{k^2-1}\left[\frac{\pi}{2} - \frac{1}{k}\tan^{-1}tk\right]_0^{\infty}$$

$$= \frac{2k}{k^2-1}\left[\frac{\pi}{2} - \frac{\pi}{2k}\right]$$

$$\frac{dI_1}{dk} = \frac{2k}{k^2-1}\cdot\frac{\pi}{2}\left(\frac{k-1}{k}\right) = \frac{\pi}{k+1}$$

$I_1 = \pi \ln(1+k) + C$

if $k = 1$; $I_1 = 0$ $\Rightarrow$ $C = -\pi \ln 2$

$$I_1 = \pi \ln\left(\frac{1+k}{2}\right) \quad \text{or} \quad \pi \ln\left(\frac{1+(b/a)}{2}\right) = \pi \ln\left(\frac{a+b}{2a}\right) \quad (\because k = b/a)$$

$\therefore \quad I = \pi\, l\text{na} + \pi\, l\text{n}\left(\frac{a+b}{2a}\right)$ or $\pi\, l\text{n}\left(\frac{a.(a+b)}{2a}\right)$ or $\pi\, l\text{n}\left(\frac{a+b}{2}\right)$ Ans]